

11. Find the coefficient of x^{10} in the power series of each of these functions.
 - a) $1/(1 - 2x)$
 - b) $1/(1 + x)^2$
 - c) $1/(1 - x)^3$
 - d) $1/(1 + 2x)^4$
 - e) $x^4/(1 - 3x)^3$
12. Find the coefficient of x^{12} in the power series of each of these functions.
 - a) $1/(1 + 3x)$
 - b) $1/(1 - 2x)^2$
 - c) $1/(1 + x)^8$
 - d) $1/(1 - 4x)^3$
 - e) $x^3/(1 + 4x)^2$
13. Use generating functions to determine the number of different ways 10 identical balloons can be given to four children if each child receives at least two balloons.
14. Use generating functions to determine the number of different ways 12 identical action figures can be given to five children so that each child receives at most three action figures.
15. Use generating functions to determine the number of different ways 15 identical stuffed animals can be given to six children so that each child receives at least one but no more than three stuffed animals.
16. Use generating functions to find the number of ways to choose a dozen bagels from three varieties—egg, salty, and plain—if at least two bagels of each kind but no more than three salty bagels are chosen.
17. In how many ways can 25 identical donuts be distributed to four police officers so that each officer gets at least three but no more than seven donuts?
18. Use generating functions to find the number of ways to select 14 balls from a jar containing 100 red balls, 100 blue balls, and 100 green balls so that no fewer than 3 and no more than 10 blue balls are selected. Assume that the order in which the balls are drawn does not matter.
19. What is the generating function for the sequence $\{c_k\}$, where c_k is the number of ways to make change for k dollars using \$1 bills, \$2 bills, \$5 bills, and \$10 bills?
20. What is the generating function for the sequence $\{c_k\}$, where c_k represents the number of ways to make change for k pesos using bills worth 10 pesos, 20 pesos, 50 pesos, and 100 pesos?
21. Give a combinatorial interpretation of the coefficient of x^4 in the expansion $(1 + x + x^2 + x^3 + \cdots)^3$. Use this interpretation to find this number.
22. Give a combinatorial interpretation of the coefficient of x^6 in the expansion $(1 + x + x^2 + x^3 + \cdots)^n$. Use this interpretation to find this number.
23. a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1 + x_2 + x_3 = k$ when x_1, x_2 , and x_3 are integers with $x_1 \geq 2, 0 \leq x_2 \leq 3$, and $2 \leq x_3 \leq 5$?
b) Use your answer to part (a) to find a_6 .
24. a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1 + x_2 + x_3 + x_4 = k$ when x_1, x_2, x_3 , and x_4 are integers with $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$, and $x_4 \geq 1$?
b) Use your answer to part (a) to find a_7 .
25. Explain how generating functions can be used to find the number of ways in which postage of r cents can be pasted on an envelope using 3-cent, 4-cent, and 20-cent stamps.
 - a) Assume that the order the stamps are pasted on does not matter.
 - b) Assume that the order in which the stamps are pasted on matters.
 - c) Use your answer to part (a) to determine the number of ways 46 cents of postage can be pasted on an envelope using 3-cent, 4-cent, and 20-cent stamps when the order the stamps are pasted on does not matter. (Use of a computer algebra program is advised.)
 - d) Use your answer to part (b) to determine the number of ways 46 cents of postage can be pasted in a row on an envelope using 3-cent, 4-cent, and 20-cent stamps when the order in which the stamps are pasted on matters. (Use of a computer algebra program is advised.)
26. Explain how generating functions can be used to find the number of ways in which postage of r cents can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps.
 - a) Assume that the order the stamps are pasted on does not matter.
 - b) Assume that the order the stamps are pasted on matters.
 - c) Use your answer to part (a) to determine the number of ways 49 cents of postage can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps when the order the stamps are pasted on does not matter. (Use of a computer algebra program is advised.)
 - d) Use your answer to part (b) to determine the number of ways 49 cents of postage can be pasted on an envelope using 2-cent, 7-cent, 13-cent, and 32-cent stamps when the order in which the stamps are pasted on matters. (Use of a computer algebra program is advised.)
27. Customers at a quirky tropical fruit stand can buy at most four mangos, at most two passion fruit, any even number of papayas, three or more coconuts, and carambolas in groups of five.
 - a) Explain how generating functions can be used to find the number of ways a customer can buy n pieces of these fruits, following the restrictions listed.
 - b) Use your answer in part (a) to determine the number of ways you can buy a dozen pieces of these fruits.
28. a) Show that $1/(1 - x - x^2 - x^3 - x^4 - x^5 - x^6)$ is the generating function for the number of ways that the sum n can be obtained when a die is rolled repeatedly and the order of the rolls matters.
b) Use part (a) to find the number of ways to roll a total of 8 when a die is rolled repeatedly, and the order of the rolls matters. (Use of a computer algebra package is advised.)
29. Use generating functions (and a computer algebra package, if available) to find the number of ways to make change for \$1 using
 - a) dimes and quarters.
 - b) nickels, dimes, and quarters.
 - c) pennies, dimes, and quarters.
 - d) pennies, nickels, dimes, and quarters.

30. Use generating functions (and a computer algebra package, if available) to find the number of ways to make change for \$1 using pennies, nickels, dimes, and quarters with
- no more than 10 pennies.
 - no more than 10 pennies and no more than 10 nickels.
 - * no more than 10 coins.
31. Use generating functions to find the number of ways to make change for \$100 using
- \$10, \$20, and \$50 bills.
 - \$5, \$10, \$20, and \$50 bills.
 - \$5, \$10, \$20, and \$50 bills if at least one bill of each denomination is used.
 - \$5, \$10, and \$20 bills if at least one and no more than four of each denomination is used.
32. If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for each of these sequences?
- $2a_0, 2a_1, 2a_2, 2a_3, \dots$
 - $0, a_0, a_1, a_2, a_3, \dots$ (assuming that terms follow the pattern of all but the first term)
 - $0, 0, 0, 0, a_2, a_3, \dots$ (assuming that terms follow the pattern of all but the first four terms)
 - $a_2, a_3, a_4, \dots$
 - $a_1, 2a_2, 3a_3, 4a_4, \dots$ [Hint: Calculus required here.]
 - $a_0^2, 2a_0a_1, a_1^2 + 2a_0a_2, 2a_0a_3 + 2a_1a_2, 2a_0a_4 + 2a_1a_3 + a_2^2, \dots$
33. If $G(x)$ is the generating function for the sequence $\{a_k\}$, what is the generating function for each of these sequences?
- $0, 0, 0, a_3, a_4, a_5, \dots$ (assuming that terms follow the pattern of all but the first three terms)
 - $a_0, 0, a_1, 0, a_2, 0, \dots$
 - $0, 0, 0, 0, a_0, a_1, a_2, \dots$ (assuming that terms follow the pattern of all but the first four terms)
 - $a_0, 2a_1, 4a_2, 8a_3, 16a_4, \dots$
 - $0, a_0, a_1/2, a_2/3, a_3/4, \dots$ [Hint: Calculus required here.]
 - $a_0, a_0 + a_1, a_0 + a_1 + a_2, a_0 + a_1 + a_2 + a_3, \dots$
34. Use generating functions to solve the recurrence relation $a_k = 7a_{k-1}$ with the initial condition $a_0 = 5$.
35. Use generating functions to solve the recurrence relation $a_k = 3a_{k-1} + 2$ with the initial condition $a_0 = 1$.
36. Use generating functions to solve the recurrence relation $a_k = 3a_{k-1} + 4^{k-1}$ with the initial condition $a_0 = 1$.
37. Use generating functions to solve the recurrence relation $a_k = 5a_{k-1} - 6a_{k-2}$ with initial conditions $a_0 = 6$ and $a_1 = 30$.
38. Use generating functions to solve the recurrence relation $a_k = a_{k-1} + 2a_{k-2} + 2^k$ with initial conditions $a_0 = 4$ and $a_1 = 12$.
39. Use generating functions to solve the recurrence relation $a_k = 4a_{k-1} - 4a_{k-2} + k^2$ with initial conditions $a_0 = 2$ and $a_1 = 5$.
40. Use generating functions to solve the recurrence relation $a_k = 2a_{k-1} + 3a_{k-2} + 4^k + 6$ with initial conditions $a_0 = 20, a_1 = 60$.

41. Use generating functions to find an explicit formula for the Fibonacci numbers.

*42. a) Show that if n is a positive integer, then

$$\binom{-1/2}{n} = \binom{2n}{n} / (-4)^n.$$

- b) Use the extended binomial theorem and part (a) to show that the coefficient of x^n in the expansion of $(1 - 4x)^{-1/2}$ is $\binom{2n}{n}$ for all nonnegative integers n .

*43. (Calculus required) Let $\{C_n\}$ be the sequence of Catalan numbers, that is, the solution to the recurrence relation $C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1}$ with $C_0 = C_1 = 1$ (see Example 5 in Section 8.1).

- a) Show that if $G(x)$ is the generating function for the sequence of Catalan numbers, then $xG(x)^2 - G(x) + 1 = 0$. Conclude (using the initial conditions) that $G(x) = (1 - \sqrt{1 - 4x})/(2x)$.

- b) Use Exercise 42 to conclude that

$$G(x) = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n,$$

so that

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

- c) Show that $C_n \geq 2^{n-1}$ for all positive integers n .

44. Use generating functions to prove Pascal's identity: $C(n, r) = C(n-1, r) + C(n-1, r-1)$ when n and r are positive integers with $r < n$. [Hint: Use the identity $(1+x)^n = (1+x)^{n-1} + x(1+x)^{n-1}$.]

45. Use generating functions to prove Vandermonde's identity: $C(m+n, r) = \sum_{k=0}^r C(m, r-k)C(n, k)$, whenever m, n , and r are nonnegative integers with r not exceeding either m or n . [Hint: Look at the coefficient of x^r in both sides of $(1+x)^{m+n} = (1+x)^m(1+x)^n$.]

46. This exercise shows how to use generating functions to derive a formula for the sum of the first n squares.

- a) Show that $(x^2 + x)/(1-x)^4$ is the generating function for the sequence $\{a_n\}$, where $a_n = 1^2 + 2^2 + \dots + n^2$.

- b) Use part (a) to find an explicit formula for the sum $1^2 + 2^2 + \dots + n^2$.

The **exponential generating function** for the sequence $\{a_n\}$ is the series

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n.$$

For example, the exponential generating function for the sequence $1, 1, 1, \dots$ is the function $\sum_{n=0}^{\infty} x^n/n! = e^x$. (You will find this particular series useful in these exercises.) Note that e^x is the (ordinary) generating function for the sequence $1, 1, 1/2!, 1/3!, 1/4!, \dots$.

47. Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where
- $a_n = 2$.
 - $a_n = (-1)^n$.
 - $a_n = 3^n$.
 - $a_n = n + 1$.
 - $a_n = 1/(n + 1)$.
48. Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where
- $a_n = (-2)^n$.
 - $a_n = -1$.
 - $a_n = n$.
 - $a_n = n(n - 1)$.
 - $a_n = 1/((n + 1)(n + 2))$.
49. Find the sequence with each of these functions as its exponential generating function.
- $f(x) = e^{-x}$
 - $f(x) = 3x^{2x}$
 - $f(x) = e^{3x} - 3e^{2x}$
 - $f(x) = (1 - x) + e^{-2x}$
 - $f(x) = e^{-2x} - (1/(1 - x))$
 - $f(x) = e^{-3x} - (1 + x) + (1/(1 - 2x))$
 - $f(x) = e^{x^2}$
50. Find the sequence with each of these functions as its exponential generating function.
- $f(x) = e^{3x}$
 - $f(x) = 2e^{-3x+1}$
 - $f(x) = e^{4x} + e^{-4x}$
 - $f(x) = (1 + 2x) + e^{3x}$
 - $f(x) = e^x - (1/(1 + x))$
 - $f(x) = xe^x$
 - $f(x) = e^{x^3}$
51. A coding system encodes messages using strings of octal (base 8) digits. A codeword is considered valid if and only if it contains an even number of 7s.
- Find a linear nonhomogeneous recurrence relation for the number of valid codewords of length n . What are the initial conditions?
 - Solve this recurrence relation using Theorem 6 in Section 8.2.
 - Solve this recurrence relation using generating functions.
- *52. A coding system encodes messages using strings of base 4 digits (that is, digits from the set $\{0, 1, 2, 3\}$). A codeword is valid if and only if it contains an even number of 0s and an even number of 1s. Let a_n equal the number of valid codewords of length n . Furthermore, let b_n , c_n , and d_n equal the number of strings of base 4 digits of length n with an even number of 0s and an odd number of 1s, with an odd number of 0s and an even number of 1s, and with an odd number of 0s and an odd number of 1s, respectively.
- Show that $d_n = 4^n - a_n - b_n - c_n$. Use this to show that $a_{n+1} = 2a_n + b_n + c_n$, $b_{n+1} = b_n - c_n + 4^n$, and $c_{n+1} = c_n - b_n + 4^n$.
 - What are a_1 , b_1 , c_1 , and d_1 ?
 - Use parts (a) and (b) to find a_3 , b_3 , c_3 , and d_3 .
 - Use the recurrence relations in part (a), together with the initial conditions in part (b), to set up three equations relating the generating functions $A(x)$, $B(x)$, and $C(x)$ for the sequences $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$, respectively.
 - Solve the system of equations from part (d) to get explicit formulae for $A(x)$, $B(x)$, and $C(x)$ and use these to get explicit formulae for a_n , b_n , c_n , and d_n .

Generating functions are useful in studying the number of different types of partitions of an integer n . A **partition** of a positive integer is a way to write this integer as the sum of positive integers where repetition is allowed and the order of the integers in the sum does not matter. For example, the partitions of 5 (with no restrictions) are $1 + 1 + 1 + 1 + 1$, $1 + 1 + 1 + 2$, $1 + 1 + 3$, $1 + 2 + 2$, $1 + 4$, $2 + 3$, and 5. Exercises 53–58 illustrate some of these uses.

53. Show that the coefficient $p(n)$ of x^n in the formal power series expansion of $1/((1-x)(1-x^2)(1-x^3)\cdots)$ equals the number of partitions of n .
54. Show that the coefficient $p_o(n)$ of x^n in the formal power series expansion of $1/((1-x)(1-x^3)(1-x^5)\cdots)$ equals the number of partitions of n into odd integers, that is, the number of ways to write n as the sum of odd positive integers, where the order does not matter and repetitions are allowed.
55. Show that the coefficient $p_d(n)$ of x^n in the formal power series expansion of $(1+x)(1+x^2)(1+x^3)\cdots$ equals the number of partitions of n into distinct parts, that is, the number of ways to write n as the sum of positive integers, where the order does not matter but no repetitions are allowed.
56. Find $p_o(n)$, the number of partitions of n into odd parts with repetitions allowed, and $p_d(n)$, the number of partitions of n into distinct parts, for $1 \leq n \leq 8$, by writing each partition of each type for each integer.
57. Show that if n is a positive integer, then the number of partitions of n into distinct parts equals the number of partitions of n into odd parts with repetitions allowed; that is, $p_o(n) = p_d(n)$. [Hint: Show that the generating functions for $p_o(n)$ and $p_d(n)$ are equal.]
- **58. (Requires calculus) Use the generating function of $p(n)$ to show that $p(n) \leq e^{C\sqrt{n}}$ for some constant C . [Hardy and Ramanujan showed that $p(n) \sim e^{\pi\sqrt{2/3}\sqrt{n}}/(4\sqrt{3}n)$, which means that the ratio of $p(n)$ and the right-hand side approaches 1 as n approaches infinity.]

Suppose that X is a random variable on a sample space S such that $X(s)$ is a nonnegative integer for all $s \in S$. The **probability generating function** for X is

Links 
$$G_X(x) = \sum_{k=0}^{\infty} p(X(s) = k)x^k.$$

59. (Requires calculus) Show that if G_X is the probability generating function for a random variable X such that $X(s)$ is a nonnegative integer for all $s \in S$, then
- $G_X(1) = 1$.
 - $E(X) = G'_X(1)$.
 - $V(X) = G''_X(1) + G'_X(1) - (G'_X(1))^2$.
60. Let X be the random variable whose value is n if the first success occurs on the n th trial when independent Bernoulli trials are performed, each with probability of success p .
- Find a closed formula for the probability generating function G_X .
 - Find the expected value and the variance of X using Exercise 59 and the closed form for the probability generating function found in part (a).

61. Let m be a positive integer. Let X_m be the random variable whose value is n if the m th success occurs on the $(n + m)$ th trial when independent Bernoulli trials are performed, each with probability of success p .
- a) Using Exercise 32 in the Supplementary Exercises of Chapter 7, show that the probability generating function G_{X_m} is given by $G_{X_m}(x) = p^m / (1 - qx)^m$, where $q = 1 - p$.
- b) Find the expected value and the variance of X_m using Exercise 59 and the closed form for the probability generating function in part (a).
62. Show that if X and Y are independent random variables on a sample space S such that $X(s)$ and $Y(s)$ are nonnegative integers for all $s \in S$, then $G_{X+Y}(x) = G_X(x)G_Y(x)$.

8.5 Inclusion–Exclusion

8.5.1 Introduction

A discrete mathematics class contains 30 women and 50 sophomores. How many students in the class are either women or sophomores? This question cannot be answered unless more information is provided. Adding the number of women in the class and the number of sophomores probably does not give the correct answer, because women sophomores are counted twice. This observation shows that the number of students in the class that are either sophomores or women is the sum of the number of women and the number of sophomores in the class minus the number of women sophomores. A technique for solving such counting problems was introduced in Section 6.1. In this section we will generalize the ideas introduced in that section to solve problems that require us to count the number of elements in the union of more than two sets.

8.5.2 The Principle of Inclusion–Exclusion

How many elements are in the union of two finite sets? In Section 2.2 we showed that the number of elements in the union of the two sets A and B is the sum of the numbers of elements in the sets minus the number of elements in their intersection. That is,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

As we showed in Section 6.1, the formula for the number of elements in the union of two sets is useful in counting problems. Examples 1–3 provide additional illustrations of the usefulness of this formula.

EXAMPLE 1 In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

Solution: Let A be the set of students in the class majoring in computer science and B be the set of students in the class majoring in mathematics. Then $A \cap B$ is the set of students in the class who are joint mathematics and computer science majors. Because every student in the class is majoring in either computer science or mathematics (or both), it follows that the number of students in the class is $|A \cup B|$. Therefore,

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 25 + 13 - 8 = 30. \end{aligned}$$

Therefore, there are 30 students in the class. This computation is illustrated in Figure 1. ◀

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$$

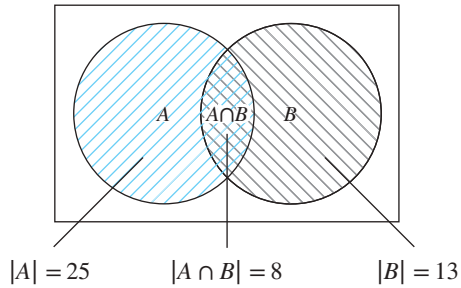


FIGURE 1 The set of students in a discrete mathematics class.

$$|A \cup B| = |A| + |B| - |A \cap B| = 142 + 90 - 12 = 220$$

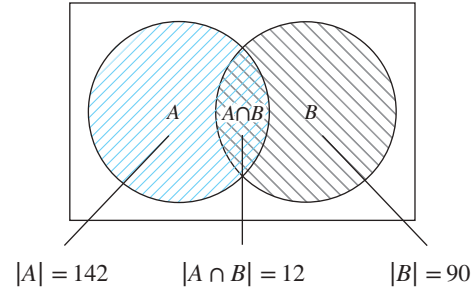


FIGURE 2 The set of positive integers not exceeding 1000 divisible by either 7 or 11.

EXAMPLE 2 How many positive integers not exceeding 1000 are divisible by 7 or 11?

Solution: Let A be the set of positive integers not exceeding 1000 that are divisible by 7, and let B be the set of positive integers not exceeding 1000 that are divisible by 11. Then $A \cup B$ is the set of integers not exceeding 1000 that are divisible by either 7 or 11, and $A \cap B$ is the set of integers not exceeding 1000 that are divisible by both 7 and 11. From Example 2 of Section 4.1, we know that among the positive integers not exceeding 1000 there are $\lfloor 1000/7 \rfloor$ integers divisible by 7 and $\lfloor 1000/11 \rfloor$ divisible by 11. Because 7 and 11 are relatively prime, the integers divisible by both 7 and 11 are those divisible by $7 \cdot 11$. Consequently, there are $\lfloor 1000/(11 \cdot 7) \rfloor$ positive integers not exceeding 1000 that are divisible by both 7 and 11. It follows that there are

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= \left\lfloor \frac{1000}{7} \right\rfloor + \left\lfloor \frac{1000}{11} \right\rfloor - \left\lfloor \frac{1000}{7 \cdot 11} \right\rfloor \\ &= 142 + 90 - 12 = 220 \end{aligned}$$

positive integers not exceeding 1000 that are divisible by either 7 or 11. This computation is illustrated in Figure 2. ▶

Example 3 shows how to find the number of elements in a finite universal set that are outside the union of two sets.

EXAMPLE 3 Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and mathematics. How many are not taking a course either in computer science or in mathematics?

Solution: To find the number of freshmen who are not taking a course in either mathematics or computer science, subtract the number that are taking a course in either of these subjects from the total number of freshmen. Let A be the set of all freshmen taking a course in computer science, and let B be the set of all freshmen taking a course in mathematics. It follows that $|A| = 453$, $|B| = 567$, and $|A \cap B| = 299$. The number of freshmen taking a course in either computer science or mathematics is

$$|A \cup B| = |A| + |B| - |A \cap B| = 453 + 567 - 299 = 721.$$

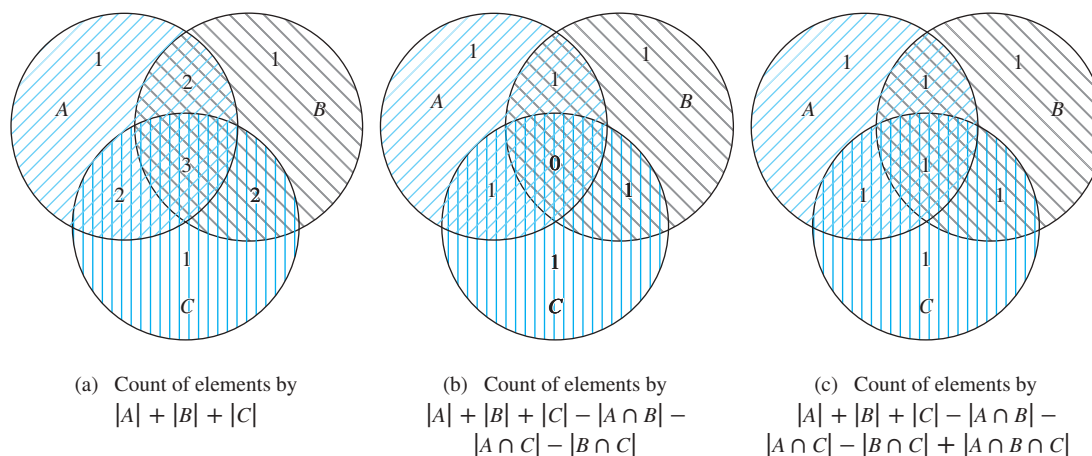


FIGURE 3 Finding a formula for the number of elements in the union of three sets.

Consequently, there are $1807 - 721 = 1086$ freshmen who are not taking a course in computer science or mathematics. ▶

We will now begin our development of a formula for the number of elements in the union of a finite number of sets. The formula we will develop is called the **principle of inclusion–exclusion**. For concreteness, before we consider unions of n sets, where n is any positive integer, we will derive a formula for the number of elements in the union of three sets A , B , and C . To construct this formula, we note that $|A| + |B| + |C|$ counts each element that is in exactly one of the three sets once, elements that are in exactly two of the sets twice, and elements in all three sets three times. This is illustrated in the first panel in Figure 3.

To remove the overcount of elements in more than one of the sets, we subtract the number of elements in the intersections of all pairs of the three sets. We obtain

$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C|.$$

This expression still counts elements that occur in exactly one of the sets once. An element that occurs in exactly two of the sets is also counted exactly once, because this element will occur in one of the three intersections of sets taken two at a time. However, those elements that occur in all three sets will be counted zero times by this expression, because they occur in all three intersections of sets taken two at a time. This is illustrated in the second panel in Figure 3.

To remedy this undercount, we add the number of elements in the intersection of all three sets. This final expression counts each element once, whether it is in one, two, or three of the sets. Thus,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

This formula is illustrated in the third panel of Figure 3.

Example 4 illustrates how this formula can be used.

EXAMPLE 4 A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both

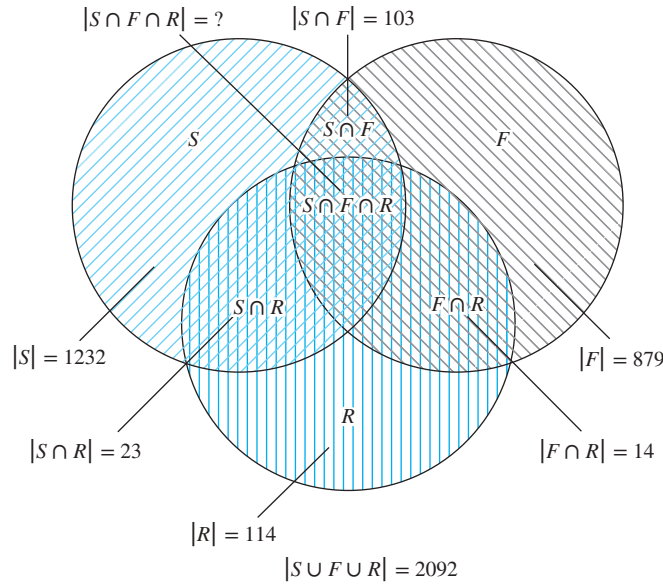


FIGURE 4 The set of students who have taken courses in Spanish, French, and Russian.

French and Russian. If 2092 students have taken at least one of Spanish, French, and Russian, how many students have taken a course in all three languages?

Solution: Let S be the set of students who have taken a course in Spanish, F the set of students who have taken a course in French, and R the set of students who have taken a course in Russian. Then

$$\begin{aligned} |S| &= 1232, & |F| &= 879, & |R| &= 114, \\ |S \cap F| &= 103, & |S \cap R| &= 23, & |F \cap R| &= 14, \end{aligned}$$

and

$$|S \cup F \cup R| = 2092.$$

When we insert these quantities into the equation

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|$$

we obtain

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|.$$

We now solve for $|S \cap F \cap R|$. We find that $|S \cap F \cap R| = 7$. Therefore, there are seven students who have taken courses in Spanish, French, and Russian. This is illustrated in Figure 4. ▶

We will now state and prove the **inclusion–exclusion principle** for n sets, where n is a positive integer. This principle tells us that we can count the elements in a union of n sets by adding the number of elements in the sets, then subtracting the sum of the number of elements in all intersections of two of these sets, then adding the number of elements in all intersections

of three of these sets, and so on, until we reach the number of elements in the intersection of all the sets. It is added when there is an odd number of sets and added when there is an even number of sets.

THEOREM 1 THE PRINCIPLE OF INCLUSION–EXCLUSION Let $A_1, A_2, \dots, A_n$ be finite sets. Then

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| &= \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

Proof: We will prove the formula by showing that an element in the union is counted exactly once by the right-hand side of the equation. Suppose that a is a member of exactly r of the sets $A_1, A_2, \dots, A_n$ where $1 \leq r \leq n$. This element is counted $C(r, 1)$ times by $\sum |A_i|$. It is counted $C(r, 2)$ times by $\sum |A_i \cap A_j|$. In general, it is counted $C(r, m)$ times by the summation involving m of the sets A_i . Thus, this element is counted exactly

$$C(r, 1) - C(r, 2) + C(r, 3) - \dots + (-1)^{r+1} C(r, r)$$

times by the expression on the right-hand side of this equation. Our goal is to evaluate this quantity. By Corollary 2 of Section 6.4, we have

$$C(r, 0) - C(r, 1) + C(r, 2) - \dots + (-1)^r C(r, r) = 0.$$

Hence,

$$1 = C(r, 0) = C(r, 1) - C(r, 2) + \dots + (-1)^{r+1} C(r, r).$$

Therefore, each element in the union is counted exactly once by the expression on the right-hand side of the equation. This proves the principle of inclusion–exclusion. 

The inclusion–exclusion principle gives a formula for the number of elements in the union of n sets for every positive integer n . There are terms in this formula for the number of elements in the intersection of every nonempty subset of the collection of the n sets. Hence, there are $2^n - 1$ terms in this formula.

EXAMPLE 5 Give a formula for the number of elements in the union of four sets.

Extra Examples 

Solution: The inclusion–exclusion principle shows that

$$\begin{aligned} |A_1 \cup A_2 \cup A_3 \cup A_4| &= |A_1| + |A_2| + |A_3| + |A_4| \\ &\quad - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_1 \cap A_4| - |A_2 \cap A_3| - |A_2 \cap A_4| \\ &\quad - |A_3 \cap A_4| + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + |A_1 \cap A_3 \cap A_4| \\ &\quad + |A_2 \cap A_3 \cap A_4| - |A_1 \cap A_2 \cap A_3 \cap A_4|. \end{aligned}$$

Note that this formula contains 15 different terms, one for each nonempty subset of $\{A_1, A_2, A_3, A_4\}$. 

Exercises

- How many elements are in $A_1 \cup A_2$ if there are 12 elements in A_1 , 18 elements in A_2 , and
 - $A_1 \cap A_2 = \emptyset$
 - $|A_1 \cap A_2| = 1$
 - $|A_1 \cap A_2| = 6$
 - $A_1 \subseteq A_2$
- There are 345 students at a college who have taken a course in calculus, 212 who have taken a course in discrete mathematics, and 188 who have taken courses in both calculus and discrete mathematics. How many students have taken a course in either calculus or discrete mathematics?
- A survey of households in the United States reveals that 96% have at least one television set, 98% have telephone service, and 95% have telephone service and at least one television set. What percentage of households in the United States have neither telephone service nor a television set?
- A marketing report concerning personal computers states that 650,000 owners will buy a printer for their machines next year and 1,250,000 will buy at least one software package. If the report states that 1,450,000 owners will buy either a printer or at least one software package, how many will buy both a printer and at least one software package?
- Find the number of elements in $A_1 \cup A_2 \cup A_3$ if there are 100 elements in each set and if
 - the sets are pairwise disjoint.
 - there are 50 common elements in each pair of sets and no elements in all three sets.
 - there are 50 common elements in each pair of sets and 25 elements in all three sets.
 - the sets are equal.
- Find the number of elements in $A_1 \cup A_2 \cup A_3$ if there are 100 elements in A_1 , 1000 in A_2 , and 10,000 in A_3 if
 - $A_1 \subseteq A_2$ and $A_2 \subseteq A_3$.
 - the sets are pairwise disjoint.
 - there are two elements common to each pair of sets and one element in all three sets.
- There are 2504 computer science students at a school. Of these, 1876 have taken a course in Java, 999 have taken a course in Linux, and 345 have taken a course in C. Further, 876 have taken courses in both Java and Linux, 231 have taken courses in both Linux and C, and 290 have taken courses in both Java and C. If 189 of these students have taken courses in Linux, Java, and C, how many of these 2504 students have not taken a course in any of these three programming languages?
- In a survey of 270 college students, it is found that 64 like Brussels sprouts, 94 like broccoli, 58 like cauliflower, 26 like both Brussels sprouts and broccoli, 28 like both Brussels sprouts and cauliflower, 22 like both broccoli and cauliflower, and 14 like all three vegetables. How many of the 270 students do not like any of these vegetables?
- How many students are enrolled in a course either in calculus, discrete mathematics, data structures, or programming languages at a school if there are 507, 292, 312, and 344 students in these courses, respectively; 14 in both calculus and data structures; 213 in both calculus and programming languages; 211 in both discrete mathematics and data structures; 43 in both discrete mathematics and programming languages; and no student may take calculus and discrete mathematics, or data structures and programming languages, concurrently?
- Find the number of positive integers not exceeding 100 that are not divisible by 5 or by 7.
- Find the number of positive integers not exceeding 1000 that are not divisible by 3, 17, or 35.
- Find the number of positive integers not exceeding 10,000 that are not divisible by 3, 4, 7, or 11.
- Find the number of positive integers not exceeding 100 that are either odd or the square of an integer.
- Find the number of positive integers not exceeding 1000 that are either the square or the cube of an integer.
- How many bit strings of length eight do not contain six consecutive 0s?
- * How many permutations of the 26 letters of the English alphabet do not contain any of the strings *fish*, *rat* or *bird*?
- How many permutations of the 10 digits either begin with the 3 digits 987, contain the digits 45 in the fifth and sixth positions, or end with the 3 digits 123?
- How many elements are in the union of four sets if each of the sets has 100 elements, each pair of the sets shares 50 elements, each three of the sets share 25 elements, and there are 5 elements in all four sets?
- How many elements are in the union of four sets if the sets have 50, 60, 70, and 80 elements, respectively, each pair of the sets has 5 elements in common, each triple of the sets has 1 common element, and no element is in all four sets?
- How many terms are there in the formula for the number of elements in the union of 10 sets given by the principle of inclusion–exclusion?
- Write out the explicit formula given by the principle of inclusion–exclusion for the number of elements in the union of five sets.
- How many elements are in the union of five sets if the sets contain 10,000 elements each, each pair of sets has 1000 common elements, each triple of sets has 100 common elements, every four of the sets have 10 common elements, and there is 1 element in all five sets?
- Write out the explicit formula given by the principle of inclusion–exclusion for the number of elements in the union of six sets when it is known that no three of these sets have a common intersection.

- *24. Prove the principle of inclusion–exclusion using mathematical induction.
25. Let E_1, E_2 , and E_3 be three events from a sample space S . Find a formula for the probability of $E_1 \cup E_2 \cup E_3$.
26. Find the probability that when a fair coin is flipped five times tails comes up exactly three times, the first and last flips come up tails, or the second and fourth flips come up heads.
27. Find the probability that when four numbers from 1 to 100, inclusive, are picked at random with no repetitions allowed, either all are odd, all are divisible by 3, or all are divisible by 5.
28. Find a formula for the probability of the union of four events in a sample space if no three of them can occur at the same time.
29. Find a formula for the probability of the union of five events in a sample space if no four of them can occur at the same time.
30. Find a formula for the probability of the union of n events in a sample space when no two of these events can occur at the same time.
31. Find a formula for the probability of the union of n events in a sample space.

8.6 Applications of Inclusion–Exclusion

8.6.1 Introduction

Many counting problems can be solved using the principle of inclusion–exclusion. For instance, we can use this principle to find the number of primes less than a positive integer. Many problems can be solved by counting the number of onto functions from one finite set to another. The inclusion–exclusion principle can be used to find the number of such functions. The well-known hatcheck problem can be solved using the principle of inclusion–exclusion. This problem asks for the probability that no person is given the correct hat back by a hatcheck person who gives the hats back randomly.

8.6.2 An Alternative Form of Inclusion–Exclusion

There is an alternative form of the principle of inclusion–exclusion that is useful in counting problems. In particular, this form can be used to solve problems that ask for the number of elements in a set that have none of n properties $P_1, P_2, \dots, P_n$.

Let A_i be the subset containing the elements that have property P_i . The number of elements with all the properties $P_{i_1}, P_{i_2}, \dots, P_{i_k}$ will be denoted by $N(P_{i_1}P_{i_2} \dots P_{i_k})$. Writing these quantities in terms of sets, we have

$$|A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}| = N(P_{i_1}P_{i_2} \dots P_{i_k}).$$

If the number of elements with none of the properties $P_1, P_2, \dots, P_n$ is denoted by $N(P'_1P'_2 \dots P'_n)$ and the number of elements in the set is denoted by N , it follows that

$$N(P'_1P'_2 \dots P'_n) = N - |A_1 \cup A_2 \cup \dots \cup A_n|.$$

From the inclusion–exclusion principle, we see that

$$\begin{aligned} N(P'_1P'_2 \dots P'_n) &= N - \sum_{1 \leq i \leq n} N(P_i) + \sum_{1 \leq i < j \leq n} N(P_iP_j) \\ &\quad - \sum_{1 \leq i < j < k \leq n} N(P_iP_jP_k) + \dots + (-1)^n N(P_1P_2 \dots P_n). \end{aligned}$$

Example 1 shows how the principle of inclusion–exclusion can be used to determine the number of solutions in integers of an equation with constraints.

EXAMPLE 1 How many solutions does

$$x_1 + x_2 + x_3 = 11$$

have, where x_1, x_2 , and x_3 are nonnegative integers with $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$?

Solution: To apply the principle of inclusion–exclusion, let a solution have property P_1 if $x_1 > 3$, property P_2 if $x_2 > 4$, and property P_3 if $x_3 > 6$. The number of solutions satisfying the inequalities $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$ is

$$\begin{aligned} N(P'_1 P'_2 P'_3) &= N - N(P_1) - N(P_2) - N(P_3) + N(P_1 P_2) \\ &\quad + N(P_1 P_3) + N(P_2 P_3) - N(P_1 P_2 P_3). \end{aligned}$$

Using the same techniques as in Example 5 of Section 6.5, it follows that

- ▶ N = total number of solutions $= C(3 + 11 - 1, 11) = 78$,
- ▶ $N(P_1)$ = (number of solutions with $x_1 \geq 4$) $= C(3 + 7 - 1, 7) = C(9, 7) = 36$,
- ▶ $N(P_2)$ = (number of solutions with $x_2 \geq 5$) $= C(3 + 6 - 1, 6) = C(8, 6) = 28$,
- ▶ $N(P_3)$ = (number of solutions with $x_3 \geq 7$) $= C(3 + 4 - 1, 4) = C(6, 4) = 15$,
- ▶ $N(P_1 P_2)$ = (number of solutions with $x_1 \geq 4$ and $x_2 \geq 5$) $= C(3 + 2 - 1, 2) = C(4, 2) = 6$,
- ▶ $N(P_1 P_3)$ = (number of solutions with $x_1 \geq 4$ and $x_3 \geq 7$) $= C(3 + 0 - 1, 0) = 1$,
- ▶ $N(P_2 P_3)$ = (number of solutions with $x_2 \geq 5$ and $x_3 \geq 7$) $= 0$,
- ▶ $N(P_1 P_2 P_3)$ = (number of solutions with $x_1 \geq 4$, $x_2 \geq 5$, and $x_3 \geq 7$) $= 0$.

Inserting these quantities into the formula for $N(P'_1 P'_2 P'_3)$ shows that the number of solutions with $x_1 \leq 3$, $x_2 \leq 4$, and $x_3 \leq 6$ equals

$$N(P'_1 P'_2 P'_3) = 78 - 36 - 28 - 15 + 6 + 1 + 0 - 0 = 6.$$

8.6.3 The Sieve of Eratosthenes

In Section 4.3 we showed how to use the sieve of Eratosthenes to find all primes less than a specified positive integer n . Using the principle of inclusion–exclusion, we can find the number of primes not exceeding a specified positive integer with the same reasoning as is used in the sieve of Eratosthenes. Recall that a composite integer is divisible by a prime not exceeding its square root. So, to find the number of primes not exceeding 100, first note that composite integers not exceeding 100 must have a prime factor not exceeding 10. Because the only primes not exceeding 10 are 2, 3, 5, and 7, the primes not exceeding 100 are these four primes and those positive integers greater than 1 and not exceeding 100 that are divisible by none of 2, 3, 5, or 7. To apply the principle of inclusion–exclusion, let P_1 be the property that an integer is divisible by 2, let P_2 be the property that an integer is divisible by 3, let P_3 be the property that an integer is divisible by 5, and let P_4 be the property that an integer is divisible by 7. Thus, the number of primes not exceeding 100 is

$$4 + N(P'_1 P'_2 P'_3 P'_4).$$

Because there are 99 positive integers greater than 1 and not exceeding 100, the principle of inclusion–exclusion shows that

$$\begin{aligned}
 N(P'_1 P'_2 P'_3 P'_4) &= 99 - N(P_1) - N(P_2) - N(P_3) - N(P_4) \\
 &\quad + N(P_1 P_2) + N(P_1 P_3) + N(P_1 P_4) + N(P_2 P_3) + N(P_2 P_4) + N(P_3 P_4) \\
 &\quad - N(P_1 P_2 P_3) - N(P_1 P_2 P_4) - N(P_1 P_3 P_4) - N(P_2 P_3 P_4) \\
 &\quad + N(P_1 P_2 P_3 P_4).
 \end{aligned}$$

The number of integers not exceeding 100 (and greater than 1) that are divisible by all the primes in a subset of $\{2, 3, 5, 7\}$ is $\lfloor 100/N \rfloor$, where N is the product of the primes in this subset. (This follows because any two of these primes have no common factor.) Consequently,

$$\begin{aligned}
 N(P'_1 P'_2 P'_3 P'_4) &= 99 - \left\lfloor \frac{100}{2} \right\rfloor - \left\lfloor \frac{100}{3} \right\rfloor - \left\lfloor \frac{100}{5} \right\rfloor - \left\lfloor \frac{100}{7} \right\rfloor \\
 &\quad + \left\lfloor \frac{100}{2 \cdot 3} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 5} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{3 \cdot 5} \right\rfloor + \left\lfloor \frac{100}{3 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{5 \cdot 7} \right\rfloor \\
 &\quad - \left\lfloor \frac{100}{2 \cdot 3 \cdot 5} \right\rfloor - \left\lfloor \frac{100}{2 \cdot 3 \cdot 7} \right\rfloor - \left\lfloor \frac{100}{2 \cdot 5 \cdot 7} \right\rfloor - \left\lfloor \frac{100}{3 \cdot 5 \cdot 7} \right\rfloor + \left\lfloor \frac{100}{2 \cdot 3 \cdot 5 \cdot 7} \right\rfloor \\
 &= 99 - 50 - 33 - 20 - 14 + 16 + 10 + 7 + 6 + 4 + 2 - 3 - 2 - 1 - 0 + 0 \\
 &= 21.
 \end{aligned}$$

Hence, there are $4 + 21 = 25$ primes not exceeding 100.

8.6.4 The Number of Onto Functions

The principle of inclusion–exclusion can also be used to determine the number of onto functions from a set with m elements to a set with n elements. First consider Example 2.

EXAMPLE 2 How many onto functions are there from a set with six elements to a set with three elements?

Solution: Suppose that the elements in the codomain are b_1, b_2 , and b_3 . Let P_1, P_2 , and P_3 be the properties that b_1, b_2 , and b_3 are not in the range of the function, respectively. Note that a function is onto if and only if it has none of the properties P_1, P_2 , or P_3 . By the inclusion–exclusion principle it follows that the number of onto functions from a set with six elements to a set with three elements is

$$\begin{aligned}
 N(P'_1 P'_2 P'_3) &= N - [N(P_1) + N(P_2) + N(P_3)] \\
 &\quad + [N(P_1 P_2) + N(P_1 P_3) + N(P_2 P_3)] - N(P_1 P_2 P_3),
 \end{aligned}$$

where N is the total number of functions from a set with six elements to one with three elements. We will evaluate each of the terms on the right-hand side of this equation.

From Example 6 of Section 6.1, it follows that $N = 3^6$. Note that $N(P_i)$ is the number of functions that do not have b_i in their range. Hence, there are two choices for the value of the function at each element of the domain. Therefore, $N(P_i) = 2^6$. Furthermore, there are $C(3, 1)$ terms of this kind. Note that $N(P_i P_j)$ is the number of functions that do not have b_i and b_j in their range. Hence, there is only one choice for the value of the function at each element of the domain. Therefore, $N(P_i P_j) = 1^6 = 1$. Furthermore, there are $C(3, 2)$ terms of this kind. Also, note that $N(P_1 P_2 P_3) = 0$, because this term is the number of functions that have none

of b_1 , b_2 , and b_3 in their range. Clearly, there are no such functions, so the number of onto functions from a set with six elements to one with three elements is

$$3^6 - C(3, 1)2^6 + C(3, 2)1^6 = 729 - 192 + 3 = 540.$$

The general result that tells us how many onto functions there are from a set with m elements to one with n elements will now be stated. The proof of this result is left as an exercise for the reader.

THEOREM 1

Let m and n be positive integers with $m \geq n$. Then, there are

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1}C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Counting onto functions is much harder than counting one-to-one functions!

An onto function from a set with m elements to a set with n elements corresponds to a way to distribute the m elements in the domain to n indistinguishable boxes so that no box is empty, and then to associate each of the n elements of the codomain to a box. This means that the number of onto functions from a set with m elements to a set with n elements is the number of ways to distribute m distinguishable objects to n indistinguishable boxes so that no box is empty multiplied by the number of permutations of a set with n elements. Consequently, the number of onto functions from a set with m elements to a set with n elements equals $n!S(m, n)$, where $S(m, n)$ is a *Stirling number of the second kind* defined in Section 6.5. This means that we can use Theorem 1 to deduce the formula given in Section 6.5 for $S(m, n)$. (See Chapter 6 of [MiRo91] for more details about Stirling numbers of the second kind.)

One of the many different applications of Theorem 1 will now be described.

EXAMPLE 3 How many ways are there to assign five different jobs to four different employees if every employee is assigned at least one job?

Solution: Consider the assignment of jobs as a function from the set of five jobs to the set of four employees. An assignment where every employee gets at least one job is the same as an onto function from the set of jobs to the set of employees. Hence, by Theorem 1 it follows that there are

$$4^5 - C(4, 1)3^5 + C(4, 2)2^5 - C(4, 3)1^5 = 1024 - 972 + 192 - 4 = 240$$

ways to assign the jobs so that each employee is assigned at least one job.

8.6.5 Derangements

The principle of inclusion–exclusion will be used to count the permutations of n objects that leave no objects in their original positions. Consider Example 4.

EXAMPLE 4 The Hatcheck Problem A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. What is the probability that no one receives the correct hat?

Remark: The answer is the number of ways the hats can be arranged so that there is no hat in its original position divided by $n!$, the number of permutations of n hats. We will return to this example after we find the number of permutations of n objects that leave no objects in their original position.



A **derangement** is a permutation of objects that leaves no object in its original position. To solve the problem posed in Example 4 we will need to determine the number of derangements of a set of n objects.

EXAMPLE 5 The permutation 21453 is a derangement of 12345 because no number is left in its original position. However, 21543 is not a derangement of 12345, because this permutation leaves 4 fixed. 

Let D_n denote the number of derangements of n objects. For instance, $D_3 = 2$, because the derangements of 123 are 231 and 312. We will evaluate D_n , for all positive integers n , using the principle of inclusion–exclusion.

THEOREM 2

The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right].$$

Proof: Let a permutation have property P_i if it fixes element i . The number of derangements is the number of permutations having none of the properties P_i for $i = 1, 2, \dots, n$. This means that

$$D_n = N(P'_1 P'_2 \cdots P'_n).$$

Using the principle of inclusion–exclusion, it follows that

$$D_n = N - \sum_i N(P_i) + \sum_{i < j} N(P_i P_j) - \sum_{i < j < k} N(P_i P_j P_k) + \cdots + (-1)^n N(P_1 P_2 \cdots P_n),$$

where N is the number of permutations of n elements. This equation states that the number of permutations that fix no elements equals the total number of permutations, less the number that fix at least one element, plus the number that fix at least two elements, less the number that fix at least three elements, and so on. All the quantities that occur on the right-hand side of this equation will now be found.

First, note that $N = n!$, because N is simply the total number of permutations of n elements. Also, $N(P_i) = (n-1)!$. This follows from the product rule, because $N(P_i)$ is the number of permutations that fix element i , so the i th position of the permutation is determined, but each of the remaining positions can be filled arbitrarily. Similarly,

$$N(P_i P_j) = (n-2)!,$$



HISTORICAL NOTE In *rencontres* (matches), an old French card game, the 52 cards in a deck are laid out in a row. The cards of a second deck are laid out with one card of the second deck on top of each card of the first deck. The score is determined by counting the number of matching cards in the two decks. In 1708 Pierre Raymond de Montmort (1678–1719) posed *le problème de rencontres*: What is the probability that no matches take place in the game of rencontres? The solution to Montmort's problem is the probability that a randomly selected permutation of 52 objects is a derangement, namely, $D_{52}/52!$, which, as we will see, is approximately $1/e$.

because this is the number of permutations that fix elements i and j , but where the other $n - 2$ elements can be arranged arbitrarily. In general, note that

$$N(P_{i_1} P_{i_2} \dots P_{i_m}) = (n - m)!,$$

because this is the number of permutations that fix elements $i_1, i_2, \dots, i_m$, but where the other $n - m$ elements can be arranged arbitrarily. Because there are $C(n, m)$ ways to choose m elements from n , it follows that

$$\begin{aligned} \sum_{1 \leq i \leq n} N(P_i) &= C(n, 1)(n - 1)!, \\ \sum_{1 \leq i < j \leq n} N(P_i P_j) &= C(n, 2)(n - 2)!, \end{aligned}$$

and in general,

$$\sum_{1 \leq i_1 < i_2 < \dots < i_m \leq n} N(P_{i_1} P_{i_2} \dots P_{i_m}) = C(n, m)(n - m)!.$$

Consequently, inserting these quantities into our formula for D_n gives

$$\begin{aligned} D_n &= n! - C(n, 1)(n - 1)! + C(n, 2)(n - 2)! - \dots + (-1)^n C(n, n)(n - n)! \\ &= n! - \frac{n!}{1!(n - 1)!}(n - 1)! + \frac{n!}{2!(n - 2)!}(n - 2)! - \dots + (-1)^n \frac{n!}{n! 0!} 0!. \end{aligned}$$

Simplifying this expression gives

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!} \right].$$

It is now straightforward to find D_n for a given positive integer n . For instance, using Theorem 2, it follows that

$$D_3 = 3! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right] = 6 \left(1 - 1 + \frac{1}{2} - \frac{1}{6} \right) = 2,$$

as we have previously remarked.

The solution of the problem in Example 4 can now be given.

Solution: The probability that no one receives the correct hat is $D_n/n!$. By Theorem 2, this probability is

$$\frac{D_n}{n!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \dots + (-1)^n \frac{1}{n!}.$$

The values of this probability for $2 \leq n \leq 7$ are displayed in Table 1.

TABLE 1 The Probability of a Derangement.						
n	2	3	4	5	6	7
$D_n/n!$	0.50000	0.33333	0.37500	0.36667	0.36806	0.36786

By the identity $e^x = \sum_{j=0}^{\infty} x^j/j!$ for all real numbers x (from calculus), we know that

$$e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!} + \cdots \approx 0.368.$$

Because this is an alternating series with terms tending to zero, it follows that as n grows without bound, the probability that no one receives the correct hat converges to $e^{-1} \approx 0.368$. In fact, this probability can be shown to be within $1/(n+1)!$ of e^{-1} . ◀

Exercises

- Suppose that in a bushel of 100 apples there are 20 that have worms in them and 15 that have bruises. Only those apples with neither worms nor bruises can be sold. If there are 10 bruised apples that have worms in them, how many of the 100 apples can be sold?
- Of 1000 applicants for a mountain-climbing trip in the Himalayas, 450 get altitude sickness, 622 are not in good enough shape, and 30 have allergies. An applicant qualifies if and only if this applicant does not get altitude sickness, is in good shape, and does not have allergies. If there are 111 applicants who get altitude sickness and are not in good enough shape, 14 who get altitude sickness and have allergies, 18 who are not in good enough shape and have allergies, and 9 who get altitude sickness, are not in good enough shape, and have allergies, how many applicants qualify?
- How many solutions does the equation $x_1 + x_2 + x_3 = 13$ have where x_1, x_2 , and x_3 are nonnegative integers less than 6?
- Find the number of solutions of the equation $x_1 + x_2 + x_3 + x_4 = 17$, where $x_i, i = 1, 2, 3, 4$, are nonnegative integers such that $x_1 \leq 3, x_2 \leq 4, x_3 \leq 5$, and $x_4 \leq 8$.
- Find the number of primes less than 200 using the principle of inclusion–exclusion.
- An integer is called **squarefree** if it is not divisible by the square of a positive integer greater than 1. Find the number of squarefree positive integers less than 100.
- How many positive integers less than 10,000 are not the second or higher power of an integer?
- How many onto functions are there from a set with seven elements to one with five elements?
- How many ways are there to distribute six different toys to three different children such that each child gets at least one toy?
- In how many ways can eight distinct balls be distributed into three distinct urns if each urn must contain at least one ball?
- In how many ways can seven different jobs be assigned to four different employees so that each employee is assigned at least one job and the most difficult job is assigned to the best employee?
- List all the derangements of $\{1, 2, 3, 4\}$.
- How many derangements are there of a set with seven elements?
- What is the probability that none of 10 people receives the correct hat if a hatcheck person hands their hats back randomly?
- A machine that inserts letters into envelopes goes haywire and inserts letters randomly into envelopes. What is the probability that in a group of 100 letters
 - no letter is put into the correct envelope?
 - exactly one letter is put into the correct envelope?
 - exactly 98 letters are put into the correct envelopes?
 - exactly 99 letters are put into the correct envelopes?
 - all letters are put into the correct envelopes?
- A group of n students is assigned seats for each of two classes in the same classroom. How many ways can these seats be assigned if no student is assigned the same seat for both classes?
- How many ways can the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 be arranged so that no even digit is in its original position?
- Use a combinatorial argument to show that the sequence $\{D_n\}$, where D_n denotes the number of derangements of n objects, satisfies the recurrence relation

$$D_n = (n-1)(D_{n-1} + D_{n-2})$$
 for $n \geq 2$. [Hint: Note that there are $n-1$ choices for the first element k of a derangement. Consider separately the derangements that start with k that do and do not have 1 in the k th position.]
- Use Exercise 18 to show that

$$D_n = nD_{n-1} + (-1)^n$$
 for $n \geq 1$.
- Use Exercise 19 to find an explicit formula for D_n .
- For which positive integers n is D_n , the number of derangements of n objects, even?
- Suppose that p and q are distinct primes. Use the principle of inclusion–exclusion to find $\phi(pq)$, the number of positive integers not exceeding pq that are relatively prime to pq .
- Use the principle of inclusion–exclusion to derive a formula for $\phi(n)$ when the prime factorization of n is

$$n = p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}.$$

*24. Show that if n is a positive integer, then

$$n! = C(n, 0)D_n + C(n, 1)D_{n-1} + \cdots + C(n, n-1)D_1 + C(n, n)D_0,$$

where D_k is the number of derangements of k objects.

25. How many derangements of $\{1, 2, 3, 4, 5, 6\}$ begin with the integers 1, 2, and 3, in some order?

26. How many derangements of $\{1, 2, 3, 4, 5, 6\}$ end with the integers 1, 2, and 3, in some order?

27. Prove Theorem 1.

Key Terms and Results

TERMS

recurrence relation: a formula expressing terms of a sequence, except for some initial terms, as a function of one or more previous terms of the sequence

initial conditions for a recurrence relation: the values of the terms of a sequence satisfying the recurrence relation before this relation takes effect

dynamic programming: an algorithmic paradigm that finds the solution to an optimization problem by recursively breaking down the problem into overlapping subproblems and combining their solutions with the help of a recurrence relation

linear homogeneous recurrence relation with constant coefficients: a recurrence relation that expresses the terms of a sequence, except initial terms, as a linear combination of previous terms

characteristic roots of a linear homogeneous recurrence relation with constant coefficients: the roots of the polynomial associated with a linear homogeneous recurrence relation with constant coefficients

linear nonhomogeneous recurrence relation with constant coefficients: a recurrence relation that expresses the terms of a sequence, except for initial terms, as a linear combination of previous terms plus a function that is not identically zero that depends only on the index

divide-and-conquer algorithm: an algorithm that solves a problem recursively by splitting it into a fixed number of smaller nonoverlapping subproblems of the same type

generating function of a sequence: the formal series that has the n th term of the sequence as the coefficient of x^n

sieve of Eratosthenes: a procedure for finding the primes less than a specified positive integer

derangement: a permutation of objects such that no object is in its original place

RESULTS

the formula for the number of elements in the union of two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

the formula for the number of elements in the union of three finite sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

the principle of inclusion–exclusion:

$$\begin{aligned} |A_1 \cup A_2 \cup \cdots \cup A_n| &= \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| \\ &\quad + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| \\ &\quad - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n| \end{aligned}$$

the number of onto functions from a set with m elements to a set with n elements:

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1} C(n, n-1) \cdot 1^m$$

the number of derangements of n objects:

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!} \right]$$

Review Questions

- What is a recurrence relation?
 - Find a recurrence relation for the amount of money that will be in an account after n years if \$1,000,000 is deposited in an account yielding 9% annually.
- Explain how the Fibonacci numbers are used to solve Fibonacci's problem about rabbits.
- Find a recurrence relation for the number of steps needed to solve the Tower of Hanoi puzzle.
 - Show how this recurrence relation can be solved using iteration.
- Explain how to find a recurrence relation for the number of bit strings of length n not containing two consecutive 1s.
 - Describe another counting problem that has a solution satisfying the same recurrence relation.

5. a) What is dynamic programming and how are recurrence relations used in algorithms that follow this paradigm?
b) Explain how dynamic programming can be used to schedule talks in a lecture hall from a set of possible talks to maximize overall attendance.
6. Define a linear homogeneous recurrence relation of degree k .
7. a) Explain how to solve linear homogeneous recurrence relations of degree 2.
b) Solve the recurrence relation $a_n = 13a_{n-1} - 22a_{n-2}$ for $n \geq 2$ if $a_0 = 3$ and $a_1 = 15$.
c) Solve the recurrence relation $a_n = 14a_{n-1} - 49a_{n-2}$ for $n \geq 2$ if $a_0 = 3$ and $a_1 = 35$.
8. a) Explain how to find $f(b^k)$ where k is a positive integer if $f(n)$ satisfies the divide-and-conquer recurrence relation $f(n) = af(n/b) + g(n)$ whenever b divides the positive integer n .
b) Find $f(256)$ if $f(n) = 3f(n/4) + 5n/4$ and $f(1) = 7$.
9. a) Derive a divide-and-conquer recurrence relation for the number of comparisons used to find a number in a list using a binary search.
b) Give a big- O estimate for the number of comparisons used by a binary search from the divide-and-conquer recurrence relation you gave in (a) using Theorem 1 in Section 8.3.
10. a) Give a formula for the number of elements in the union of three sets.
b) Explain why this formula is valid.
c) Explain how to use the formula from (a) to find the number of integers not exceeding 1000 that are divisible by 6, 10, or 15.
- d) Explain how to use the formula from (a) to find the number of solutions in nonnegative integers to the equation $x_1 + x_2 + x_3 + x_4 = 22$ with $x_1 < 8$, $x_2 < 6$, and $x_3 < 5$.
11. a) Give a formula for the number of elements in the union of four sets and explain why it is valid.
b) Suppose the sets A_1, A_2, A_3 , and A_4 each contain 25 elements, the intersection of any two of these sets contains 5 elements, the intersection of any three of these sets contains 2 elements, and 1 element is in all four of the sets. How many elements are in the union of the four sets?
12. a) State the principle of inclusion–exclusion.
b) Outline a proof of this principle.
13. Explain how the principle of inclusion–exclusion can be used to count the number of onto functions from a set with m elements to a set with n elements.
14. a) How can you count the number of ways to assign m jobs to n employees so that each employee is assigned at least one job?
b) How many ways are there to assign seven jobs to three employees so that each employee is assigned at least one job?
15. Explain how the inclusion–exclusion principle can be used to count the number of primes not exceeding the positive integer n .
16. a) Define a derangement.
b) Why is counting the number of ways a hatcheck person can return hats to n people, so that no one receives the correct hat, the same as counting the number of derangements of n objects?
c) Explain how to count the number of derangements of n objects.

Supplementary Exercises

1. A group of 10 people begin a chain letter, with each person sending the letter to four other people. Each of these people sends the letter to four additional people.
 - a) Find a recurrence relation for the number of letters sent at the n th stage of this chain letter, if no person ever receives more than one letter.
 - b) What are the initial conditions for the recurrence relation in part (a)?
 - c) How many letters are sent at the n th stage of the chain letter?
2. A nuclear reactor has created 18 grams of a particular radioactive isotope. Every hour 1% of this radioactive isotope decays.
 - a) Set up a recurrence relation for the amount of this isotope left n hours after its creation.
 - b) What are the initial conditions for the recurrence relation in part (a)?
 - c) Solve this recurrence relation.
3. Every hour the U.S. government prints 10,000 more \$1 bills, 4000 more \$5 bills, 3000 more \$10 bills, 2500 more \$20 bills, 1000 more \$50 bills, and the same number of \$100 bills as it did the previous hour. In the initial hour 1000 of each bill were produced.
 - a) Set up a recurrence relation for the amount of money produced in the n th hour.
 - b) What are the initial conditions for the recurrence relation in part (a)?
 - c) Solve the recurrence relation for the amount of money produced in the n th hour.
 - d) Set up a recurrence relation for the total amount of money produced in the first n hours.
 - e) Solve the recurrence relation for the total amount of money produced in the first n hours.
4. Suppose that every hour there are two new bacteria in a colony for each bacterium that was present the previous hour, and that all bacteria 2 hours old die. The colony starts with 100 new bacteria.

- a) Set up a recurrence relation for the number of bacteria present after n hours.
 - b) What is the solution of this recurrence relation?
 - c) When will the colony contain more than 1 million bacteria?
5. Messages are sent over a communications channel using two different signals. One signal requires 2 microseconds for transmittal, and the other signal requires 3 microseconds for transmittal. Each signal of a message is followed immediately by the next signal.
- a) Find a recurrence relation for the number of different signals that can be sent in n microseconds.
 - b) What are the initial conditions of the recurrence relation in part (a)?
 - c) How many different messages can be sent in 12 microseconds?
6. A small post office has only 4-cent stamps, 6-cent stamps, and 10-cent stamps. Find a recurrence relation for the number of ways to form postage of n cents with these stamps if the order that the stamps are used matters. What are the initial conditions for this recurrence relation?
7. How many ways are there to form these postages using the rules described in Exercise 6?
- a) 12 cents b) 14 cents
 - c) 18 cents d) 22 cents
8. Find the solutions of the simultaneous system of recurrence relations

$$\begin{aligned}a_n &= a_{n-1} + b_{n-1} \\ b_n &= a_{n-1} - b_{n-1}\end{aligned}$$

with $a_0 = 1$ and $b_0 = 2$.

9. Solve the recurrence relation $a_n = a_{n-1}^2/a_{n-2}$ if $a_0 = 1$ and $a_1 = 2$. [Hint: Take logarithms of both sides to obtain a recurrence relation for the sequence $\log a_n$, $n = 0, 1, 2, \dots$]
- *10. Solve the recurrence relation $a_n = a_{n-1}^3 a_{n-2}^2$ if $a_0 = 2$ and $a_1 = 2$. (See the hint for Exercise 9.)
11. Find the solution of the recurrence relation $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3} + 1$ if $a_0 = 2$, $a_1 = 4$, and $a_2 = 8$.
12. Find the solution of the recurrence relation $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$ if $a_0 = 2$, $a_1 = 2$, and $a_2 = 4$.
- *13. Suppose that in Example 1 of Section 8.1 a pair of rabbits leaves the island after reproducing twice. Find a recurrence relation for the number of rabbits on the island in the middle of the n th month.
- *14. In this exercise we construct a dynamic programming algorithm for solving the problem of finding a subset S of items chosen from a set of n items where item i has a weight w_i , which is a positive integer, so that the total weight of the items in S is a maximum but does not exceed a fixed weight limit W . Let $M(j, w)$ denote the maximum total weight of the items in a subset of the first j items such that this total weight does not exceed w . This problem is known as the **knapsack problem**.
 - a) Show that if $w_j > w$, then $M(j, w) = M(j-1, w)$.

- b) Show that if $w_j \leq w$, then $M(j, w) = \max(M(j-1, w), w_j + M(j-1, w - w_j))$.
- c) Use (a) and (b) to construct a dynamic programming algorithm for determining the maximum total weight of items so that this total weight does not exceed W . In your algorithm store the values $M(j, w)$ as they are found.
- d) Explain how you can use the values $M(j, w)$ computed by the algorithm in part (c) to find a subset of items with maximum total weight not exceeding W .

In Exercises 15–18 we develop a dynamic programming algorithm for finding a longest common subsequence of two sequences $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_n$, an important problem in the comparison of DNA of different organisms.

15. Suppose that $c_1, c_2, \dots, c_p$ is a longest common subsequence of the sequences $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_n$.
 - a) Show that if $a_m = b_n$, then $c_p = a_m = b_n$ and $c_1, c_2, \dots, c_{p-1}$ is a longest common subsequence of $a_1, a_2, \dots, a_{m-1}$ and $b_1, b_2, \dots, b_{n-1}$ when $p > 1$.
 - b) Suppose that $a_m \neq b_n$. Show that if $c_p \neq a_m$, then $c_1, c_2, \dots, c_p$ is a longest common subsequence of $a_1, a_2, \dots, a_{m-1}$ and $b_1, b_2, \dots, b_n$ and also show that if $c_p \neq b_n$, then $c_1, c_2, \dots, c_p$ is a longest common subsequence of $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_{n-1}$.
16. Let $L(i, j)$ denote the length of a longest common subsequence of $a_1, a_2, \dots, a_i$ and $b_1, b_2, \dots, b_j$, where $0 \leq i \leq m$ and $0 \leq j \leq n$. Use parts (a) and (b) of Exercise 15 to show that $L(i, j)$ satisfies the recurrence relation $L(i, j) = L(i-1, j-1) + 1$ if both i and j are nonzero and $a_i = b_j$, and $L(i, j) = \max(L(i, j-1), L(i-1, j))$ if both i and j are nonzero and $a_i \neq b_j$, and the initial condition $L(i, j) = 0$ if $i = 0$ or $j = 0$.
17. Use Exercise 16 to construct a dynamic programming algorithm for computing the length of a longest common subsequence of two sequences $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_n$, storing the values of $L(i, j)$ as they are found.
18. Develop an algorithm for finding a longest common subsequence of two sequences $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_n$ using the values $L(i, j)$ found by the algorithm in Exercise 17.
19. Find the solution to the recurrence relation $f(n) = f(n/2) + n^2$ for $n = 2^k$ where k is a positive integer and $f(1) = 1$.
20. Find the solution to the recurrence relation $f(n) = 3f(n/5) + 2n^4$, when n is divisible by 5, for $n = 5^k$, where k is a positive integer and $f(1) = 1$.
21. Give a big- O estimate for the size of f in Exercise 20 if f is an increasing function.

22. Find a recurrence relation that describes the number of comparisons used by the following algorithm: Find the largest and second largest elements of a sequence of n numbers recursively by splitting the sequence into two subsequences with an equal number of terms, or where there is one more term in one subsequence than in the other, at each stage. Stop when subsequences with two terms are reached.
23. Give a big- O estimate for the number of comparisons used by the algorithm described in Exercise 22.
24. A sequence $a_1, a_2, \dots, a_n$ is **unimodal** if and only if there is an index m , $1 \leq m \leq n$, such that $a_i < a_{i+1}$ when $1 \leq i < m$ and $a_i > a_{i+1}$ when $m \leq i < n$. That is, the terms of the sequence strictly increase until the m th term and they strictly decrease after it, which implies that a_m is the largest term. In this exercise, a_m will always denote the largest term of the unimodal sequence $a_1, a_2, \dots, a_n$.
- Show that a_m is the unique term of the sequence that is greater than both the term immediately preceding it and the term immediately following it.
 - Show that if $a_i < a_{i+1}$ where $1 \leq i < n$, then $i + 1 \leq m \leq n$.
 - Show that if $a_i > a_{i+1}$ where $1 \leq i < n$, then $1 \leq m \leq i$.
 - Develop a divide-and-conquer algorithm for locating the index m . [Hint: Suppose that $i < m < j$. Use parts (a), (b), and (c) to determine whether $\lfloor (i+j)/2 \rfloor + 1 \leq m \leq n$, $1 \leq m \leq \lfloor (i+j)/2 \rfloor - 1$, or $m = \lfloor (i+j)/2 \rfloor$.]
25. Show that the algorithm from Exercise 24 has worst-case time complexity $O(\log n)$ in terms of the number of comparisons.
- Let $\{a_n\}$ be a sequence of real numbers. The **forward differences** of this sequence are defined recursively as follows: The **first forward difference** is $\Delta a_n = a_{n+1} - a_n$; the **$(k+1)$ st forward difference** $\Delta^{k+1} a_n$ is obtained from $\Delta^k a_n$ by $\Delta^{k+1} a_n = \Delta^k a_{n+1} - \Delta^k a_n$.
26. Find Δa_n , where
- $a_n = 3$.
 - $a_n = 4n + 7$.
 - $a_n = n^2 + n + 1$.
27. Let $a_n = 3n^3 + n + 2$. Find $\Delta^k a_n$, where k equals
- 2.
 - 3.
 - 4.
- *28. Suppose that $a_n = P(n)$, where P is a polynomial of degree d . Prove that $\Delta^{d+1} a_n = 0$ for all nonnegative integers n .
29. Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers. Show that
- $$\Delta(a_n b_n) = a_{n+1}(\Delta b_n) + b_n(\Delta a_n).$$
30. Show that if $F(x)$ and $G(x)$ are the generating functions for the sequences $\{a_k\}$ and $\{b_k\}$, respectively, and c and d are real numbers, then $(cF + dG)(x)$ is the generating function for $\{ca_k + db_k\}$.
31. (Requires calculus) This exercise shows how generating functions can be used to solve the recurrence relation
- $$(n+1)a_{n+1} = a_n + (1/n!) \text{ for } n \geq 0 \text{ with initial condition } a_0 = 1.$$
- Let $G(x)$ be the generating function for $\{a_n\}$. Show that $G'(x) = G(x) + e^x$ and $G(0) = 1$.
 - Show from part (a) that $(e^{-x}G(x))' = 1$, and conclude that $G(x) = xe^x + e^x$.
 - Use part (b) to find a closed form for a_n .
32. Suppose that 14 students receive an A on the first exam in a discrete mathematics class, and 18 receive an A on the second exam. If 22 students received an A on either the first exam or the second exam, how many students received an A on both exams?
33. There are 323 farms in Monmouth County that have at least one of horses, cows, and sheep. If 224 have horses, 85 have cows, 57 have sheep, and 18 farms have all three types of animals, how many farms have exactly two of these three types of animals?
34. Queries to a database of student records at a college produced the following data: There are 2175 students at the college, 1675 of these are not freshmen, 1074 students have taken a course in calculus, 444 students have taken a course in discrete mathematics, 607 students are not freshmen and have taken calculus, 350 students have taken calculus and discrete mathematics, 201 students are not freshmen and have taken discrete mathematics, and 143 students are not freshmen and have taken both calculus and discrete mathematics. Can all the responses to the queries be correct?
35. Students in the school of mathematics at a university major in one or more of the following four areas: applied mathematics (AM), pure mathematics (PM), operations research (OR), and computer science (CS). How many students are in this school if (including joint majors) there are 23 students majoring in AM; 17 in PM; 44 in OR; 63 in CS; 5 in AM and PM; 8 in AM and CS; 4 in AM and OR; 6 in PM and CS; 5 in PM and OR; 14 in OR and CS; 2 in PM, OR, and CS; 2 in AM, OR, and CS; 1 in PM, AM, and OR; 1 in PM, AM, and CS; and 1 in all four fields.
36. How many terms are needed when the inclusion-exclusion principle is used to express the number of elements in the union of seven sets if no more than five of these sets have a common element?
37. How many solutions in positive integers are there to the equation $x_1 + x_2 + x_3 = 20$ with $2 < x_1 < 6$, $6 < x_2 < 10$, and $0 < x_3 < 5$?
38. How many positive integers less than 1,000,000 are
- divisible by 2, 3, or 5?
 - not divisible by 7, 11, or 13?
 - divisible by 3 but not by 7?
39. How many positive integers less than 200 are
- second or higher powers of integers?
 - either primes or second or higher powers of integers?
 - not divisible by the square of an integer greater than 1?
 - not divisible by the cube of an integer greater than 1?
 - not divisible by three or more primes?

- *40. How many ways are there to assign six different jobs to three different employees if the hardest job is assigned to the most experienced employee and the easiest job is assigned to the least experienced employee?
41. What is the probability that exactly one person is given back the correct hat by a hatcher person who gives n people their hats back at random?

42. How many bit strings of length six do not contain four consecutive 1s?
43. What is the probability that a bit string of length six chosen at random contains at least four 1s?

Computer Projects

Write programs with these input and output.

- Given a positive integer n , list all the moves required in the Tower of Hanoi puzzle to move n disks from one peg to another according to the rules of the puzzle.
- Given a positive integer n and an integer k with $1 \leq k \leq n$, list all the moves used by the Frame–Stewart algorithm (described in the preamble to Exercise 38 of Section 8.1) to move n disks from one peg to another using four pegs according to the rules of the puzzle.
- Given a positive integer n , list all the bit sequences of length n that do not have a pair of consecutive 0s.
- Given an integer n greater than 1, write out all ways to parenthesize the product of $n + 1$ variables.
- Given a set of n talks, their start and end times, and the number of attendees at each talk, use dynamic programming to schedule a subset of these talks in a single lecture hall to maximize total attendance.
- Given matrices $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_n$, with dimensions $m_1 \times m_2, m_2 \times m_3, \dots, m_n \times m_{n+1}$, respectively, each with integer entries, use dynamic programming, as outlined in Exercise 57 in Section 8.1, to find the minimum number of multiplications of integers needed to compute $\mathbf{A}_1 \mathbf{A}_2 \cdots \mathbf{A}_n$.
- Given a recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, where c_1 and c_2 are real numbers, initial conditions $a_0 = C_0$ and $a_1 = C_1$, and a positive integer k , find a_k using iteration.
- Given a recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ and initial conditions $a_0 = C_0$ and $a_1 = C_1$, determine the unique solution.
- Given a recurrence relation of the form $f(n) = af(n/b) + c$, where a is a real number, b is a positive integer, and c is a real number, and a positive integer k , find $f(b^k)$ using iteration.
- Given the number of elements in the intersection of three sets, the number of elements in each pairwise intersection of these sets, and the number of elements in each set, find the number of elements in their union.
- Given a positive integer n , produce the formula for the number of elements in the union of n sets.
- Given positive integers m and n , find the number of onto functions from a set with m elements to a set with n elements.
- Given a positive integer n , list all the derangements of the set $\{1, 2, 3, \dots, n\}$.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

- Find the exact value of f_{100} , f_{500} , and f_{1000} , where f_n is the n th Fibonacci number.
- Find the smallest Fibonacci number greater than 1,000,000, greater than 1,000,000,000, and greater than 1,000,000,000,000.
- Find as many prime Fibonacci numbers as you can. It is unknown whether there are infinitely many of these.
- Write out all the moves required to solve the Tower of Hanoi puzzle with 10 disks.
- Write out all the moves required to use the Frame–Stewart algorithm to move 20 disks from one peg to another peg using four pegs according to the rules of the Reve’s puzzle.
- Verify the Frame conjecture for solving the Reve’s puzzle for n disks for as many integers n as possible by showing that the puzzle cannot be solved using fewer moves than are made by the Frame–Stewart algorithm with the optimal choice of k .
- Compute the number of operations required to multiply two integers with n bits for various integers n including 16, 64, 256, and 1024 using the fast multiplication described in Example 4 of Section 8.3 and the standard algorithm for multiplying integers (Algorithm 3 in Section 4.2).
- Compute the number of operations required to multiply two $n \times n$ matrices for various integers n including 4, 16, 64, and 128 using the fast matrix multiplication described

- in Example 5 of Section 8.3 and the standard algorithm for multiplying matrices (Algorithm 1 in Section 3.3).
9. Find the number of primes not exceeding 10,000 using the method described in Section 8.6 to find the number of primes not exceeding 100.
 10. List all the derangements of $\{1, 2, 3, 4, 5, 6, 7, 8\}$.
 11. Compute the probability that a permutation of n objects is a derangement for all positive integers not exceeding 20 and determine how quickly these probabilities approach the number $1/e$.

Writing Projects

Respond to these with essays using outside sources.

1. Find the original source where Fibonacci presented his puzzle about modeling rabbit populations. Discuss this problem and other problems posed by Fibonacci and give some information about Fibonacci himself.
2. Explain how the Fibonacci numbers arise in a variety of applications, such as in phyllotaxis, the study of arrangement of leaves in plants, in the study of reflections by mirrors, and so on.
3. Describe different variations of the Tower of Hanoi puzzle, including those with more than three pegs (including the Reve's puzzle discussed in the text and exercises), those where disk moves are restricted, and those where disks may have the same size. Include what is known about the number of moves required to solve each variation.
4. Discuss as many different problems as possible where the Catalan numbers arise.
5. Discuss some of the problems in which Richard Bellman first used dynamic programming.
6. Describe the role dynamic programming algorithms play in bioinformatics including for DNA sequence comparison, gene comparison, and RNA structure prediction.
7. Describe the use of dynamic programming in economics including its use to study optimal consumption and saving.
8. Explain how dynamic programming can be used to solve the egg-dropping puzzle which determines which floors of a multistory building it is safe to drop eggs from without breaking.
9. Describe the solution of Ulam's problem (see Exercise 28 in Section 8.3) involving searching with one lie found by Andrzej Pelc.
10. Discuss variations of Ulam's problem (see Exercise 28 in Section 8.3) involving searching with more than one lie and what is known about this problem.
11. Define the convex hull of a set of points in the plane and describe three different algorithms, including a divide-and-conquer algorithm, for finding the convex hull of a set of points in the plane.
12. Describe how sieve methods are used in number theory. What kind of results have been established using such methods?
13. Look up the rules of the old French card game of *rencontres*. Describe these rules and describe the work of Pierre Raymond de Montmort on *le problème de rencontres*.
14. Describe how exponential generating functions can be used to solve a variety of counting problems.
15. Describe the Polyá theory of counting and the kind of counting problems that can be solved using this theory.
16. The *problème des ménages* (the problem of the households) asks for the number of ways to arrange n couples around a table so that the sexes alternate and no husband and wife are seated together. Explain the method used by E. Lucas to solve this problem.
17. Explain how *rook polynomials* can be used to solve counting problems.

9

Relations

- 9.1 Relations and Their Properties
- 9.2 n -ary Relations and Their Applications
- 9.3 Representing Relations
- 9.4 Closures of Relations
- 9.5 Equivalence Relations
- 9.6 Partial Orderings

Relationships between elements of sets occur in many contexts. Every day we deal with relationships such as those between a business and its telephone number, an employee and his or her salary, a person and a relative, and so on. In mathematics we study relationships such as those between a positive integer and one that it divides, an integer and one that it is congruent to modulo 5, a real number and one that is larger than it, a real number x and the value $f(x)$ where f is a function, and so on. Relationships such as that between a program and a variable it uses, and that between a computer language and a valid statement in this language, often arise in computer science. Relationships between elements of two sets are represented using the structure called a binary relation, which is just a subset of the Cartesian product of the sets. Relations can be used to solve problems such as determining which pairs of cities are linked by airline flights in a network, or finding a viable order for the different phases of a complicated project. We will introduce a number of different properties binary relations may enjoy.

Relationships between elements of more than two sets arise in many contexts. These relationships can be represented by n -ary relations, which are collections of n -tuples. Such relations are the basis of the relational data model, the most common way to store information in computer databases. We will introduce the terminology used to study relational databases, define some important operations on them, and introduce the database query language SQL. We will conclude our brief study of n -ary relations and databases with an important application from data mining. In particular, we will show how databases of transactions, represented by n -ary relations, are used to measure the likelihood that someone buys a particular product from a store when they buy one or more other products.

Two methods for representing relations, using square matrices and using directed graphs, consisting of vertices and directed edges, will be introduced and used in later sections of the chapter. We will also study relationships that have certain collections of properties that relations may enjoy. For example, in some computer languages, only the first 31 characters of the name of a variable matter. The relation consisting of ordered pairs of strings in which the first string has the same initial 31 characters as the second string is an example of a special type of relation, known as an equivalence relation. Equivalence relations arise throughout mathematics and computer science. Finally, we will study relations called partial orderings, which generalize the notion of the less than or equal to relation. For example, the set of all pairs of strings of English letters in which the second string is the same as the first string or comes after the first in dictionary order is a partial ordering.

9.1 Relations and Their Properties

9.1.1 Introduction



The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. For this reason, sets of ordered pairs are called binary relations. In this section we introduce the basic terminology used to describe binary relations. Later in this chapter we will use relations to solve problems involving communications networks, project scheduling, and identifying elements in sets with common properties.

Definition 1

Let A and B be sets. A *binary relation from A to B* is a subset of $A \times B$.

In other words, a binary relation from A to B is a set R of ordered pairs, where the first element of each ordered pair comes from A and the second element comes from B . We use the notation $a R b$ to denote that $(a, b) \in R$ and $a \not R b$ to denote that $(a, b) \notin R$. Moreover, when (a, b) belongs to R , a is said to be **related to** b by R .

Binary relations represent relationships between the elements of two sets. We will introduce n -ary relations, which express relationships among elements of more than two sets, later in this chapter. We will omit the word *binary* when there is no danger of confusion.

Examples 1–3 illustrate the notion of a relation.

EXAMPLE 1 Let A be the set of students in your school, and let B be the set of courses. Let R be the relation that consists of those pairs (a, b) , where a is a student enrolled in course b . For instance, if Jason Goodfriend and Deborah Sherman are enrolled in CS518, the pairs (Jason Goodfriend, CS518) and (Deborah Sherman, CS518) belong to R . If Jason Goodfriend is also enrolled in CS510, then the pair (Jason Goodfriend, CS510) is also in R . However, if Deborah Sherman is not enrolled in CS510, then the pair (Deborah Sherman, CS510) is not in R .

Note that if a student is not currently enrolled in any courses there will be no pairs in R that have this student as the first element. Similarly, if a course is not currently being offered there will be no pairs in R that have this course as their second element. ◀

EXAMPLE 2 Let A be the set of cities in the U.S.A., and let B be the set of the 50 states in the U.S.A. Define the relation R by specifying that (a, b) belongs to R if a city with name a is in the state b . For instance, (Boulder, Colorado), (Bangor, Maine), (Ann Arbor, Michigan), (Middletown, New Jersey), (Middletown, New York), (Cupertino, California), and (Red Bank, New Jersey) are in R . ▶

EXAMPLE 3 Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B . This means, for instance, that $0 R a$, but that $1 \not R b$. Relations can be represented graphically, as shown in Figure 1, using arrows to represent ordered pairs. Another way to represent this relation is to use a table, which is also done in Figure 1. We will discuss representations of relations in more detail in Section 9.3. ▶

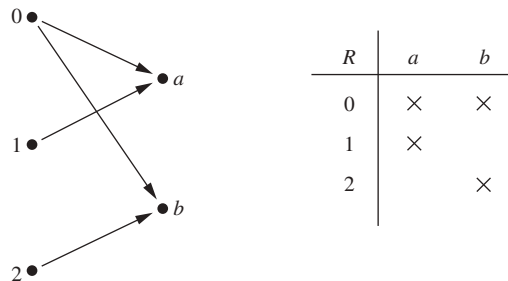


FIGURE 1 Displaying the ordered pairs in the relation R from Example 3.

9.1.2 Functions as Relations

Recall that a function f from a set A to a set B (as defined in Section 2.3) assigns exactly one element of B to each element of A . The graph of f is the set of ordered pairs (a, b) such

that $b = f(a)$. Because the graph of f is a subset of $A \times B$, it is a relation from A to B . Moreover, the graph of a function has the property that every element of A is the first element of exactly one ordered pair of the graph.

Conversely, if R is a relation from A to B such that every element in A is the first element of exactly one ordered pair of R , then a function can be defined with R as its graph. This can be done by assigning to an element a of A the unique element $b \in B$ such that $(a, b) \in R$. (Note that the relation R in Example 2 is not the graph of a function because Middletown occurs more than once as the first element of an ordered pair in R .)

A relation can be used to express a one-to-many relationship between the elements of the sets A and B (as in Example 2), where an element of A may be related to more than one element of B . A function represents a relation where exactly one element of B is related to each element of A .

Relations are a generalization of graphs of functions; they can be used to express a much wider class of relationships between sets. (Recall that the graph of the function f from A to B is the set of ordered pairs $(a, f(a))$ for $a \in A$.)

9.1.3 Relations on a Set

Relations from a set A to itself are of special interest.

Definition 2

A relation on a set A is a relation from A to A .

In other words, a relation on a set A is a subset of $A \times A$.

EXAMPLE 4

Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}.$$

The pairs in this relation are displayed both graphically and in tabular form in Figure 2. ◀

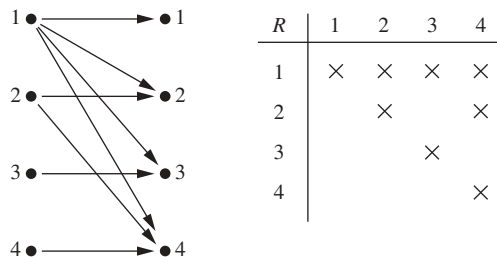


FIGURE 2 Displaying the ordered pairs in the relation R from Example 4.

Next, some examples of relations on the set of integers will be given in Example 5.

EXAMPLE 5 Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of these relations contain each of the pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, $(1, -1)$, and $(2, 2)$?

Remark: Unlike the relations in Examples 1–4, these are relations on an infinite set.

Solution: The pair $(1, 1)$ is in R_1 , R_3 , R_4 , and R_6 ; $(1, 2)$ is in R_1 and R_6 ; $(2, 1)$ is in R_2 , R_5 , and R_6 ; $(1, -1)$ is in R_2 , R_3 , and R_6 ; and finally, $(2, 2)$ is in R_1 , R_3 , and R_4 . ◀

It is not hard to determine the number of relations on a finite set, because a relation on a set A is simply a subset of $A \times A$.

EXAMPLE 6 How many relations are there on a set with n elements?

Solution: A relation on a set A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements, and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$. Thus, there are 2^{n^2} relations on a set with n elements. For example, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$. ◀

9.1.4 Properties of Relations

There are several properties that are used to classify relations on a set. We will introduce the most important of these here. (You may find it instructive to study this material with the contents of Section 9.3. In that section, several methods for representing relations will be introduced that can help you understand each of the properties that we introduce here.)

In some relations an element is always related to itself. For instance, let R be the relation on the set of all people consisting of pairs (x, y) where x and y have the same mother and the same father. Then xRx for every person x .

Definition 3 A relation R on a set A is called *reflexive* if $(a, a) \in R$ for every element $a \in A$.

Remark: Using quantifiers we see that the relation R on the set A is reflexive if $\forall a((a, a) \in R)$, where the universe of discourse is the set of all elements in A .

We see that a relation on A is reflexive if every element of A is related to itself. Examples 7–9 illustrate the concept of a reflexive relation.

EXAMPLE 7 Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

Solution: The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive because $(3, 3)$ is not in any of these relations. 

EXAMPLE 8 Which of the relations from Example 5 are reflexive?

Solution: The reflexive relations from Example 5 are R_1 (because $a \leq a$ for every integer a), R_3 , and R_4 . For each of the other relations in this example it is easy to find a pair of the form (a, a) that is not in the relation. (This is left as an exercise for the reader.) 

EXAMPLE 9 Is the “divides” relation on the set of positive integers reflexive?

Solution: Because $a \mid a$ whenever a is a positive integer, the “divides” relation is reflexive. (Note that if we replace the set of positive integers with the set of all integers the relation is not reflexive because by definition 0 does not divide 0.) 

In some relations an element is related to a second element if and only if the second element is also related to the first element. The relation consisting of pairs (x, y) , where x and y are students at your school with at least one common class has this property. Other relations have the property that if an element is related to a second element, then this second element is not related to the first. The relation consisting of the pairs (x, y) , where x and y are students at your school, where x has a higher grade point average than y has this property.

Definition 4

A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$. A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.

Remark: Using quantifiers, we see that the relation R on the set A is symmetric if $\forall a \forall b ((a, b) \in R \rightarrow (b, a) \in R)$. Similarly, the relation R on the set A is antisymmetric if $\forall a \forall b (((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b))$.

In other words, a relation is symmetric if and only if a is related to b always implies that b is related to a . For instance, the equality relation is symmetric because $a = b$ if and only if $b = a$. A relation is antisymmetric if and only if there are no pairs of distinct elements a and b with a related to b and b related to a . That is, the only way to have a related to b and b related to a is for a and b to be the same element. For instance, the less than or equal to relation is



antisymmetric. To see this, note that $a \leq b$ and $b \leq a$ implies that $a = b$. The terms *symmetric* and *antisymmetric* are not opposites, because a relation can have both of these properties or may lack both of them (see Exercise 10). A relation cannot be both symmetric and antisymmetric if it contains some pair of the form (a, b) in which $a \neq b$.

Remark: Although relatively few of the 2^{n^2} relations on a set with n elements are symmetric or antisymmetric, as counting arguments can show, many important relations have one of these properties. (See Exercise 49.)

EXAMPLE 10 Which of the relations from Example 7 are symmetric and which are antisymmetric?

Extra Examples ➤

Solution: The relations R_2 and R_3 are symmetric, because in each case (b, a) belongs to the relation whenever (a, b) does. For R_2 , the only thing to check is that both $(2, 1)$ and $(1, 2)$ are in the relation. For R_3 , it is necessary to check that both $(1, 2)$ and $(2, 1)$ belong to the relation, and $(1, 4)$ and $(4, 1)$ belong to the relation. The reader should verify that none of the other relations is symmetric. This is done by finding a pair (a, b) such that it is in the relation but (b, a) is not.

R_4 , R_5 , and R_6 are all antisymmetric. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a, b) and (b, a) belong to the relation. The reader should verify that none of the other relations is antisymmetric. This is done by finding a pair (a, b) with $a \neq b$ such that (a, b) and (b, a) are both in the relation. ◀

EXAMPLE 11 Which of the relations from Example 5 are symmetric and which are antisymmetric?

Solution: The relations R_3 , R_4 , and R_6 are symmetric. R_3 is symmetric, for if $a = b$ or $a = -b$, then $b = a$ or $b = -a$. R_4 is symmetric because $a = b$ implies that $b = a$. R_6 is symmetric because $a + b \leq 3$ implies that $b + a \leq 3$. The reader should verify that none of the other relations is symmetric.

The relations R_1 , R_2 , R_4 , and R_5 are antisymmetric. R_1 is antisymmetric because the inequalities $a \leq b$ and $b \leq a$ imply that $a = b$. R_2 is antisymmetric because it is impossible that $a > b$ and $b > a$. R_4 is antisymmetric, because two elements are related with respect to R_4 if and only if they are equal. R_5 is antisymmetric because it is impossible that $a = b + 1$ and $b = a + 1$. The reader should verify that none of the other relations is antisymmetric. ◀

EXAMPLE 12 Is the “divides” relation on the set of positive integers symmetric? Is it antisymmetric?

Solution: This relation is not symmetric because $1 \mid 2$, but $2 \nmid 1$. However, it is antisymmetric. To see this, note that if a and b are positive integers with $a \mid b$ and $b \mid a$, then $a = b$ (the verification of this is left as an exercise for the reader). ◀

Let R be the relation consisting of all pairs (x, y) of students at your school, where x has taken more credits than y . Suppose that x is related to y and y is related to z . This means that x has taken more credits than y and y has taken more credits than z . We can conclude that x has taken more credits than z , so that x is related to z . What we have shown is that R has the transitive property, which is defined as follows.

Definition 5

A relation R on a set A is called *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

Remark: Using quantifiers we see that the relation R on a set A is transitive if we have $\forall a \forall b \forall c ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$.

EXAMPLE 13 Which of the relations in Example 7 are transitive?

**Extra
Examples** 

Solution: R_4, R_5 , and R_6 are transitive. For each of these relations, we can show that it is transitive by verifying that if (a, b) and (b, c) belong to this relation, then (a, c) also does. For instance, R_4 is transitive, because $(3, 2)$ and $(2, 1)$, $(4, 2)$ and $(2, 1)$, $(4, 3)$ and $(3, 1)$, and $(4, 3)$ and $(3, 2)$ are the only such sets of pairs, and $(3, 1)$, $(4, 1)$, and $(4, 2)$ belong to R_4 . The reader should verify that R_5 and R_6 are transitive.

R_1 is not transitive because $(3, 4)$ and $(4, 1)$ belong to R_1 , but $(3, 1)$ does not. R_2 is not transitive because $(2, 1)$ and $(1, 2)$ belong to R_2 , but $(2, 2)$ does not. R_3 is not transitive because $(4, 1)$ and $(1, 2)$ belong to R_3 , but $(4, 2)$ does not. 

EXAMPLE 14 Which of the relations in Example 5 are transitive?

Solution: The relations R_1, R_2, R_3 , and R_4 are transitive. R_1 is transitive because $a \leq b$ and $b \leq c$ imply that $a \leq c$. R_2 is transitive because $a > b$ and $b > c$ imply that $a > c$. R_3 is transitive because $a = \pm b$ and $b = \pm c$ imply that $a = \pm c$. R_4 is clearly transitive, as the reader should verify. R_5 is not transitive because $(2, 1)$ and $(1, 0)$ belong to R_5 , but $(2, 0)$ does not. R_6 is not transitive because $(2, 1)$ and $(1, 2)$ belong to R_6 , but $(2, 2)$ does not. 

EXAMPLE 15 Is the “divides” relation on the set of positive integers transitive?

Solution: Suppose that a divides b and b divides c . Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c . It follows that this relation is transitive. 

We can use counting techniques to determine the number of relations with specific properties. Finding the number of relations with a particular property provides information about how common this property is in the set of all relations on a set with n elements.

EXAMPLE 16 How many reflexive relations are there on a set with n elements?

Solution: A relation R on a set A is a subset of $A \times A$. Consequently, a relation is determined by specifying whether each of the n^2 ordered pairs in $A \times A$ is in R . However, if R is reflexive, each of the n ordered pairs (a, a) for $a \in A$ must be in R . Each of the other $n(n - 1)$ ordered pairs of the form (a, b) , where $a \neq b$, may or may not be in R . Hence, by the product rule for counting, there are $2^{n(n-1)}$ reflexive relations [this is the number of ways to choose whether each element (a, b) , with $a \neq b$, belongs to R]. 

Formulas for the number of symmetric relations and the number of antisymmetric relations on a set with n elements can be found using reasoning similar to that in Example 16 (see Exercise 49). However, no general formula is known that counts the transitive relations on a set with n elements. Currently, $T(n)$, the number of transitive relations on a set with n elements, is known only for $0 \leq n \leq 18$. For example, $T(4) = 3,994$, $T(5) = 154,303$, and $T(6) = 9,415,189$. (The values of $T(n)$ for $n = 0, 1, 2, \dots, 18$, are the terms of the sequence A006905 in the OEIS, which is discussed in Section 2.4.)

9.1.5 Combining Relations

Because relations from A to B are subsets of $A \times B$, two relations from A to B can be combined in any way two sets can be combined. Consider Examples 17–19.

EXAMPLE 17 Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. The relations $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain

$$R_1 \cup R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\},$$

$$R_1 \cap R_2 = \{(1, 1)\},$$

$$R_1 - R_2 = \{(2, 2), (3, 3)\},$$

$$R_2 - R_1 = \{(1, 2), (1, 3), (1, 4)\}.$$

EXAMPLE 18 Let A and B be the set of all students and the set of all courses at a school, respectively. Suppose that R_1 consists of all ordered pairs (a, b) , where a is a student who has taken course b , and R_2 consists of all ordered pairs (a, b) , where a is a student who requires course b to graduate. What are the relations $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 \oplus R_2$, $R_1 - R_2$, and $R_2 - R_1$?

Solution: The relation $R_1 \cup R_2$ consists of all ordered pairs (a, b) , where a is a student who either has taken course b or needs course b to graduate, and $R_1 \cap R_2$ is the set of all ordered pairs (a, b) , where a is a student who has taken course b and needs this course to graduate. Also, $R_1 \oplus R_2$ consists of all ordered pairs (a, b) , where student a has taken course b but does not need it to graduate or needs course b to graduate but has not taken it. $R_1 - R_2$ is the set of ordered pairs (a, b) , where a has taken course b but does not need it to graduate; that is, b is an elective course that a has taken. $R_2 - R_1$ is the set of all ordered pairs (a, b) , where b is a course that a needs to graduate but has not taken.

EXAMPLE 19 Let R_1 be the less than relation on the set of real numbers and let R_2 be the greater than relation on the set of real numbers, that is, $R_1 = \{(x, y) \mid x < y\}$ and $R_2 = \{(x, y) \mid x > y\}$. What are $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$, and $R_1 \oplus R_2$?

Solution: We note that $(x, y) \in R_1 \cup R_2$ if and only if $(x, y) \in R_1$ or $(x, y) \in R_2$. Hence, $(x, y) \in R_1 \cup R_2$ if and only if $x < y$ or $x > y$. Because the condition $x < y$ or $x > y$ is the same as the condition $x \neq y$, it follows that $R_1 \cup R_2 = \{(x, y) \mid x \neq y\}$. In other words, the union of the less than relation and the greater than relation is the not equals relation.

Next, note that it is impossible for a pair (x, y) to belong to both R_1 and R_2 because it is impossible that $x < y$ and $x > y$. It follows that $R_1 \cap R_2 = \emptyset$. We also see that $R_1 - R_2 = R_1$, $R_2 - R_1 = R_2$, and $R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2 = \{(x, y) \mid x \neq y\}$.

There is another way that relations are combined that is analogous to the composition of functions.

Definition 6

Let R be a relation from a set A to a set B and S a relation from B to a set C . The *composite* of R and S is the relation consisting of ordered pairs (a, c) , where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

Computing the composite of two relations requires that we find elements that are the second element of ordered pairs in the first relation and the first element of ordered pairs in the second relation, as Examples 20 and 21 illustrate.

EXAMPLE 20 What is the composite of the relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?

Solution: $S \circ R$ is constructed using all ordered pairs in R and ordered pairs in S , where the second element of the ordered pair in R agrees with the first element of the ordered pair in S . For example, the ordered pairs $(2, 3)$ in R and $(3, 1)$ in S produce the ordered pair $(2, 1)$ in $S \circ R$. Computing all the ordered pairs in the composite, we find

$$S \circ R = \{(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)\}.$$

Figure 3 illustrates how this composition is found. In the figure, we examine all paths that travel via two directed edges from the leftmost elements to the rightmost elements via an element in the middle.

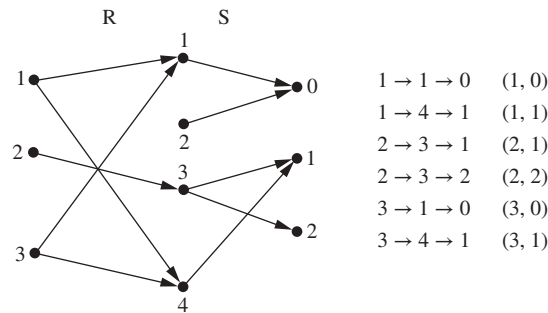


FIGURE 3 Constructing $S \circ R$.

EXAMPLE 21 **Composing the Parent Relation with Itself** Let R be the relation on the set of all people such that $(a, b) \in R$ if person a is a parent of person b . Then $(a, c) \in R \circ R$ if and only if there is a person b such that $(a, b) \in R$ and $(b, c) \in R$, that is, if and only if there is a person b such that a is a parent of b and b is a parent of c . In other words, $(a, c) \in R \circ R$ if and only if a is a grandparent of c .

The powers of a relation R can be recursively defined from the definition of a composite of two relations.

Definition 7

Let R be a relation on the set A . The powers R^n , $n = 1, 2, 3, \dots$, are defined recursively by

$$R^1 = R \quad \text{and} \quad R^{n+1} = R^n \circ R.$$

The definition shows that $R^2 = R \circ R$, $R^3 = R^2 \circ R = (R \circ R) \circ R$, and so on.

EXAMPLE 22 Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers R^n , $n = 2, 3, 4, \dots$.

Solution: Because $R^2 = R \circ R$, we find that $R^2 = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$. Furthermore, because $R^3 = R^2 \circ R$, $R^3 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$. Additional computation shows that R^4

is the same as R^3 , so $R^4 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$. It also follows that $R^n = R^3$ for $n = 5, 6, 7, \dots$. The reader should verify this. 

The following theorem shows that the powers of a transitive relation are subsets of this relation. It will be used in Section 9.4.

THEOREM 1

The relation R on a set A is transitive if and only if $R^n \subseteq R$ for $n = 1, 2, 3, \dots$.

Proof: We first prove the “if” part of the theorem. We suppose that $R^n \subseteq R$ for $n = 1, 2, 3, \dots$. In particular, $R^2 \subseteq R$. To see that this implies R is transitive, note that if $(a, b) \in R$ and $(b, c) \in R$, then by the definition of composition, $(a, c) \in R^2$. Because $R^2 \subseteq R$, this means that $(a, c) \in R$. Hence, R is transitive.



We will use mathematical induction to prove the only if part of the theorem. Note that this part of the theorem is trivially true for $n = 1$.

Assume that $R^n \subseteq R$, where n is a positive integer. This is the inductive hypothesis. To complete the inductive step we must show that this implies that R^{n+1} is also a subset of R . To show this, assume that $(a, b) \in R^{n+1}$. Then, because $R^{n+1} = R^n \circ R$, there is an element x with $x \in A$ such that $(a, x) \in R$ and $(x, b) \in R^n$. The inductive hypothesis, namely, that $R^n \subseteq R$, implies that $(x, b) \in R$. Furthermore, because R is transitive, and $(a, x) \in R$ and $(x, b) \in R$, it follows that $(a, b) \in R$. This shows that $R^{n+1} \subseteq R$, completing the proof. 

Exercises

- List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if
 - $a = b$.
 - $a + b = 4$.
 - $a > b$.
 - $a \mid b$.
 - $\gcd(a, b) = 1$.
 - $\text{lcm}(a, b) = 2$.
- List all the ordered pairs in the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set $\{1, 2, 3, 4, 5, 6\}$.
 - Display this relation graphically, as was done in Example 4.
 - Display this relation in tabular form, as was done in Example 4.
- For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.
 - $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$
 - $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 - $\{(2, 4), (4, 2)\}$
 - $\{(1, 2), (2, 3), (3, 4)\}$
 - $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$
 - $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$
- Determine whether the relation R on the set of all people is reflexive, symmetric, antisymmetric, and/or transitive, where $(a, b) \in R$ if and only if
 - a is taller than b .
 - a and b were born on the same day.
 - a has the same first name as b .
 - a and b have a common grandparent.
- Determine whether the relation R on the set of all Web pages is reflexive, symmetric, antisymmetric, and/or transitive, where $(a, b) \in R$ if and only if
 - everyone who has visited Web page a has also visited Web page b .
 - there are no common links found on both Web page a and Web page b .
 - there is at least one common link on Web page a and Web page b .
 - there is a Web page that includes links to both Web page a and Web page b .
- Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if
 - $x + y = 0$.
 - $x = \pm y$.
 - $x - y$ is a rational number.
 - $x = 2y$.
 - $xy \geq 0$.
 - $xy = 0$.
 - $x = 1$.
 - $x = 1$ or $y = 1$.
- Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if
 - $x \neq y$.
 - $xy \geq 1$.
 - $x = y + 1$ or $x = y - 1$.
 - $x \equiv y \pmod{7}$.
 - x is a multiple of y .
 - x and y are both negative or both nonnegative.
 - $x = y^2$.
 - $x \geq y^2$.
- Show that the relation $R = \emptyset$ on a nonempty set S is symmetric and transitive, but not reflexive.
- Show that the relation $R = \emptyset$ on the empty set $S = \emptyset$ is reflexive, symmetric, and transitive.

10. Give an example of a relation on a set that is

- a) both symmetric and antisymmetric.
- b) neither symmetric nor antisymmetric.

A relation R on the set A is **irreflexive** if for every $a \in A$, $(a, a) \notin R$. That is, R is irreflexive if no element in A is related to itself.

- 11. Which relations in Exercise 3 are irreflexive?
- 12. Which relations in Exercise 4 are irreflexive?
- 13. Which relations in Exercise 5 are irreflexive?
- 14. Which relations in Exercise 6 are irreflexive?
- 15. Can a relation on a set be neither reflexive nor irreflexive?
- 16. Use quantifiers to express what it means for a relation to be irreflexive.
- 17. Give an example of an irreflexive relation on the set of all people.

A relation R is called **asymmetric** if $(a, b) \in R$ implies that $(b, a) \notin R$. Exercises 18–24 explore the notion of an asymmetric relation. Exercise 22 focuses on the difference between asymmetry and antisymmetry.

- 18. Which relations in Exercise 3 are asymmetric?
- 19. Which relations in Exercise 4 are asymmetric?
- 20. Which relations in Exercise 5 are asymmetric?
- 21. Which relations in Exercise 6 are asymmetric?
- 22. Must an asymmetric relation also be antisymmetric? Must an antisymmetric relation be asymmetric? Give reasons for your answers.
- 23. Use quantifiers to express what it means for a relation to be asymmetric.
- 24. Give an example of an asymmetric relation on the set of all people.
- 25. How many different relations are there from a set with m elements to a set with n elements?

 Let R be a relation from a set A to a set B . The **inverse relation** from B to A , denoted by R^{-1} , is the set of ordered pairs $\{(b, a) \mid (a, b) \in R\}$. The **complementary relation** $\bar{R}$ is the set of ordered pairs $\{(a, b) \mid (a, b) \notin R\}$.

- 26. Let R be the relation $R = \{(a, b) \mid a < b\}$ on the set of integers. Find
 - a) R^{-1} .
 - b) $\bar{R}$.
- 27. Let R be the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set of positive integers. Find
 - a) R^{-1} .
 - b) $\bar{R}$.
- 28. Let R be the relation on the set of all states in the United States consisting of pairs (a, b) where state a borders state b . Find
 - a) R^{-1} .
 - b) $\bar{R}$.
- 29. Suppose that the function f from A to B is a one-to-one correspondence. Let R be the relation that equals the graph of f . That is, $R = \{(a, f(a)) \mid a \in A\}$. What is the inverse relation R^{-1} ?
- 30. Let $R_1 = \{(1, 2), (2, 3), (3, 4)\}$ and $R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4)\}$ be relations from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$. Find

- a) $R_1 \cup R_2$.
- b) $R_1 \cap R_2$.
- c) $R_1 - R_2$.
- d) $R_2 - R_1$.

31. Let A be the set of students at your school and B the set of books in the school library. Let R_1 and R_2 be the relations consisting of all ordered pairs (a, b) , where student a is required to read book b in a course, and where student a has read book b , respectively. Describe the ordered pairs in each of these relations.

- a) $R_1 \cup R_2$
- b) $R_1 \cap R_2$
- c) $R_1 \oplus R_2$
- d) $R_1 - R_2$
- e) $R_2 - R_1$

32. Let R be the relation $\{(1, 2), (1, 3), (2, 3), (2, 4), (3, 1)\}$, and let S be the relation $\{(2, 1), (3, 1), (3, 2), (4, 2)\}$. Find $S \circ R$.

33. Let R be the relation on the set of people consisting of pairs (a, b) , where a is a parent of b . Let S be the relation on the set of people consisting of pairs (a, b) , where a and b are siblings (brothers or sisters). What are $S \circ R$ and $R \circ S$?

Exercises 34–38 deal with these relations on the set of real numbers:

$R_1 = \{(a, b) \in \mathbf{R}^2 \mid a > b\}$, the greater than relation,

$R_2 = \{(a, b) \in \mathbf{R}^2 \mid a \geq b\}$, the greater than or equal to relation,

$R_3 = \{(a, b) \in \mathbf{R}^2 \mid a < b\}$, the less than relation,

$R_4 = \{(a, b) \in \mathbf{R}^2 \mid a \leq b\}$, the less than or equal to relation,

$R_5 = \{(a, b) \in \mathbf{R}^2 \mid a = b\}$, the equal to relation,

$R_6 = \{(a, b) \in \mathbf{R}^2 \mid a \neq b\}$, the unequal to relation.

34. Find

- a) $R_1 \cup R_3$.
- b) $R_1 \cup R_5$.
- c) $R_2 \cap R_4$.
- d) $R_3 \cap R_5$.
- e) $R_1 - R_2$.
- f) $R_2 - R_1$.
- g) $R_1 \oplus R_3$.
- h) $R_2 \oplus R_4$.

35. Find

- a) $R_2 \cup R_4$.
- b) $R_3 \cup R_6$.
- c) $R_3 \cap R_6$.
- d) $R_4 \cap R_6$.
- e) $R_3 - R_6$.
- f) $R_6 - R_3$.
- g) $R_2 \oplus R_6$.
- h) $R_3 \oplus R_5$.

36. Find

- a) $R_1 \circ R_1$.
- b) $R_1 \circ R_2$.
- c) $R_1 \circ R_3$.
- d) $R_1 \circ R_4$.
- e) $R_1 \circ R_5$.
- f) $R_1 \circ R_6$.
- g) $R_2 \circ R_3$.
- h) $R_3 \circ R_3$.

37. Find

- a) $R_2 \circ R_1$.
- b) $R_2 \circ R_2$.
- c) $R_3 \circ R_5$.
- d) $R_4 \circ R_1$.
- e) $R_5 \circ R_3$.
- f) $R_3 \circ R_6$.
- g) $R_4 \circ R_6$.
- h) $R_6 \circ R_6$.

38. Find the relations R_i^2 for $i = 1, 2, 3, 4, 5, 6$.

39. Find the relations S_i^2 for $i = 1, 2, 3, 4, 5, 6$ where

$S_1 = \{(a, b) \in \mathbf{Z}^2 \mid a > b\}$, the greater than relation,

$S_2 = \{(a, b) \in \mathbf{Z}^2 \mid a \geq b\}$, the greater than or equal to relation,

$S_3 = \{(a, b) \in \mathbf{Z}^2 \mid a < b\}$, the less than relation,

$S_4 = \{(a, b) \in \mathbf{Z}^2 \mid a \leq b\}$, the less than or equal to relation,

$S_5 = \{(a, b) \in \mathbf{Z}^2 \mid a = b\}$, the equal to relation,

$S_6 = \{(a, b) \in \mathbf{Z}^2 \mid a \neq b\}$, the unequal to relation.

40. Let R be the parent relation on the set of all people (see Example 21). When is an ordered pair in the relation R^3 ?

41. Let R be the relation on the set of people with doctorates such that $(a, b) \in R$ if and only if a was the thesis advisor of b . When is an ordered pair (a, b) in R^2 ? When is an ordered pair (a, b) in R^n , when n is a positive integer? (Assume that every person with a doctorate has a thesis advisor.)

42. Let R_1 and R_2 be the “divides” and “is a multiple of” relations on the set of all positive integers, respectively. That is, $R_1 = \{(a, b) \mid a \text{ divides } b\}$ and $R_2 = \{(a, b) \mid a \text{ is a multiple of } b\}$. Find

- a) $R_1 \cup R_2$. b) $R_1 \cap R_2$.
c) $R_1 - R_2$. d) $R_2 - R_1$.
e) $R_1 \oplus R_2$.

43. Let R_1 and R_2 be the “congruent modulo 3” and the “congruent modulo 4” relations, respectively, on the set of integers. That is, $R_1 = \{(a, b) \mid a \equiv b \pmod{3}\}$ and $R_2 = \{(a, b) \mid a \equiv b \pmod{4}\}$. Find

- a) $R_1 \cup R_2$. b) $R_1 \cap R_2$.
c) $R_1 - R_2$. d) $R_2 - R_1$.
e) $R_1 \oplus R_2$.

44. List the 16 different relations on the set $\{0, 1\}$.

45. How many of the 16 different relations on $\{0, 1\}$ contain the pair $(0, 1)$?

46. Which of the 16 relations on $\{0, 1\}$, which you listed in Exercise 44, are

- a) reflexive? b) irreflexive?
c) symmetric? d) antisymmetric?
e) asymmetric? f) transitive?

47. a) How many relations are there on the set $\{a, b, c, d\}$?

b) How many relations are there on the set $\{a, b, c, d\}$ that contain the pair (a, a) ?

48. Let S be a set with n elements and let a and b be distinct elements of S . How many relations R are there on S such that

- a) $(a, b) \in R$? b) $(a, b) \notin R$?
c) no ordered pair in R has a as its first element?
d) at least one ordered pair in R has a as its first element?
e) no ordered pair in R has a as its first element or b as its second element?

f) at least one ordered pair in R either has a as its first element or has b as its second element?

*49. How many relations are there on a set with n elements that are

- a) symmetric? b) antisymmetric?
c) asymmetric? d) irreflexive?
e) reflexive and symmetric?
f) neither reflexive nor irreflexive?

*50. How many transitive relations are there on a set with n elements if

- a) $n = 1$? b) $n = 2$? c) $n = 3$?

51. Find the error in the “proof” of the following “theorem.”

“Theorem”: Let R be a relation on a set A that is symmetric and transitive. Then R is reflexive.

“Proof”: Let $a \in A$. Take an element $b \in A$ such that $(a, b) \in R$. Because R is symmetric, we also have $(b, a) \in R$. Now using the transitive property, we can conclude that $(a, a) \in R$ because $(a, b) \in R$ and $(b, a) \in R$.

52. Suppose that R and S are reflexive relations on a set A . Prove or disprove each of these statements.

- a) $R \cup S$ is reflexive.
b) $R \cap S$ is reflexive.
c) $R \oplus S$ is irreflexive.
d) $R - S$ is irreflexive.
e) $S \circ R$ is reflexive.

53. Show that the relation R on a set A is symmetric if and only if $R = R^{-1}$, where R^{-1} is the inverse relation.

54. Show that the relation R on a set A is antisymmetric if and only if $R \cap R^{-1}$ is a subset of the diagonal relation $\Delta = \{(a, a) \mid a \in A\}$.

55. Show that the relation R on a set A is reflexive if and only if the inverse relation R^{-1} is reflexive.

56. Show that the relation R on a set A is reflexive if and only if the complementary relation $\bar{R}$ is irreflexive.

57. Let R be a relation that is reflexive and transitive. Prove that $R^n = R$ for all positive integers n .

58. Let R be the relation on the set $\{1, 2, 3, 4, 5\}$ containing the ordered pairs $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 3)$, $(2, 4)$, $(3, 1)$, $(3, 4)$, $(3, 5)$, $(4, 2)$, $(4, 5)$, $(5, 1)$, $(5, 2)$, and $(5, 4)$. Find
a) R^2 . b) R^3 . c) R^4 . d) R^5 .

59. Let R be a reflexive relation on a set A . Show that R^n is reflexive for all positive integers n .

*60. Let R be a symmetric relation. Show that R^n is symmetric for all positive integers n .

61. Suppose that the relation R is irreflexive. Is R^2 necessarily irreflexive? Give a reason for your answer.

62. Derive a big- O estimate for the number of integer comparisons needed to count all transitive relations on a set with n elements using the brute force approach of checking every relation of this set for transitivity.

9.2 n -ary Relations and Their Applications

9.2.1 Introduction

Relationships among elements of more than two sets often arise. For instance, there is a relationship involving the name of a student, the student's major, and the student's grade point average. Similarly, there is a relationship involving the airline, flight number, starting point, destination, departure time, and arrival time of a flight. An example of such a relationship in mathematics involves three integers, where the first integer is larger than the second integer, which is larger than the third. Another example is the betweenness relationship involving points on a line, such that three points are related when the second point is between the first and the third.

We will study relationships among elements from more than two sets in this section. These relationships are called **n -ary relations**. These relations are used to represent computer databases. These representations help us answer queries about the information stored in databases, such as: Which flights land at O'Hare Airport between 3 A.M. and 4 A.M.? Which students at your school are sophomores majoring in mathematics or computer science and have greater than a 3.0 average? Which employees of a company have worked for the company less than 5 years and make more than \$50,000?

9.2.2 n -ary Relations

We begin with the basic definition on which the theory of relational databases rests.

Definition 1 Let $A_1, A_2, \dots, A_n$ be sets. An n -ary relation on these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$. The sets $A_1, A_2, \dots, A_n$ are called the *domains* of the relation, and n is called its *degree*.

EXAMPLE 1 Let R be the relation on $\mathbf{N} \times \mathbf{N} \times \mathbf{N}$ consisting of triples (a, b, c) , where a, b , and c are integers with $a < b < c$. Then $(1, 2, 3) \in R$, but $(2, 4, 3) \notin R$. The degree of this relation is 3. Its domains are all equal to the set of natural numbers. 

EXAMPLE 2 Let R be the relation on $\mathbf{Z} \times \mathbf{Z} \times \mathbf{Z}$ consisting of all triples of integers (a, b, c) in which a, b , and c form an arithmetic progression. That is, $(a, b, c) \in R$ if and only if there is an integer k such that $b = a + k$ and $c = a + 2k$, or equivalently, such that $b - a = k$ and $c - b = k$. Note that $(1, 3, 5) \in R$ because $3 = 1 + 2$ and $5 = 1 + 2 \cdot 2$, but $(2, 5, 9) \notin R$ because $5 - 2 = 3$ while $9 - 5 = 4$. This relation has degree 3 and its domains are all equal to the set of integers. 

EXAMPLE 3 Let R be the relation on $\mathbf{Z} \times \mathbf{Z} \times \mathbf{Z}^+$ consisting of triples (a, b, m) , where a, b , and m are integers with $m \geq 1$ and $a \equiv b \pmod{m}$. Then $(8, 2, 3)$, $(-1, 9, 5)$, and $(14, 0, 7)$ all belong to R , but $(7, 2, 3)$, $(-2, -8, 5)$, and $(11, 0, 6)$ do not belong to R because $8 \not\equiv 2 \pmod{3}$, $-1 \not\equiv 9 \pmod{5}$, and $14 \not\equiv 0 \pmod{7}$, but $7 \not\equiv 2 \pmod{3}$, $-2 \not\equiv -8 \pmod{5}$, and $11 \not\equiv 0 \pmod{6}$. This relation has degree 3 and its first two domains are the set of all integers and its third domain is the set of positive integers. 

EXAMPLE 4 Let R be the relation consisting of 5-tuples (A, N, S, D, T) representing airplane flights, where A is the airline, N is the flight number, S is the starting point, D is the destination, and T is the departure time. For instance, if Nadir Express Airlines has flight 963 from Newark to Bangor

at 15:00, then (Nadir, 963, Newark, Bangor, 15:00) belongs to R . The degree of this relation is 5, and its domains are the set of all airlines, the set of flight numbers, the set of cities, the set of cities (again), and the set of times. ◀

9.2.3 Databases and Relations

Links ▶

The time required to manipulate information in a database depends on how this information is stored. The operations of adding and deleting records, updating records, searching for records, and combining records from overlapping databases are performed millions of times each day in a large database. Because of the importance of these operations, various methods for representing databases have been developed. We will discuss one of these methods, called the **relational data model**, based on the concept of a relation.

A database consists of **records**, which are n -tuples, made up of **fields**. The fields are the entries of the n -tuples. For instance, a database of student records may be made up of fields containing the name, student number, major, and grade point average of the student. The relational data model represents a database of records as an n -ary relation. Thus, student records are represented as 4-tuples of the form (*Student_name*, *ID_number*, *Major*, *GPA*). A sample database of six such records is

(Ackermann, 231455, Computer Science, 3.88)
 (Adams, 888323, Physics, 3.45)
 (Chou, 102147, Computer Science, 3.49)
 (Goodfriend, 453876, Mathematics, 3.45)
 (Rao, 678543, Mathematics, 3.90)
 (Stevens, 786576, Psychology, 2.99).

Relations used to represent databases are also called **tables**, because these relations are often displayed as tables. Each column of the table corresponds to an *attribute* of the database. For instance, the same database of students is displayed in Table 1. The attributes of this database are Student Name, ID Number, Major, and GPA.

A domain of an n -ary relation is called a **primary key** when the value of the n -tuple from this domain determines the n -tuple. That is, a domain is a primary key when no two n -tuples in the relation have the same value from this domain.

Records are often added to or deleted from databases. Because of this, the property that a domain is a primary key is time-dependent. Consequently, a primary key should be chosen that remains one whenever the database is changed. The current collection of n -tuples in a relation is called the **extension** of the relation. The more permanent part of a database, including the name and attributes of the database, is called its **intension**. When selecting a primary key, the goal should be to select a key that can serve as a primary key for all possible extensions of the database. To do this, it is necessary to examine the intension of the database to understand the set of possible n -tuples that can occur in an extension.

TABLE 1 Students.

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

EXAMPLE 5 Which domains are primary keys for the n -ary relation displayed in Table 1, assuming that no n -tuples will be added in the future?

Extra
Examples

Solution: Because there is only one 4-tuple in this table for each student name, the domain of student names is a primary key. Similarly, the ID numbers in this table are unique, so the domain of ID numbers is also a primary key. However, the domain of major fields of study is not a primary key, because more than one 4-tuple contains the same major field of study. The domain of grade point averages is also not a primary key, because there are two 4-tuples containing the same GPA. ◀

Combinations of domains can also uniquely identify n -tuples in an n -ary relation. When the values of a set of domains determine an n -tuple in a relation, the Cartesian product of these domains is called a **composite key**.

EXAMPLE 6 Is the Cartesian product of the domain of major fields of study and the domain of GPAs a composite key for the n -ary relation from Table 1, assuming that no n -tuples are ever added?

Solution: Because no two 4-tuples from this table have both the same major and the same GPA, this Cartesian product is a composite key. ◀

Because primary and composite keys are used to identify records uniquely in a database, it is important that keys remain valid when new records are added to the database. Hence, checks should be made to ensure that every new record has values that are different in the appropriate field, or fields, from all other records in this table. For instance, it makes sense to use the student identification number as a key for student records if no two students ever have the same student identification number. A university should not use the name field as a key, because two students may have the same name (such as John Smith).

9.2.4 Operations on n -ary Relations

There are a variety of operations on n -ary relations that can be used to form new n -ary relations. Applied together, these operations can answer queries on databases that ask for all n -tuples that satisfy certain conditions.

The most basic operation on an n -ary relation is determining all n -tuples in the n -ary relation that satisfy certain conditions. For example, we may want to find all the records of all computer science majors in a database of student records. We may want to find all students who have a grade point average above 3.5. We may want to find the records of all computer science majors who have a grade point average above 3.5. To perform such tasks we use the selection operator.

Definition 2

Let R be an n -ary relation and C a condition that elements in R may satisfy. Then the *selection operator* s_C maps the n -ary relation R to the n -ary relation of all n -tuples from R that satisfy the condition C .

EXAMPLE 7 To find the records of computer science majors in the n -ary relation R shown in Table 1, we use the operator s_{C_1} , where C_1 is the condition Major = “Computer Science.” The result is the two 4-tuples (Ackermann, 231455, Computer Science, 3.88) and (Chou, 102147, Computer Science, 3.49). Similarly, to find the records of students who have a grade point average above 3.5 in this database, we use the operator s_{C_2} , where C_2 is the condition GPA > 3.5. The result is the two 4-tuples (Ackermann, 231455, Computer Science, 3.88) and (Rao, 678543, Mathematics, 3.9).

Extra
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3.90). Finally, to find the records of computer science majors who have a GPA above 3.5, we use the operator s_{C_3} , where C_3 is the condition (Major = "Computer Science" $\wedge$ GPA > 3.5). The result consists of the single 4-tuple (Ackermann, 231455, Computer Science, 3.88). ◀

Projections are used to form new n -ary relations by deleting the same fields in every record of the relation.

Definition 3

The *projection* $P_{i_1, i_2, \dots, i_m}$ where $i_1 < i_2 < \dots < i_m$, maps the n -tuple $(a_1, a_2, \dots, a_n)$ to the m -tuple $(a_{i_1}, a_{i_2}, \dots, a_{i_m})$, where $m \leq n$.

In other words, the projection $P_{i_1, i_2, \dots, i_m}$ deletes $n - m$ of the components of an n -tuple, leaving the i_1 th, i_2 th, $\dots$, and i_m th components.

EXAMPLE 8 What results when the projection $P_{1,3}$ is applied to the 4-tuples (2, 3, 0, 4), (Jane Doe, 234111001, Geography, 3.14), and (a_1, a_2, a_3, a_4) ?

Solution: The projection $P_{1,3}$ sends these 4-tuples to (2, 0), (Jane Doe, Geography), and (a_1, a_3) , respectively. ◀

Example 9 illustrates how new relations are produced using projections.

EXAMPLE 9 What relation results when the projection $P_{1,4}$ is applied to the relation in Table 1?

Solution: When the projection $P_{1,4}$ is used, the second and third columns of the table are deleted, and pairs representing student names and grade point averages are obtained. Table 2 displays the results of this projection. ◀

Fewer rows may result when a projection is applied to the table for a relation. This happens when some of the n -tuples in the relation have identical values in each of the m components of the projection, and only disagree in components deleted by the projection. For instance, consider the following example.

EXAMPLE 10 What is the table obtained when the projection $P_{1,2}$ is applied to the relation in Table 3?

Solution: Table 4 displays the relation obtained when $P_{1,2}$ is applied to Table 3. Note that there are fewer rows after this projection is applied. ◀

TABLE 2 GPAs.	
<i>Student_name</i>	<i>GPA</i>
Ackermann	3.88
Adams	3.45
Chou	3.49
Goodfriend	3.45
Rao	3.90
Stevens	2.99

TABLE 3 Enrollments.		
<i>Student</i>	<i>Major</i>	<i>Course</i>
Glauser	Biology	BI 290
Glauser	Biology	MS 475
Glauser	Biology	PY 410
Marcus	Mathematics	MS 511
Marcus	Mathematics	MS 603
Marcus	Mathematics	CS 322
Miller	Computer Science	MS 575
Miller	Computer Science	CS 455

TABLE 4 Majors.	
<i>Student</i>	<i>Major</i>
Glauser	Biology
Marcus	Mathematics
Miller	Computer Science

TABLE 5 Teaching_assignments.

<i>Professor</i>	<i>Department</i>	<i>Course_number</i>
Cruz	Zoology	335
Cruz	Zoology	412
Farber	Psychology	501
Farber	Psychology	617
Grammer	Physics	544
Grammer	Physics	551
Rosen	Computer Science	518
Rosen	Mathematics	575

TABLE 6 Class_schedule.

<i>Department</i>	<i>Course_number</i>	<i>Room</i>	<i>Time</i>
Computer Science	518	N521	2:00 P.M.
Mathematics	575	N502	3:00 P.M.
Mathematics	611	N521	4:00 P.M.
Physics	544	B505	4:00 P.M.
Psychology	501	A100	3:00 P.M.
Psychology	617	A110	11:00 A.M.
Zoology	335	A100	9:00 A.M.
Zoology	412	A100	8:00 A.M.

The **join** operation is used to combine two tables into one when these tables share some identical fields. For instance, a table containing fields for airline, flight number, and gate, and another table containing fields for flight number, gate, and departure time can be combined into a table containing fields for airline, flight number, gate, and departure time.

Definition 4

Let R be a relation of degree m and S a relation of degree n . The *join* $J_p(R, S)$, where $p \leq m$ and $p \leq n$, is a relation of degree $m + n - p$ that consists of all $(m + n - p)$ -tuples $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p, b_1, b_2, \dots, b_{n-p})$, where the m -tuple $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p)$ belongs to R and the n -tuple $(c_1, c_2, \dots, c_p, b_1, b_2, \dots, b_{n-p})$ belongs to S .

In other words, the join operator J_p produces a new relation from two relations by combining all m -tuples of the first relation with all n -tuples of the second relation, where the last p components of the m -tuples agree with the first p components of the n -tuples.

EXAMPLE 11 What relation results when the join operator J_2 is used to combine the relation displayed in Tables 5 and 6?

Solution: The join J_2 produces the relation shown in Table 7. 

There are other operators besides projections and joins that produce new relations from existing relations. A description of these operations can be found in books on database theory.

TABLE 7 Teaching_schedule.

<i>Professor</i>	<i>Department</i>	<i>Course_number</i>	<i>Room</i>	<i>Time</i>
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.
Farber	Psychology	617	A110	11:00 A.M.
Grammer	Physics	544	B505	4:00 P.M.
Rosen	Computer Science	518	N521	2:00 P.M.
Rosen	Mathematics	575	N502	3:00 P.M.

TABLE 8 Flights.				
<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44

9.2.5 SQL



The database query language SQL (short for Structured Query Language) can be used to carry out the operations we have described in this section. Example 12 illustrates how SQL commands are related to operations on n -ary relations.

EXAMPLE 12 We will illustrate how SQL is used to express queries by showing how SQL can be employed to make a query about airline flights using Table 8. The SQL statement

```
SELECT Departure_time
FROM Flights
WHERE Destination='Detroit'
```

is used to find the projection P_5 (on the *Departure_time* attribute) of the selection of 5-tuples in the *Flights* database that satisfy the condition: *Destination* = 'Detroit'. The output would be a list containing the times of flights that have Detroit as their destination, namely, 08:10, 08:47, and 09:44. SQL uses the *FROM* clause to identify the n -ary relation the query is applied to, the *WHERE* clause to specify the condition of the selection operation, and the *SELECT* clause to specify the projection operation that is to be applied. (*Beware:* SQL uses *SELECT* to represent a projection, rather than a selection operation. This is an unfortunate example of conflicting terminology.)

Example 13 shows how SQL queries can be made involving more than one table.

EXAMPLE 13 The SQL statement

```
SELECT Professor, Time
FROM Teaching_assignments, Class_schedule
WHERE Department='Mathematics'
```

is used to find the projection $P_{1,5}$ of the 5-tuples in the database (shown in Table 7), which is the join J_2 of the *Teaching_assignments* and *Class_schedule* databases in Tables 5 and 6, respectively, which satisfy the condition: *Department* = Mathematics. The output would consist of the single 2-tuple (Rosen, 3:00 P.M.). The SQL *FROM* clause is used here to find the join of two different databases.

We have only touched on the basic concepts of relational databases in this section. More information can be found in [AhU195].

9.2.6 Association Rules from Data Mining



We will now introduce concepts from **data mining**, the discipline with the goal of gaining useful information from data. In particular, we will discuss information that can be gleaned from databases of transactions. We will focus on supermarket transactions, but the ideas we present are relevant in a wide range of applications.

By a **transaction** we mean a set of items bought by a customer during a visit to the store, such as {milk, eggs, bread} or {orange juice, bananas, yogurt, cream}. Stores collect large databases of transactions that can be used to help them manage their businesses. We will discuss how these databases can be used to address the question: How likely is it that a customer buys a product given that they also buy a collection of one or more specified products?

We call each product in the store an **item**. A collection of items is known as an **itemset**. A **k -itemset** is an itemset that contains exactly k items. The terms **transaction** and **basket** are used synonymously with the word itemset. When a store has n items, $a_1, a_2, \dots, a_n$, for sale, each transaction can be represented by an n -tuple $b_1, b_2, \dots, b_n$, where b_i is a binary variable that tells us whether a_i occurs in this transaction. That is, $b_i = 1$ if a_i is in this transaction and $b_i = 0$ otherwise. (Note that we only care whether an item occurs in a transaction and not how many times it occurs.) We can represent a transaction by an $(n + 1)$ -tuple of the form (*transaction number*, $b_1, b_2, \dots, b_n$). The collection of all these $(n + 1)$ -tuples forms a database of transactions.

We now define additional terms used in the study of questions relating to the purchase of particular itemsets.

Definition 5

The *count* of an itemset I , denoted by $\sigma(I)$, in a set of transactions $T = \{t_1, t_2, \dots, t_k\}$, where k is a positive integer, is the number of transactions that contain this itemset. That is,

$$\sigma(I) = |\{t_i \in T \mid I \subseteq t_i\}|.$$

The *support* of an itemset I is the probability that I is included in a randomly selected transaction from T . That is,

$$\text{support}(I) = \frac{\sigma(I)}{|T|}.$$

The **support threshold** s is specified for a particular application. **Frequent itemset mining** is the process of finding itemsets I with support greater than or equal to s . Such itemsets are said to be **frequent**.

EXAMPLE 14



The morning transactions at a market stand that sells apples, pears, cider, donuts, and mangos are shown in Table 9. In Table 10 we display the corresponding binary database, where each record is an $(n + 1)$ -tuple consisting of the transaction number followed by binary entries that represent this itemset. Because apples occurs in five of the eight transactions, we see that $\sigma(\{\text{apples}\}) = 5$ and $\text{support}(\{\text{apples}\}) = 5/8$. Similarly, because the itemset {apples, cider} is a subset of four of the eight transactions, we have $\sigma(\{\text{apples, cider}\}) = 4$ and $\text{support}(\{\text{apples, cider}\}) = 4/8 = 1/2$.

If we set the support threshold to be 0.5, an item is frequent if it occurs in at least four of the eight transactions. Consequently, with this support threshold, apples, pears, donuts and cider are the frequent items. The itemset {apples, cider} is a frequent itemset, but the itemset {donuts pears} is not a frequent itemset. ◀

TABLE 9 A Set of Transactions.

<i>Transaction Number</i>	<i>Items</i>
1	{apples, pears, mangos}
2	{pears, cider}
3	{apples, cider, donuts, mangos}
4	{apples, pears, cider, donuts}
5	{apples, cider, donuts}
6	{pears, cider, donuts}
7	{pears, donuts}
8	{apples, pears, cider}

TABLE 10 Binary Database for the Transactions in Table 9.

<i>Transaction Number</i>	<i>Apples</i>	<i>Pears</i>	<i>Cider</i>	<i>Donuts</i>	<i>Mangos</i>
1	1	1	0	0	1
2	0	1	1	0	0
3	1	0	1	1	1
4	1	1	1	1	0
5	1	0	1	1	0
6	0	1	1	1	0
7	0	1	0	1	0
8	1	1	1	0	0

We can use sets of transactions to help us predict whether a customer will buy a particular item given that they also buy all the items in an itemset (which might just be one item). Before we address a question of this type, we introduce some terminology. If S is a set of items and T is a set of transactions, an **association rule** is an implication of the form $I \rightarrow J$, where I and J are disjoint subsets of S . Although this notation borrows the notation for logical implications, its meaning is not entirely analagous. The association rule $I \rightarrow J$ is not the statement that whenever I is a subset of a transaction, then J must also be one. Instead, the strength of an association rule is measured in terms of its **support**, the frequency of transactions that contain both I and J , and its **confidence**, the frequency of transactions that contain J when they also contain I .

Definition 6

If I and J are subsets of a set T of transactions, then

$$\text{support}(I \rightarrow J) = \frac{\sigma(I \cup J)}{|T|}$$

and

$$\text{confidence}(I \rightarrow J) = \frac{\sigma(I \cup J)}{\sigma(I)}.$$

The support of the association rule $I \rightarrow J$, the fraction of transactions that contain both I and J , is a useful measure because a low support value tells us that the basket containing all items in

I and all items in J is seldom purchased, whereas a high value tells us that they are purchased together in a large fraction of transactions. Note that the confidence of the association rule $I \rightarrow J$ is the conditional probability that a transaction will contain all the items in I and in J given that it contains all the items in I . So, the larger the confidence of $I \rightarrow J$, the more likely it is for J to be a subset of a transaction that contains I .

EXAMPLE 15

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What are the support and the confidence of the association rule $\{\text{cider, donuts}\} \rightarrow \{\text{apples}\}$ for the set of transactions in Example 14?

Solution: The support of this association rule is $\sigma(\{\text{cider, donuts}\} \cup \{\text{apples}\})/|T|$. Because $\sigma(\{\text{cider, donuts}\} \cup \{\text{apples}\}) = \sigma(\{\text{cider, donuts, apples}\}) = 3$ and $|T| = 8$, we see that the support of this rule is $3/8 = 0.375$.

The confidence of this rule is $\sigma(\{\text{cider, donuts}\} \cup \{\text{apples}\})/\sigma(\{\text{cider, donuts}\}) = 3/4 = 0.75$.

An important problem in data mining is to find **strong association rules**, which have support greater than or equal to a minimum support level and confidence greater than or equal to a minimum confidence level. It is important to have efficient algorithms to find strong association rules because the number of available items can be extremely large. For instance, a supermarket may have tens of thousands, or even hundreds of thousands, of items in stock. The brute-force approach of finding association rules with sufficiently large support and confidence by computing the support and confidence of all possible association rules is infeasible because there are an exponential number of such association rules (see Exercise 41). Several widely used algorithms have been developed to solve this problem much more efficiently than brute force. Such algorithms first find frequent itemsets and then turn their attention to finding all the association rules with high confidence from the frequent itemsets that have been found. Consult data mining texts such as [Ag15] for details.

Although we have presented association rules in the context of market baskets, they are useful in a surprisingly wide variety of applications. For instance, association rules can be used to improve medical diagnoses, in which itemsets are collections of test results or symptoms and transactions are the collections of test results and symptoms found on patient records. Association rules, in which itemsets are baskets of key words and transactions are the collections of words on web pages, are used by search engines. Cases of plagiarism can be found using association rules, in which itemsets are collections of sentences and transactions are the contents of documents. Association rules also play helpful roles in various aspects of computer security, including intrusion detection, in which the itemsets are collections of patterns and transactions are the strings transmitted during network attacks. The interested reader will be able to find many more such applications by searching the web.

Exercises

- List the triples in the relation $\{(a, b, c) \mid a, b, \text{ and } c \text{ are integers with } 0 < a < b < c < 5\}$.
- Which 4-tuples are in the relation $\{(a, b, c, d) \mid a, b, c, \text{ and } d \text{ are positive integers with } abcd = 6\}$?
- List the 5-tuples in the relation in Table 8.
- Assuming that no new n -tuples are added, find all the primary keys for the relations displayed in
 - Table 3.
 - Table 5.
 - Table 6.
 - Table 8.
- Assuming that no new n -tuples are added, find a composite key with two fields containing the *Airline* field for the database in Table 8.
- Assuming that no new n -tuples are added, find a composite key with two fields containing the *Professor* field for the database in Table 7.
- The 3-tuples in a 3-ary relation represent the following attributes of a student database: student ID number, name, phone number.
 - Is student ID number likely to be a primary key?
 - Is name likely to be a primary key?
 - Is phone number likely to be a primary key?

8. The 4-tuples in a 4-ary relation represent these attributes of published books: title, ISBN, publication date, number of pages.
 - a) What is a likely primary key for this relation?
 - b) Under what conditions would (title, publication date) be a composite key?
 - c) Under what conditions would (title, number of pages) be a composite key?
9. The 5-tuples in a 5-ary relation represent these attributes of all people in the United States: name, Social Security number, street address, city, state.
 - a) Determine a primary key for this relation.
 - b) Under what conditions would (name, street address) be a composite key?
 - c) Under what conditions would (name, street address, city) be a composite key?
10. What do you obtain when you apply the selection operator s_C , where C is the condition $\text{Room} = \text{A100}$, to the database in Table 7?
11. What do you obtain when you apply the selection operator s_C , where C is the condition $\text{Destination} = \text{Detroit}$, to the database in Table 8?
12. What do you obtain when you apply the selection operator s_C , where C is the condition $(\text{Project} = 2) \wedge (\text{Quantity} \geq 50)$, to the database in Table 10?
13. What do you obtain when you apply the selection operator s_C , where C is the condition $(\text{Airline} = \text{Nadir}) \vee (\text{Destination} = \text{Denver})$, to the database in Table 8?
14. What do you obtain when you apply the projection $P_{2,3,5}$ to the 5-tuple (a, b, c, d, e) ?
15. Which projection mapping is used to delete the first, second, and fourth components of a 6-tuple?
16. Display the table produced by applying the projection $P_{1,2,4}$ to Table 8.
17. Display the table produced by applying the projection $P_{1,4}$ to Table 8.
18. How many components are there in the n -tuples in the table obtained by applying the join operator J_3 to two tables with 5-tuples and 8-tuples, respectively?
19. Construct the table obtained by applying the join operator J_2 to the relations in Tables 11 and 12.
20. Show that if C_1 and C_2 are conditions that elements of the n -ary relation R may satisfy, then $s_{C_1 \wedge C_2}(R) = s_{C_1}(s_{C_2}(R))$.
21. Show that if C_1 and C_2 are conditions that elements of the n -ary relation R may satisfy, then $s_{C_1}(s_{C_2}(R)) = s_{C_2}(s_{C_1}(R))$.
22. Show that if C is a condition that elements of the n -ary relations R and S may satisfy, then $s_C(R \cup S) = s_C(R) \cup s_C(S)$.
23. Show that if C is a condition that elements of the n -ary relations R and S may satisfy, then $s_C(R \cap S) = s_C(R) \cap s_C(S)$.
24. Show that if C is a condition that elements of the n -ary relations R and S may satisfy, then $s_C(R - S) = s_C(R) - s_C(S)$.

25. Show that if R and S are both n -ary relations, then $P_{i_1, i_2, \dots, i_m}(R \cup S) = P_{i_1, i_2, \dots, i_m}(R) \cup P_{i_1, i_2, \dots, i_m}(S)$.
26. Give an example to show that if R and S are both n -ary relations, then $P_{i_1, i_2, \dots, i_m}(R \cap S)$ may be different from $P_{i_1, i_2, \dots, i_m}(R) \cap P_{i_1, i_2, \dots, i_m}(S)$.
27. Give an example to show that if R and S are both n -ary relations, then $P_{i_1, i_2, \dots, i_m}(R - S)$ may be different from $P_{i_1, i_2, \dots, i_m}(R) - P_{i_1, i_2, \dots, i_m}(S)$.
28. a) What are the operations that correspond to the query expressed using this SQL statement?

```
SELECT Supplier
FROM Part_needs
WHERE 1000 ≤ Part_number ≤ 5000
```

- b) What is the output of this query given the database in Table 11 as input?
29. a) What are the operations that correspond to the query expressed using this SQL statement?

```
SELECT Supplier, Project
FROM Part_needs, Parts_inventory
WHERE Quantity ≤ 10
```

- b) What is the output of this query given the databases in Tables 11 and 12 as input?
30. Determine whether there is a primary key for the relation in Example 2.
31. Determine whether there is a primary key for the relation in Example 3.
32. Show that an n -ary relation with a primary key can be thought of as the graph of a function that maps values of the primary key to $(n - 1)$ -tuples formed from values of the other domains.
33. Suppose that the transactions at a convenience store during an evening are {bread, milk, diapers, juice}, {bread, milk, diapers, eggs}, {milk, diapers, beer, eggs}, {bread, beer}, {milk, diapers, eggs, juice}, and {milk, diapers, beer}.
 - a) Find the count and support of diapers.
 - b) Find all frequent itemsets if the threshold level is 0.6.
34. Suppose that the key words on eight different web pages are {evolution, primate, Human, Neanderthal, DNA, fossil}, {evolution, Neanderthal, Denisovan, Human, DNA}, {cave, fossil, primate}, {Human, Neanderthal, Denisovan, evolution}, {DNA, genome, evolution, fossil}, {DNA, Human, Neanderthal, Denisovan, genome}, {evolution, primate, cave, fossil}, and {Human, Neanderthal, genome}.
 - a) Find the count and support of Neanderthal.
 - b) Find all frequent itemsets if the threshold level is 0.6.
35. Find the support and confidence of the association rule {beer} $\rightarrow$ {diapers} for the set of transactions in Exercise 33. (This association rule has played an important role in the development of the subject.)
36. Find the support and confidence of the association rule {human, DNA} $\rightarrow$ {Neanderthal} for the set of transactions in Exercise 34.

TABLE 11 Part_needs.

Supplier	Part_number	Project
23	1092	1
23	1101	3
23	9048	4
31	4975	3
31	3477	2
32	6984	4
32	9191	2
33	1001	1

TABLE 12 Parts_inventory.

Part_number	Project	Quantity	Color_code
1001	1	14	8
1092	1	2	2
1101	3	1	1
3477	2	25	2
4975	3	6	2
6984	4	10	1
9048	4	12	2
9191	2	80	4

37. Suppose that I is an itemset with positive count in a set of transactions. Find the confidence of the association rule $I \rightarrow \emptyset$.
38. Suppose that I , J , and K are itemsets. Show that the six association rules $\{I, J\} \rightarrow K$, $\{J, K\} \rightarrow I$, $\{I, K\} \rightarrow J$, $I \rightarrow \{J, K\}$, $J \rightarrow \{I, K\}$, and $K \rightarrow \{I, J\}$ all have the same support.
39. The **lift** of the association rule $I \rightarrow J$, where I and J are itemsets with positive support in a set of transactions, equals $\text{support}(I \cup J) / (\text{support}(I)\text{support}(J))$.
- a) Show that the lift of $I \rightarrow J$, when $\text{support}(I)$ and $\text{support}(J)$ are both positive, equals 1 if and only if the occurrence of I in a transaction and the occurrence of J in a transaction are independent events.
- b) Find the lift of the association rule $\{\text{beer}\} \rightarrow \{\text{diapers}\}$ for the set of transactions in Exercises 33.
- c) Find the lift of the association rule $\{\text{evolution}\} \rightarrow \{\text{Neanderthals, Denisovans}\}$ for the set of transactions in Exercise 34.
40. Show that if an itemset is frequent in a set of transactions, then all its subsets are also frequent itemsets in this set of transactions.
41. Given n unique items, show that there are 3^n possible association rules of the form $I \rightarrow J$, where I and J are disjoint subsets of the set of all items. Be sure to allow the association rules where I or J , or both, are empty.

9.3 Representing Relations

9.3.1 Introduction

In this section, and in the remainder of this chapter, all relations we study will be binary relations. Because of this, in this section and in the rest of this chapter, the word relation will always refer to a binary relation. There are many ways to represent a relation between finite sets. As we have seen in Section 9.1, one way is to list its ordered pairs. Another way to represent a relation is to use a table, as we did in Example 3 in Section 9.1. In this section we will discuss two alternative methods for representing relations. One method uses zero–one matrices. The other method uses pictorial representations called directed graphs, which we will discuss later in this section.

Generally, matrices are appropriate for the representation of relations in computer programs. On the other hand, people often find the representation of relations using directed graphs useful for understanding the properties of these relations.

9.3.2 Representing Relations Using Matrices

A relation between finite sets can be represented using a zero–one matrix. Suppose that R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$. (Here the elements of the sets A and B have been listed in a particular, but arbitrary, order. Furthermore, when $A = B$ we use the same ordering for A and B .) The relation R can be represented by the matrix $\mathbf{M}_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

In other words, the zero–one matrix representing R has a 1 as its (i, j) entry when a_i is related to b_j , and a 0 in this position if a_i is not related to b_j . (Such a representation depends on the orderings used for A and B .)

The use of matrices to represent relations is illustrated in Examples 1–6.

EXAMPLE 1 Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R if $a_1 = 1$, $a_2 = 2$, and $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?

Solution: Because $R = \{(2, 1), (3, 1), (3, 2)\}$, the matrix for R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

The 1s in $\mathbf{M}_R$ show that the pairs $(2, 1)$, $(3, 1)$, and $(3, 2)$ belong to R . The 0s show that no other pairs belong to R . ▶

EXAMPLE 2 Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}?$$

Solution: Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$
 ▶

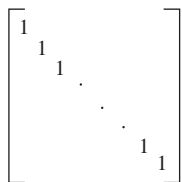


FIGURE 1 The zero–one matrix for a reflexive relation. (Off diagonal elements can be 0 or 1.)

The matrix of a relation on a set, which is a square matrix, can be used to determine whether the relation has certain properties. Recall that a relation R on A is reflexive if $(a, a) \in R$ whenever $a \in A$. Thus, R is reflexive if and only if $(a_i, a_i) \in R$ for $i = 1, 2, \dots, n$. Hence, R is reflexive if and only if $m_{ii} = 1$, for $i = 1, 2, \dots, n$. In other words, R is reflexive if all the elements on the main diagonal of $\mathbf{M}_R$ are equal to 1, as shown in Figure 1. Note that the elements off the main diagonal can be either 0 or 1.

The relation R is symmetric if $(a, b) \in R$ implies that $(b, a) \in R$. Consequently, the relation R on the set $A = \{a_1, a_2, \dots, a_n\}$ is symmetric if and only if $(a_i, a_j) \in R$ whenever $(a_j, a_i) \in R$. In terms of the entries of $\mathbf{M}_R$, R is symmetric if and only if $m_{ji} = 1$ whenever $m_{ij} = 1$. This also means $m_{ji} = 0$ whenever $m_{ij} = 0$. Consequently, R is symmetric if and only if $m_{ij} = m_{ji}$, for all pairs of integers i and j with $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$. Recalling the definition of the transpose of a matrix from Section 2.6, we see that R is symmetric if and only if

$$\mathbf{M}_R = (\mathbf{M}_R)'$$

that is, if $\mathbf{M}_R$ is a symmetric matrix. The form of the matrix for a symmetric relation is illustrated in Figure 2(a).

The relation R is antisymmetric if and only if $(a, b) \in R$ and $(b, a) \in R$ imply that $a = b$. Consequently, the matrix of an antisymmetric relation has the property that if $m_{ij} = 1$ with $i \neq j$, then $m_{ji} = 0$. Or, in other words, either $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$. The form of the matrix for an antisymmetric relation is illustrated in Figure 2(b).

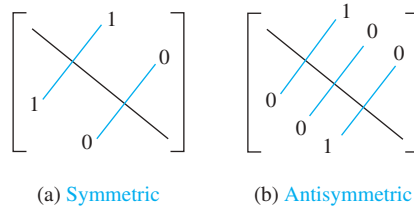


FIGURE 2 The zero-one matrices for symmetric and antisymmetric relations.

EXAMPLE 3 Suppose that the relation R on a set is represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution: Because all the diagonal elements of this matrix are equal to 1, R is reflexive. Moreover, because $\mathbf{M}_R$ is symmetric, it follows that R is symmetric. It is also easy to see that R is not antisymmetric. ▶

The Boolean operations join and meet (discussed in Section 2.6) can be used to find the matrices representing the union and the intersection of two relations. Suppose that R_1 and R_2 are relations on a set A represented by the matrices $\mathbf{M}_{R_1}$ and $\mathbf{M}_{R_2}$, respectively. The matrix representing the union of these relations has a 1 in the positions where either $\mathbf{M}_{R_1}$ or $\mathbf{M}_{R_2}$ has a 1. The matrix representing the intersection of these relations has a 1 in the positions where both $\mathbf{M}_{R_1}$ and $\mathbf{M}_{R_2}$ have a 1. Thus, the matrices representing the union and intersection of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} \quad \text{and} \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2}.$$

EXAMPLE 4 Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Solution: The matrices of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$
▶

We now turn our attention to determining the matrix for the composite of relations. This matrix can be found using the Boolean product of the matrices (discussed in Section 2.6)

for these relations. In particular, suppose that R is a relation from A to B and S is a relation from B to C . Suppose that A , B , and C have m , n , and p elements, respectively. Let the zero-one matrices for $S \circ R$, R , and S be $\mathbf{M}_{S \circ R} = [t_{ij}]$, $\mathbf{M}_R = [r_{ij}]$, and $\mathbf{M}_S = [s_{ij}]$, respectively (these matrices have sizes $m \times p$, $m \times n$, and $n \times p$, respectively). The ordered pair (a_i, c_j) belongs to $S \circ R$ if and only if there is an element b_k such that (a_i, b_k) belongs to R and (b_k, c_j) belongs to S . It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k . From the definition of the Boolean product, this means that

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S.$$

EXAMPLE 5 Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The matrix representing the composite of two relations can be used to find the matrix for $\mathbf{M}_{R^n}$. In particular,

$$\mathbf{M}_{R^n} = \mathbf{M}_R^{[n]},$$

from the definition of Boolean powers. Exercise 35 asks for a proof of this formula.

EXAMPLE 6 Find the matrix representing the relation R^2 , where the matrix representing R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Solution: The matrix for R^2 is

$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

9.3.3 Representing Relations Using Digraphs

We have shown that a relation can be represented by listing all of its ordered pairs or by using a zero-one matrix. There is another important way of representing a relation using a pictorial representation. Each element of the set is represented by a point, and each ordered pair is represented using an arc with its direction indicated by an arrow. We use such pictorial representations when we think of relations on a finite set as **directed graphs**, or **digraphs**.

Definition 1

A *directed graph*, or *digraph*, consists of a set V of *vertices* (or *nodes*) together with a set E of ordered pairs of elements of V called *edges* (or *arcs*). The vertex a is called the *initial vertex* of the edge (a, b) , and the vertex b is called the *terminal vertex* of this edge.

An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a **loop**.

EXAMPLE 7

The directed graph with vertices a, b, c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is displayed in Figure 3.

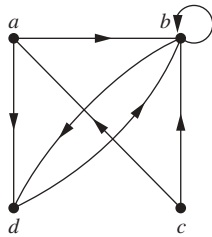


FIGURE 3
A directed graph.

The relation R on a set A is represented by the directed graph that has the elements of A as its vertices and the ordered pairs (a, b) , where $(a, b) \in R$, as edges. This assignment sets up a one-to-one correspondence between the relations on a set A and the directed graphs with A as their set of vertices. Thus, every statement about relations corresponds to a statement about directed graphs, and vice versa. Directed graphs give a visual display of information about relations. As such, they are often used to study relations and their properties. (Note that relations from a set A to a set B can be represented by a directed graph where there is a vertex for each element of A and a vertex for each element of B , as shown in Section 9.1. However, when $A = B$, such representation provides much less insight than the digraph representations described here.) The use of directed graphs to represent relations on a set is illustrated in Examples 8–10.

EXAMPLE 8

The directed graph of the relation

$$R_1 = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)\}$$

on the set $\{1, 2, 3, 4\}$ is shown in Figure 4.

EXAMPLE 9

What are the ordered pairs in the relation R_2 represented by the directed graph shown in Figure 5?

Solution: The ordered pairs (x, y) in the relation are

$$R_2 = \{(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1), (4, 3)\}.$$

Each of these pairs corresponds to an edge of the directed graph, with $(2, 2)$ and $(3, 3)$ corresponding to loops.

We will study directed graphs extensively in Chapter 10.

The directed graph representing a relation can be used to determine whether the relation has various properties. For instance, a relation is reflexive if and only if there is a loop at every vertex of the directed graph, so that every ordered pair of the form (x, x) occurs in the relation. A relation is symmetric if and only if for every edge between distinct vertices in its digraph there is an edge in the opposite direction, so that (y, x) is in the relation whenever (x, y) is in the relation. Similarly, a relation is antisymmetric if and only if there are never two edges in opposite directions between distinct vertices. Finally, a relation is transitive if and only if whenever there is an edge from a vertex x to a vertex y and an edge from a vertex y to a vertex z , there is an edge from x to z (completing a triangle where each side is a directed edge with the correct direction).

Remark: Note that a symmetric relation can be represented by an undirected graph, which is a graph where edges do not have directions. We will study undirected graphs in Chapter 10.

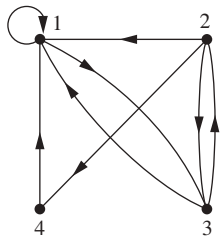


FIGURE 4 The directed graph of the relation R_1 .

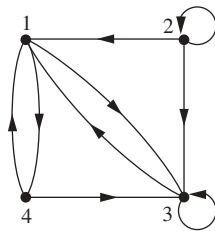


FIGURE 5 The directed graph of the relation R_2 .

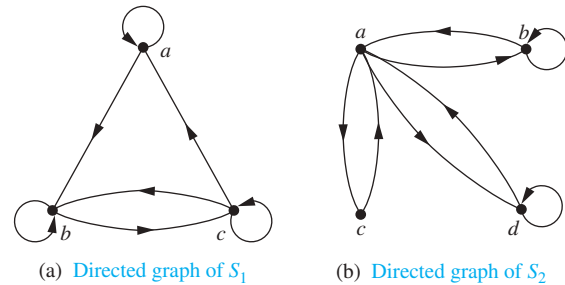


FIGURE 6 The directed graphs of the relations S_1 and S_2 .

EXAMPLE 10 Determine whether the relations for the directed graphs shown in Figure 6 are reflexive, symmetric, antisymmetric, and/or transitive.

Solution: Because there are loops at every vertex of the directed graph of S_1 , it is reflexive. The relation S_1 is neither symmetric nor antisymmetric because there is an edge from a to b but not one from b to a , but there are edges in both directions connecting b and c . Finally, S_1 is not transitive because there is an edge from a to b and an edge from b to c , but no edge from a to c .

Because loops are not present at all the vertices of the directed graph of S_2 , this relation is not reflexive. It is symmetric and not antisymmetric, because every edge between distinct vertices is accompanied by an edge in the opposite direction. It is also not hard to see from the directed graph that S_2 is not transitive, because (c, a) and (a, b) belong to S_2 , but (c, b) does not belong to S_2 . ◀

Exercises

- Represent each of these relations on $\{1, 2, 3\}$ with a matrix (with the elements of this set listed in increasing order).
 - $\{(1, 1), (1, 2), (1, 3)\}$
 - $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$
 - $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$
 - $\{(1, 3), (3, 1)\}$
- Represent each of these relations on $\{1, 2, 3, 4\}$ with a matrix (with the elements of this set listed in increasing order).
 - $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
 - $\{(1, 1), (1, 4), (2, 2), (3, 3), (4, 1)\}$
 - $\{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$
 - $\{(2, 4), (3, 1), (3, 2), (3, 4)\}$
- List the ordered pairs in the relations on $\{1, 2, 3\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a)
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- List the ordered pairs in the relations on $\{1, 2, 3, 4\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a)
$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

- How can the matrix representing a relation R on a set A be used to determine whether the relation is irreflexive?
- How can the matrix representing a relation R on a set A be used to determine whether the relation is asymmetric?
- Determine whether the relations represented by the matrices in Exercise 3 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.
- Determine whether the relations represented by the matrices in Exercise 4 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

9. How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 100\}$ consisting of the first 100 positive integers have if R is
- $\{(a, b) \mid a > b\}$
 - $\{(a, b) \mid a \neq b\}$
 - $\{(a, b) \mid a = b + 1\}$
 - $\{(a, b) \mid a = 1\}$
 - $\{(a, b) \mid ab = 1\}$
10. How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 1000\}$ consisting of the first 1000 positive integers have if R is
- $\{(a, b) \mid a \leq b\}$
 - $\{(a, b) \mid a = b \pm 1\}$
 - $\{(a, b) \mid a + b = 1000\}$
 - $\{(a, b) \mid a + b \leq 1001\}$
 - $\{(a, b) \mid a \neq 0\}$
11. How can the matrix for $\bar{R}$, the complement of the relation R , be found from the matrix representing R , when R is a relation on a finite set A ?
12. How can the matrix for R^{-1} , the inverse of the relation R , be found from the matrix representing R , when R is a relation on a finite set A ?
13. Let R be the relation represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Find the matrix representing

- R^{-1} .
 - $\bar{R}$.
 - R^2 .
14. Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

- $R_1 \cup R_2$.
 - $R_1 \cap R_2$.
 - $R_2 \circ R_1$.
 - $R_1 \circ R_1$.
 - $R_1 \oplus R_2$.
15. Let R be the relation represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

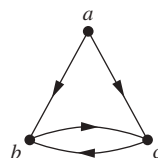
Find the matrices that represent

- R^2 .
 - R^3 .
 - R^4 .
16. Let R be a relation on a set A with n elements. If there are k nonzero entries in $\mathbf{M}_R$, the matrix representing R , how many nonzero entries are there in $\mathbf{M}_{R^{-1}}$, the matrix representing R^{-1} , the inverse of R ?
17. Let R be a relation on a set A with n elements. If there are k nonzero entries in $\mathbf{M}_R$, the matrix representing R , how many nonzero entries are there in $\mathbf{M}_{\bar{R}}$, the matrix representing $\bar{R}$, the complement of R ?
18. Draw the directed graphs representing each of the relations from Exercise 1.

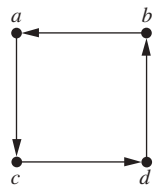
19. Draw the directed graphs representing each of the relations from Exercise 2.
20. Draw the directed graph representing each of the relations from Exercise 3.
21. Draw the directed graph representing each of the relations from Exercise 4.
22. Draw the directed graph that represents the relation $\{(a, a), (a, b), (b, c), (c, b), (c, d), (d, a), (d, b)\}$.

In Exercises 23–28 list the ordered pairs in the relations represented by the directed graphs.

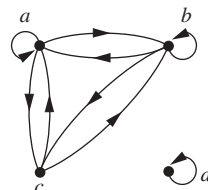
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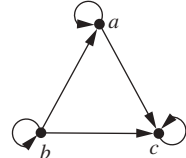
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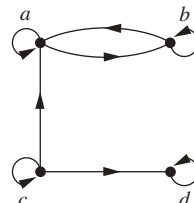
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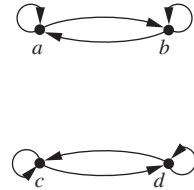
24.



26.



28.



29. How can the directed graph of a relation R on a finite set A be used to determine whether a relation is asymmetric?
30. How can the directed graph of a relation R on a finite set A be used to determine whether a relation is irreflexive?
31. Determine whether the relations represented by the directed graphs shown in Exercises 23–25 are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.
32. Determine whether the relations represented by the directed graphs shown in Exercises 26–28 are reflexive, irreflexive, symmetric, antisymmetric, asymmetric, and/or transitive.
33. Let R be a relation on a set A . Explain how to use the directed graph representing R to obtain the directed graph representing the inverse relation R^{-1} .
34. Let R be a relation on a set A . Explain how to use the directed graph representing R to obtain the directed graph representing the complementary relation $\bar{R}$.
35. Show that if $\mathbf{M}_R$ is the matrix representing the relation R , then $\mathbf{M}_R^{[n]}$ is the matrix representing the relation R^n .
36. Given the directed graphs representing two relations, how can the directed graph of the union, intersection, symmetric difference, difference, and composition of these relations be found?

9.4 Closures of Relations

9.4.1 Introduction

A computer network has data centers in Boston, Chicago, Denver, Detroit, New York, and San Diego. There are direct, one-way telephone lines from Boston to Chicago, from Boston to Detroit, from Chicago to Detroit, from Detroit to Denver, and from New York to San Diego. Let R be the relation containing (a, b) if there is a telephone line from the data center in a to that in b . How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one center to another? Because not all links are direct, such as the link from Boston to Denver that goes through Detroit, R cannot be used directly to answer this. In the language of relations, R is not transitive, so it does not contain all the pairs that can be linked. As we will show in this section, we can find all pairs of data centers that have a link by constructing a transitive relation S containing R such that S is a subset of every transitive relation containing R . Here, S is the smallest transitive relation that contains R . This relation is called the **transitive closure** of R .

9.4.2 Different Types of Closures

If R is a relation on a set A , it may or may not have some property $\mathbf{P}$, such as reflexivity, symmetry, or transitivity. When R does not enjoy property $\mathbf{P}$, we would like to find the smallest relation S on A with property $\mathbf{P}$ that contains R .

Definition 1

If R is a relation on a set A , then the **closure** of R with respect to $\mathbf{P}$, if it exists, is the relation S on A with property $\mathbf{P}$ that contains R and is a subset of every subset of $A \times A$ containing R with property $\mathbf{P}$.

If there is a relation S that is a subset of every relation containing R with property $\mathbf{P}$, it must be unique. To see this, suppose that relations S and T both have property $\mathbf{P}$ and are subsets of every relation with property $\mathbf{P}$ that contains R . Then, S and T are subsets of each other, and so are equal. Such a relation, if it exists, is the smallest relation with property $\mathbf{P}$ that contains R because it is a subset of every relation with property $\mathbf{P}$ that contains R .

We will show how reflexive, symmetric, and transitive closures of relations can be found. In Exercises 15 and 35 we give properties $\mathbf{P}$ for which the closure of a relation with respect to $\mathbf{P}$ may not exist.

The relation $R = \{(1, 1), (1, 2), (2, 1), (3, 2)\}$ on the set $A = \{1, 2, 3\}$ is not reflexive. How can we produce a reflexive relation containing R that is as small as possible? This can be done by adding $(2, 2)$ and $(3, 3)$ to R , because these are the only pairs of the form (a, a) that are not in R . This new relation contains R . Furthermore, *any* reflexive relation that contains R must also contain $(2, 2)$ and $(3, 3)$. Because this relation contains R , is reflexive, and is contained within every reflexive relation that contains R , it is called the **reflexive closure** of R .

As this example illustrates, given a relation R on a set A , the reflexive closure of R can be formed by adding to R all pairs of the form (a, a) with $a \in A$, not already in R . The addition of these pairs produces a new relation that is reflexive, contains R , and is contained within any reflexive relation containing R . We see that the reflexive closure of R equals $R \cup \Delta$, where $\Delta = \{(a, a) \mid a \in A\}$ is the **diagonal relation** on A . (The reader should verify this.)

EXAMPLE 1 What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers?

Solution: The reflexive closure of R is

$$R \cup \Delta = \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in \mathbf{Z}\} = \{(a, b) \mid a \leq b\}.$$



The relation $\{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2)\}$ on $\{1, 2, 3\}$ is not symmetric. How can we produce a symmetric relation that is as small as possible and contains R ? To do this, we need only add $(2, 1)$ and $(1, 3)$, because these are the only pairs of the form (b, a) with $(a, b) \in R$ that are not in R . This new relation is symmetric and contains R . Furthermore, *any* symmetric relation that contains R must contain this new relation, because a symmetric relation that contains R must contain $(2, 1)$ and $(1, 3)$. Consequently, this new relation is called the **symmetric closure** of R .

As this example illustrates, the symmetric closure of a relation R can be constructed by adding all ordered pairs of the form (b, a) , where (a, b) is in the relation, that are not already present in R . Adding these pairs produces a relation that is symmetric, that contains R , and that is contained in any symmetric relation that contains R . The symmetric closure of a relation can be constructed by taking the union of a relation with its inverse (defined in the preamble of Exercise 26 in Section 9.1); that is, $R \cup R^{-1}$ is the symmetric closure of R , where $R^{-1} = \{(b, a) \mid (a, b) \in R\}$. The reader should verify this statement.

EXAMPLE 2 What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

**Extra
Examples** 

Solution: The symmetric closure of R is the relation

$$R \cup R^{-1} = \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a \neq b\}.$$

This last equality follows because R contains all ordered pairs of positive integers, where the first element is greater than the second element, and R^{-1} contains all ordered pairs of positive integers, where the first element is less than the second. 

Suppose that a relation R is not transitive. How can we produce a transitive relation that contains R such that this new relation is contained within any transitive relation that contains R ? Can the transitive closure of a relation R be produced by adding all the pairs of the form (a, c) , where (a, b) and (b, c) are already in the relation? Consider the relation $R = \{(1, 3), (1, 4), (2, 1), (3, 2)\}$ on the set $\{1, 2, 3, 4\}$. This relation is not transitive because it does not contain all pairs of the form (a, c) where (a, b) and (b, c) are in R . The pairs of this form not in R are $(1, 2)$, $(2, 3)$, $(2, 4)$, and $(3, 1)$. Adding these pairs does *not* produce a transitive relation, because the resulting relation contains $(3, 1)$ and $(1, 4)$ but does not contain $(3, 4)$. This shows that constructing the transitive closure of a relation is more complicated than constructing either the reflexive or symmetric closure. The rest of this section develops algorithms for constructing transitive closures. As will be shown later in this section, the transitive closure of a relation can be found by adding new ordered pairs that must be present and then repeating this process until no new ordered pairs are needed.

9.4.3 Paths in Directed Graphs

We will see that representing relations by directed graphs helps in the construction of transitive closures. We now introduce some terminology that we will use for this purpose.

A path in a directed graph is obtained by traversing along edges (in the same direction as indicated by the arrow on the edge).

Definition 2

A *path* from a to b in the directed graph G is a sequence of edges $(x_0, x_1), (x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n)$ in G , where n is a nonnegative integer, and $x_0 = a$ and $x_n = b$, that is, a sequence of edges where the terminal vertex of an edge is the same as the initial vertex in the next edge in the path. This path is denoted by $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ and has *length* n . We view the empty set of edges as a path of length zero from a to a . A path of length $n \geq 1$ that begins and ends at the same vertex is called a *circuit* or *cycle*.

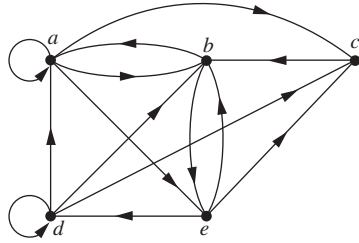


FIGURE 1 A directed graph.

A path in a directed graph can pass through a vertex more than once. Moreover, an edge in a directed graph can occur more than once in a path.

EXAMPLE 3 Which of the following are paths in the directed graph shown in Figure 1: a, b, e, d ; a, e, c, d, b ; b, a, c, b, a, a, b ; d, c ; c, b, a ; e, b, a, b, a, b, e ? What are the lengths of those that are paths? Which of the paths in this list are circuits?

Solution: Because each of (a, b) , (b, e) , and (e, d) is an edge, a, b, e, d is a path of length three. Because (c, d) is not an edge, a, e, c, d, b is not a path. Also, b, a, c, b, a, a, b is a path of length six because (b, a) , (a, c) , (c, b) , (b, a) , (a, a) , and (a, b) are all edges. We see that d, c is a path of length one, because (d, c) is an edge. Also c, b, a is a path of length two, because (c, b) and (b, a) are edges. All of (e, b) , (b, a) , (a, b) , (b, a) , (a, b) , and (b, e) are edges, so e, b, a, b, a, b, e is a path of length six.

The two paths b, a, c, b, a, a, b and e, b, a, b, a, b, e are circuits because they begin and end at the same vertex. The paths a, b, e, d ; c, b, a ; and d, c are not circuits. ◀

The term *path* also applies to relations. Carrying over the definition from directed graphs to relations, there is a **path** from a to b in R if there is a sequence of elements $a, x_1, x_2, \dots, x_{n-1}, b$ with $(a, x_1) \in R$, $(x_1, x_2) \in R, \dots$, and $(x_{n-1}, b) \in R$. Theorem 1 can be obtained from the definition of a path in a relation.

THEOREM 1 Let R be a relation on a set A . There is a path of length n , where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Proof: We will use mathematical induction. By definition, there is a path from a to b of length one if and only if $(a, b) \in R$, so the theorem is true when $n = 1$.

Assume that the theorem is true for the positive integer n . This is the inductive hypothesis. There is a path of length $n + 1$ from a to b if and only if there is an element $c \in A$ such that there is a path of length one from a to c , so $(a, c) \in R$, and a path of length n from c to b , that is, $(c, b) \in R^n$. Consequently, by the inductive hypothesis, there is a path of length $n + 1$ from a to b if and only if there is an element c with $(a, c) \in R$ and $(c, b) \in R^n$. But there is such an element if and only if $(a, b) \in R^{n+1}$. Therefore, there is a path of length $n + 1$ from a to b if and only if $(a, b) \in R^{n+1}$. This completes the proof. ◀

9.4.4 Transitive Closures

We now show that finding the transitive closure of a relation is equivalent to determining which pairs of vertices in the associated directed graph are connected by a path. With this in mind, we define a new relation.

Definition 3

Let R be a relation on a set A . The *connectivity relation* R^* consists of the pairs (a, b) such that there is a path of length at least one from a to b in R .

Because R^n consists of the pairs (a, b) such that there is a path of length n from a to b , it follows that R^* is the union of all the sets R^n . In other words,

$$R^* = \bigcup_{n=1}^{\infty} R^n.$$

The connectivity relation is useful in many models.

EXAMPLE 4 Let R be the relation on the set of all people in the world that contains (a, b) if a has met b . What is R^n , where n is a positive integer greater than one? What is R^* ?

Solution: The relation R^2 contains (a, b) if there is a person c such that $(a, c) \in R$ and $(c, b) \in R$, that is, if there is a person c such that a has met c and c has met b . Similarly, R^n consists of those pairs (a, b) such that there are people $x_1, x_2, \dots, x_{n-1}$ such that a has met x_1 , x_1 has met x_2 , $\dots$, and x_{n-1} has met b .

The relation R^* contains (a, b) if there is a sequence of people, starting with a and ending with b , such that each person in the sequence has met the next person in the sequence. (There are many interesting conjectures about R^* . Do you think that this connectivity relation includes the pair with you as the first element and the president of Mongolia as the second element? We will use graphs to model this application in Chapter 10.)

EXAMPLE 5 Let R be the relation on the set of all subway stops in New York City that contains (a, b) if it is possible to travel from stop a to stop b without changing trains. What is R^n when n is a positive integer? What is R^* ?

Solution: The relation R^n contains (a, b) if it is possible to travel from stop a to stop b by making at most $n - 1$ changes of trains. The relation R^* consists of the ordered pairs (a, b) where it is possible to travel from stop a to stop b making as many changes of trains as necessary. (The reader should verify these statements.)

EXAMPLE 6 Let R be the relation on the set of all states in the United States that contains (a, b) if state a and state b have a common border. What is R^n , where n is a positive integer? What is R^* ?

Solution: The relation R^n consists of the pairs (a, b) , where it is possible to go from state a to state b by crossing exactly n state borders. R^* consists of the ordered pairs (a, b) , where it is possible to go from state a to state b crossing as many borders as necessary. (The reader should verify these statements.) The only ordered pairs not in R^* are those containing states that are not connected to the continental United States (that is, those pairs containing Alaska or Hawaii).

Theorem 2 shows that the transitive closure of a relation and the associated connectivity relation are the same.

THEOREM 2

The transitive closure of a relation R equals the connectivity relation R^* .

Proof: Note that R^* contains R by definition. To show that R^* is the transitive closure of R we must also show that R^* is transitive and that $R^* \subseteq S$ whenever S is a transitive relation that contains R .

First, we show that R^* is transitive. If $(a, b) \in R^*$ and $(b, c) \in R^*$, then there are paths from a to b and from b to c in R . We obtain a path from a to c by starting with the path from a to b and following it with the path from b to c . Hence, $(a, c) \in R^*$. It follows that R^* is transitive.

Now suppose that S is a transitive relation containing R . Because S is transitive, S^n also is transitive (the reader should verify this) and $S^n \subseteq S$ (by Theorem 1 of Section 9.1). Furthermore, because

$$S^* = \bigcup_{k=1}^{\infty} S^k$$

and $S^k \subseteq S$, it follows that $S^* \subseteq S$. Now note that if $R \subseteq S$, then $R^* \subseteq S^*$, because any path in R is also a path in S . Consequently, $R^* \subseteq S^* \subseteq S$. Thus, any transitive relation that contains R must also contain R^* . Therefore, R^* is the transitive closure of R . $\triangleleft$

Now that we know that the transitive closure equals the connectivity relation, we turn our attention to the problem of computing this relation. We do not need to examine arbitrarily long paths to determine whether there is a path between two vertices in a finite directed graph. As Lemma 1 shows, it is sufficient to examine paths containing no more than n edges, where n is the number of elements in the set.

LEMMA 1

Let A be a set with n elements, and let R be a relation on A . If there is a path of length at least one in R from a to b , then there is such a path with length not exceeding n . Moreover, when $a \neq b$, if there is a path of length at least one in R from a to b , then there is such a path with length not exceeding $n - 1$.

Proof: Suppose there is a path from a to b in R . Let m be the length of the shortest such path. Suppose that $x_0, x_1, x_2, \dots, x_{m-1}, x_m$, where $x_0 = a$ and $x_m = b$, is such a path.

Suppose that $a \neq b$ and that $m > n$, so that $m \geq n + 1$. By the pigeonhole principle, because there are n vertices in A , among the m vertices $x_0, x_1, \dots, x_{m-1}$, at least two are equal (see Figure 2).

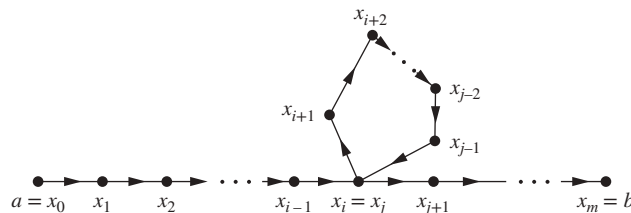


FIGURE 2 Producing a path with length not exceeding n .

Suppose that $x_i = x_j$ with $0 \leq i < j \leq m-1$. Then the path contains a circuit from x_i to itself. This circuit can be deleted from the path from a to b , leaving a path, namely, $x_0, x_1, \dots, x_i, x_{j+1}, \dots, x_{m-1}, x_m$, from a to b of shorter length. Hence, the path of shortest length must have length less than or equal to n .

The case where $a \neq b$ is left as an exercise for the reader. 

From Lemma 1, we see that the transitive closure of R is the union of $R, R^2, R^3, \dots$, and R^n . This follows because there is a path in R^* between two vertices if and only if there is a path between these vertices in R^i , for some positive integer i with $i \leq n$. Because

$$R^* = R \cup R^2 \cup R^3 \cup \dots \cup R^n$$

and the zero-one matrix representing a union of relations is the join of the zero-one matrices of these relations, the zero-one matrix for the transitive closure is the join of the zero-one matrices of the first n powers of the zero-one matrix of R .

THEOREM 3

Let $\mathbf{M}_R$ be the zero-one matrix of the relation R on a set with n elements. Then the zero-one matrix of the transitive closure R^* is

$$\mathbf{M}_{R^*} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \mathbf{M}_R^{[3]} \vee \dots \vee \mathbf{M}_R^{[n]}.$$

EXAMPLE 7 Find the zero-one matrix of the transitive closure of the relation R where

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: By Theorem 3, it follows that the zero-one matrix of R^* is

$$\mathbf{M}_{R^*} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \mathbf{M}_R^{[3]}.$$

Because

$$\mathbf{M}_R^{[2]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_R^{[3]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix},$$

it follows that

$$\mathbf{M}_{R^*} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Links 

Theorem 3 can be used as a basis for an algorithm for computing the matrix of the relation R^* . To find this matrix, the successive Boolean powers of $\mathbf{M}_R$, up to the n th power, are computed. As each power is calculated, its join with the join of all smaller powers is formed. When this is done with the n th power, the matrix for R^* has been found. This procedure is displayed as Algorithm 1.

ALGORITHM 1 A Procedure for Computing the Transitive Closure.

```

procedure transitive closure ( $\mathbf{M}_R$  : zero-one  $n \times n$  matrix)
 $\mathbf{A} := \mathbf{M}_R$ 
 $\mathbf{B} := \mathbf{A}$ 
for  $i := 2$  to  $n$ 
     $\mathbf{A} := \mathbf{A} \odot \mathbf{M}_R$ 
     $\mathbf{B} := \mathbf{B} \vee \mathbf{A}$ 
return  $\mathbf{B}$  { $\mathbf{B}$  is the zero-one matrix for  $R^*$ }

```

We can easily find the number of bit operations used by Algorithm 1 to determine the transitive closure of a relation. Computing the Boolean powers $\mathbf{M}_R, \mathbf{M}_R^{[2]}, \dots, \mathbf{M}_R^{[n]}$ requires that $n - 1$ Boolean products of $n \times n$ zero-one matrices be found. Each of these Boolean products can be found using $n^2(2n - 1)$ bit operations. Hence, these products can be computed using $n^2(2n - 1)(n - 1)$ bit operations.

To find $\mathbf{M}_{R^*}$ from the n Boolean powers of $\mathbf{M}_R$, $n - 1$ joins of zero-one matrices need to be found. Computing each of these joins uses n^2 bit operations. Hence, $(n - 1)n^2$ bit operations are used in this part of the computation. Therefore, when Algorithm 1 is used, the matrix of the transitive closure of a relation on a set with n elements can be found using $n^2(2n - 1)(n - 1) + (n - 1)n^2 = 2n^3(n - 1)$, which is $O(n^4)$ bit operations. The remainder of this section describes a more efficient algorithm for finding transitive closures.

9.4.5 Warshall's Algorithm



Warshall's algorithm, named after Stephen Warshall, who described this algorithm in 1960, is an efficient method for computing the transitive closure of a relation. Algorithm 1 can find the transitive closure of a relation on a set with n elements using $2n^3(n - 1)$ bit operations. However, the transitive closure can be found by Warshall's algorithm using only $2n^3$ bit operations.

Remark: Warshall's algorithm is sometimes called the Roy–Warshall algorithm, because Bernard Roy described this algorithm in 1959.

Suppose that R is a relation on a set with n elements. Let $v_1, v_2, \dots, v_n$ be an arbitrary listing of these n elements. The concept of the **interior vertices** of a path is used in Warshall's algorithm. If $a, x_1, x_2, \dots, x_{m-1}, b$ is a path, its interior vertices are $x_1, x_2, \dots, x_{m-1}$, that is, all the vertices of the path that occur somewhere other than as the first and last vertices in the path. For instance, the interior vertices of a path a, c, d, f, g, h, b, j in a directed graph are c, d, f, g, h , and b . The interior vertices of a, c, d, a, f, b are c, d, a , and f . (Note that the first vertex in the path is not an interior vertex unless it is visited again by the path, except as the last vertex. Similarly, the last vertex in the path is not an interior vertex unless it was visited previously by the path, except as the first vertex.)

Warshall's algorithm is based on the construction of a sequence of zero-one matrices. These matrices are $\mathbf{W}_0, \mathbf{W}_1, \dots, \mathbf{W}_n$, where $\mathbf{W}_0 = \mathbf{M}_R$ is the zero-one matrix of this relation, and $\mathbf{W}_k = [w_{ij}^{(k)}]$, where $w_{ij}^{(k)} = 1$ if there is a path from v_i to v_j such that all the interior vertices of this path are in the set $\{v_1, v_2, \dots, v_k\}$ (the first k vertices in the list) and is 0 otherwise. (The first and last vertices in the path may be outside the set of the first k vertices in the list.) Note that $\mathbf{W}_n = \mathbf{M}_{R^*}$, because the (i, j) th entry of $\mathbf{M}_{R^*}$ is 1 if and only if there is a path from v_i to v_j , with all interior

vertices in the set $\{v_1, v_2, \dots, v_n\}$ (but these are the only vertices in the directed graph). Example 8 illustrates what the matrix $\mathbf{W}_k$ represents.

EXAMPLE 8 Let R be the relation with directed graph shown in Figure 3. Let a, b, c, d be a listing of the elements of the set. Find the matrices $\mathbf{W}_0, \mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3$, and $\mathbf{W}_4$. The matrix $\mathbf{W}_4$ is the transitive closure of R .

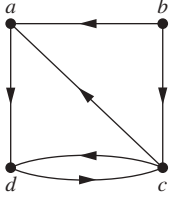


FIGURE 3
The directed
graph of the
relation R .

Solution: Let $v_1 = a, v_2 = b, v_3 = c$, and $v_4 = d$. $\mathbf{W}_0$ is the matrix of the relation. Hence,

$$\mathbf{W}_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$\mathbf{W}_1$ has 1 as its (i, j) th entry if there is a path from v_i to v_j that has only $v_1 = a$ as an interior vertex. Note that all paths of length one can still be used because they have no interior vertices. Also, there is now an allowable path from b to d , namely, b, a, d . Hence,

$$\mathbf{W}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

$\mathbf{W}_2$ has 1 as its (i, j) th entry if there is a path from v_i to v_j that has only $v_1 = a$ and/or $v_2 = b$ as its interior vertices, if any. Because there are no edges that have b as a terminal vertex, no new paths are obtained when we permit b to be an interior vertex. Hence, $\mathbf{W}_2 = \mathbf{W}_1$.

$\mathbf{W}_3$ has 1 as its (i, j) th entry if there is a path from v_i to v_j that has only $v_1 = a, v_2 = b$, and/or $v_3 = c$ as its interior vertices, if any. We now have paths from d to a , namely, d, c, a , and from d to d , namely, d, c, d . Hence,

$$\mathbf{W}_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

Finally, $\mathbf{W}_4$ has 1 as its (i, j) th entry if there is a path from v_i to v_j that has $v_1 = a, v_2 = b, v_3 = c$, and/or $v_4 = d$ as interior vertices, if any. Because these are all the vertices of the graph, this entry is 1 if and only if there is a path from v_i to v_j . Hence,

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

This last matrix, $\mathbf{W}_4$, is the matrix of the transitive closure. ◀

Warshall's algorithm computes $\mathbf{M}_{R^*}$ by efficiently computing $\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_n = \mathbf{M}_{R^*}$. This observation shows that we can compute $\mathbf{W}_k$ directly from $\mathbf{W}_{k-1}$: There is a path from v_i to v_j with no vertices other than $v_1, v_2, \dots, v_k$ as interior vertices if and only if either there is a path from v_i to v_j with its interior vertices among the first $k-1$ vertices in the list, or there are paths from v_i to v_k and from v_k to v_j that have interior vertices only among the first $k-1$ vertices in the list. That is, either a path from v_i to v_j already existed before v_k was permitted as an interior vertex, or allowing v_k as an interior vertex produces a path that goes from v_i to v_k and then from v_k to v_j . These two cases are shown in Figure 4.

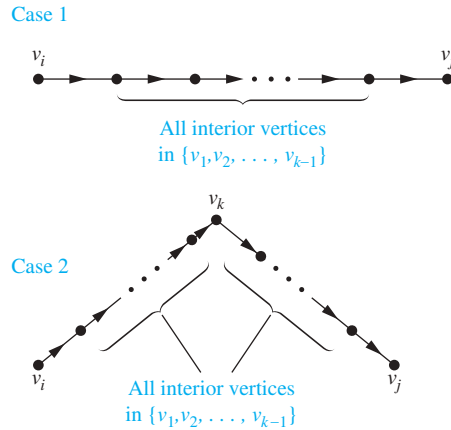


FIGURE 4 Adding v_k to the set of allowable interior vertices.

The first type of path exists if and only if $w_{ij}^{[k-1]} = 1$, and the second type of path exists if and only if both $w_{ik}^{[k-1]}$ and $w_{kj}^{[k-1]}$ are 1. Hence, $w_{ij}^{[k]}$ is 1 if and only if either $w_{ij}^{[k-1]}$ is 1 or both $w_{ik}^{[k-1]}$ and $w_{kj}^{[k-1]}$ are 1. This gives us Lemma 2.

LEMMA 2

Let $\mathbf{W}_k = [w_{ij}^{[k]}]$ be the zero–one matrix that has a 1 in its (i, j) th position if and only if there is a path from v_i to v_j with interior vertices from the set $\{v_1, v_2, \dots, v_k\}$. Then

$$w_{ij}^{[k]} = w_{ij}^{[k-1]} \vee (w_{ik}^{[k-1]} \wedge w_{kj}^{[k-1]}),$$

whenever i, j , and k are positive integers not exceeding n .

Lemma 2 gives us the means to compute efficiently the matrices $\mathbf{W}_k, k = 1, 2, \dots, n$. We display the pseudocode for Warshall’s algorithm, using Lemma 2, as Algorithm 2.

Links



Courtesy of Stephen Warshall

STEPHEN WARSHALL (1935–2006) Stephen Warshall, born in New York City, went to public school in Brooklyn. He attended Harvard University, receiving his degree in mathematics in 1956. He never received an advanced degree, because at that time no programs were available in his areas of interest. However, he took graduate courses at several different universities and contributed to the development of computer science and software engineering.

After graduating from Harvard, Warshall worked at ORO (Operation Research Office), which was set up by Johns Hopkins to do research and development for the U.S. Army. In 1958 he left ORO to take a position at a company called Technical Operations, where he helped build a research and development laboratory for military software projects. In 1961 he left Technical Operations to found Massachusetts Computer Associates. Later, this company became part of Applied Data Research (ADR). After the merger, Warshall sat on the board of directors of ADR and managed a variety of projects and organizations. He retired from ADR in 1982.

During his career Warshall carried out research and development in operating systems, compiler design, language design, and operations research. In the 1971–1972 academic year he presented lectures on software engineering at French universities. There is an interesting anecdote about his proof that the transitive closure algorithm, now known as Warshall’s algorithm, is correct. He and a colleague at Technical Operations bet a bottle of rum on who first could determine whether this algorithm always works. Warshall came up with his proof overnight, winning the bet and the rum, which he shared with the loser of the bet. Because Warshall did not like sitting at a desk, he did much of his creative work in unconventional places, such as on a sailboat in the Indian Ocean or in a Greek lemon orchard.

ALGORITHM 2 Warshall Algorithm.

```

procedure Warshall ( $\mathbf{M}_R : n \times n$  zero-one matrix)
 $\mathbf{W} := \mathbf{M}_R$ 
for  $k := 1$  to  $n$ 
    for  $i := 1$  to  $n$ 
        for  $j := 1$  to  $n$ 
             $w_{ij} := w_{ij} \vee (w_{ik} \wedge w_{kj})$ 
return  $\mathbf{W}$  {  $\mathbf{W} = [w_{ij}]$  is  $\mathbf{M}_{R^*}$  }

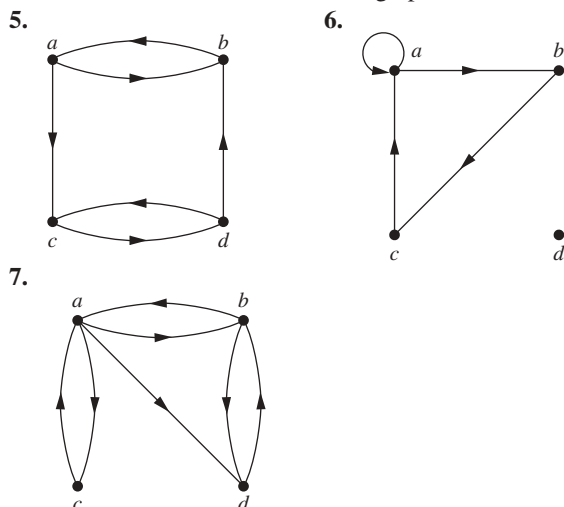
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The computational complexity of Warshall's algorithm can easily be computed in terms of bit operations. To find the entry $w_{ij}^{[k]}$ from the entries $w_{ij}^{[k-1]}$, $w_{ik}^{[k-1]}$, and $w_{kj}^{[k-1]}$ using Lemma 2 requires two bit operations. To find all n^2 entries of $\mathbf{W}_k$ from those of $\mathbf{W}_{k-1}$ requires $2n^2$ bit operations. Because Warshall's algorithm begins with $\mathbf{W}_0 = \mathbf{M}_R$ and computes the sequence of n zero-one matrices $\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_n = \mathbf{M}_{R^*}$, the total number of bit operations used is $n \cdot 2n^2 = 2n^3$.

Exercises

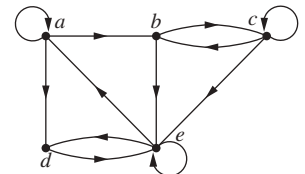
- Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the
 - reflexive closure of R .
 - symmetric closure of R .
- Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers. What is the reflexive closure of R ?
- Let R be the relation $\{(a, b) \mid a \text{ divides } b\}$ on the set of integers. What is the symmetric closure of R ?
- How can the directed graph representing the reflexive closure of a relation on a finite set be constructed from the directed graph of the relation?

In Exercises 5–7 draw the directed graph of the reflexive closure of the relations with the directed graph shown.



- How can the directed graph representing the symmetric closure of a relation on a finite set be constructed from the directed graph for this relation?
- Find the directed graphs of the symmetric closures of the relations with directed graphs shown in Exercises 5–7.

- Find the smallest relation containing the relation in Example 2 that is both reflexive and symmetric.
- Find the directed graph of the smallest relation that is both reflexive and symmetric that contains each of the relations with directed graphs shown in Exercises 5–7.
- Suppose that the relation R on the finite set A is represented by the matrix $\mathbf{M}_R$. Show that the matrix that represents the reflexive closure of R is $\mathbf{M}_R \vee \mathbf{I}_n$.
- Suppose that the relation R on the finite set A is represented by the matrix $\mathbf{M}_R$. Show that the matrix that represents the symmetric closure of R is $\mathbf{M}_R \vee \mathbf{M}_R^t$.
- Show that the closure of a relation R with respect to a property $\mathbf{P}$, if it exists, is the intersection of all the relations with property $\mathbf{P}$ that contain R .
- When is it possible to define the “irreflexive closure” of a relation R , that is, a relation that contains R , is irreflexive, and is contained in every irreflexive relation that contains R ?
- Determine whether these sequences of vertices are paths in this directed graph.
 - a, b, c, e
 - b, e, c, b, e
 - a, a, b, e, d, e
 - b, c, e, d, a, a, b
 - b, c, c, b, e, d, e, d
 - $a, a, b, b, c, c, b, e, d$



- Find all circuits of length three in the directed graph in Exercise 16.
- Determine whether there is a path in the directed graph in Exercise 16 beginning at the first vertex given and ending at the second vertex given.

a) a, b	b) b, a	c) b, b
d) a, e	e) b, d	f) c, d
g) d, d	h) e, a	i) e, c

19. Let R be the relation on the set $\{1, 2, 3, 4, 5\}$ containing the ordered pairs $(1, 3)$, $(2, 4)$, $(3, 1)$, $(3, 5)$, $(4, 3)$, $(5, 1)$, $(5, 2)$, and $(5, 4)$. Find
 - a) R^2 . b) R^3 . c) R^4 .
 - d) R^5 . e) R^6 . f) R^* .
20. Let R be the relation that contains the pair (a, b) if a and b are cities such that there is a direct nonstop airline flight from a to b . When is (a, b) in
 - a) R^2 ? b) R^3 ? c) R^* ?
21. Let R be the relation on the set of all students containing the ordered pair (a, b) if a and b are in at least one common class and $a \neq b$. When is (a, b) in
 - a) R^2 ? b) R^3 ? c) R^* ?
22. Suppose that the relation R is reflexive. Show that R^* is reflexive.
23. Suppose that the relation R is symmetric. Show that R^* is symmetric.
24. Suppose that the relation R is irreflexive. Is the relation R^2 necessarily irreflexive?
25. Use Algorithm 1 to find the transitive closures of these relations on $\{1, 2, 3, 4\}$.
 - a) $\{(1, 2), (2, 1), (2, 3), (3, 4), (4, 1)\}$
 - b) $\{(2, 1), (2, 3), (3, 1), (3, 4), (4, 1), (4, 3)\}$
 - c) $\{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$
 - d) $\{(1, 1), (1, 4), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 2)\}$
26. Use Algorithm 1 to find the transitive closures of these relations on $\{a, b, c, d, e\}$.
 - a) $\{(a, c), (b, d), (c, a), (d, b), (e, d)\}$
 - b) $\{(b, c), (b, e), (c, e), (d, a), (e, b), (e, c)\}$
 - c) $\{(a, b), (a, c), (a, e), (b, a), (b, c), (c, a), (c, b), (d, a), (e, d)\}$
 - d) $\{(a, e), (b, a), (b, d), (c, d), (d, a), (d, c), (e, a), (e, b), (e, c), (e, e)\}$
27. Use Warshall's algorithm to find the transitive closures of the relations in Exercise 25.
28. Use Warshall's algorithm to find the transitive closures of the relations in Exercise 26.
29. Find the smallest relation containing the relation $\{(1, 2), (1, 4), (3, 3), (4, 1)\}$ that is
 - a) reflexive and transitive.
 - b) symmetric and transitive.
 - c) reflexive, symmetric, and transitive.
30. Finish the proof of the case when $a \neq b$ in Lemma 1.
31. Algorithms have been devised that use $O(n^{2.8})$ bit operations to compute the Boolean product of two $n \times n$ zero-one matrices. Assuming that these algorithms can be used, give big- O estimates for the number of bit operations using Algorithm 1 and using Warshall's algorithm to find the transitive closure of a relation on a set with n elements.
- *32. Devise an algorithm using the concept of interior vertices in a path to find the length of the shortest path between two vertices in a directed graph, if such a path exists.
33. Adapt Algorithm 1 to find the reflexive closure of the transitive closure of a relation on a set with n elements.
34. Adapt Warshall's algorithm to find the reflexive closure of the transitive closure of a relation on a set with n elements.
35. Show that the closure with respect to the property $\mathbf{P}$ of the relation $R = \{(0, 0), (0, 1), (1, 1), (2, 2)\}$ on the set $\{0, 1, 2\}$ does not exist if $\mathbf{P}$ is the property
 - a) "is not reflexive."
 - b) "has an odd number of elements."
36. Give an example of a relation R on the set $\{a, b, c\}$ such that the symmetric closure of the reflexive closure of the transitive closure of R is not transitive.

9.5 Equivalence Relations

9.5.1 Introduction

In some programming languages the names of variables can contain an unlimited number of characters. However, there is a limit on the number of characters that are checked when a compiler determines whether two variables are equal. For instance, in traditional C, only the first eight characters of a variable name are checked by the compiler. (These characters are uppercase or lowercase letters, digits, or underscores.) Consequently, the compiler considers strings longer than eight characters that agree in their first eight characters the same. Let R be the relation on the set of strings of characters such that sRt , where s and t are two strings, if s and t are at least eight characters long and the first eight characters of s and t agree, or $s = t$. It is easy to see that R is reflexive, symmetric, and transitive. Moreover, R divides the set of all strings into classes, where all strings in a particular class are considered the same by a compiler for traditional C.

The integers a and b are related by the "congruence modulo 4" relation when 4 divides $a - b$. We will show later that this relation is reflexive, symmetric, and transitive. It is not hard to see that a is related to b if and only if a and b have the same remainder when divided by 4. It follows that this relation splits the set of integers into four different classes.

When we care only what remainder an integer leaves when it is divided by 4, we need only know which class it is in, not its particular value.

These two relations, R and congruence modulo 4, are examples of equivalence relations, namely, relations that are reflexive, symmetric, and transitive. In this section we will show that such relations split sets into disjoint classes of equivalent elements. Equivalence relations arise whenever we care only whether an element of a set is in a certain class of elements, instead of caring about its particular identity.

9.5.2 Equivalence Relations



In this section we will study relations with a particular combination of properties that allows them to be used to relate objects that are similar in some way.

Definition 1

A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Equivalence relations are important in every branch of mathematics!

Equivalence relations are important throughout mathematics and computer science. One reason for this is that in an equivalence relation, when two elements are related it makes sense to say they are equivalent.

Definition 2

Two elements a and b that are related by an equivalence relation are called *equivalent*. The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

For the notion of equivalent elements to make sense, every element should be equivalent to itself, as the reflexive property guarantees for an equivalence relation. It makes sense to say that a and b are related (not just that a is related to b) by an equivalence relation, because when a is related to b , by the symmetric property, b is related to a . Furthermore, because an equivalence relation is transitive, if a and b are equivalent and b and c are equivalent, it follows that a and c are equivalent.

Examples 1–5 illustrate the notion of an equivalence relation.

EXAMPLE 1

Let R be the relation on the set of integers such that aRb if and only if $a = b$ or $a = -b$. In Section 9.1 we showed that R is reflexive, symmetric, and transitive. It follows that R is an equivalence relation. ◀

EXAMPLE 2

Let R be the relation on the set of real numbers such that aRb if and only if $a - b$ is an integer. Is R an equivalence relation?



Solution: Because $a - a = 0$ is an integer for all real numbers a , aRa for all real numbers a . Hence, R is reflexive. Now suppose that aRb . Then $a - b$ is an integer, so $b - a$ is also an integer. Hence, bRa . It follows that R is symmetric. If aRb and bRc , then $a - b$ and $b - c$ are integers. Therefore, $a - c = (a - b) + (b - c)$ is also an integer. Hence, aRc . Thus, R is transitive. Consequently, R is an equivalence relation. ◀

One of the most widely used equivalence relations is congruence modulo m , where m is an integer greater than 1.

EXAMPLE 3 Congruence Modulo m Let m be an integer with $m > 1$. Show that the relation

$$R = \{(a, b) \mid a \equiv b \pmod{m}\}$$

is an equivalence relation on the set of integers.

Solution: Recall from Section 4.1 that $a \equiv b \pmod{m}$ if and only if m divides $a - b$. Note that $a - a = 0$ is divisible by m , because $0 = 0 \cdot m$. Hence, $a \equiv a \pmod{m}$, so congruence modulo m is reflexive. Now suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m , so $a - b = km$, where k is an integer. It follows that $b - a = (-k)m$, so $b \equiv a \pmod{m}$. Hence, congruence modulo m is symmetric. Next, suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. Then m divides both $a - b$ and $b - c$. Therefore, there are integers k and l with $a - b = km$ and $b - c = lm$. Adding these two equations shows that $a - c = (a - b) + (b - c) = km + lm = (k + l)m$. Thus, $a \equiv c \pmod{m}$. Therefore, congruence modulo m is transitive. It follows that congruence modulo m is an equivalence relation. 

EXAMPLE 4 Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?

Solution: Because $l(a) = l(a)$, it follows that aRa whenever a is a string, so that R is reflexive. Next, suppose that aRb , so that $l(a) = l(b)$. Then bRa , because $l(b) = l(a)$. Hence, R is symmetric. Finally, suppose that aRb and bRc . Then $l(a) = l(b)$ and $l(b) = l(c)$. Hence, $l(a) = l(c)$, so aRc . Consequently, R is transitive. Because R is reflexive, symmetric, and transitive, it is an equivalence relation. 

EXAMPLE 5 Let n be a positive integer and S a set of strings. Suppose that R_n is the relation on S such that $sR_n t$ if and only if $s = t$, or both s and t have at least n characters and the first n characters of s and t are the same. That is, a string of fewer than n characters is related only to itself; a string s with at least n characters is related to a string t if and only if t has at least n characters and t begins with the n characters at the start of s . For example, let $n = 3$ and let S be the set of all bit strings. Then $sR_3 t$ either when $s = t$ or both s and t are bit strings of length 3 or more that begin with the same three bits. For instance, $01R_3 01$ and $00111R_3 00101$, but $01R_3 010$ and $01011R_3 01110$.

Show that for every set S of strings and every positive integer n , R_n is an equivalence relation on S .

Solution: The relation R_n is reflexive because $s = s$, so that $sR_n s$ whenever s is a string in S . If $sR_n t$, then either $s = t$ or s and t are both at least n characters long that begin with the same n characters. This means that $tR_n s$. We conclude that R_n is symmetric.

Now suppose that $sR_n t$ and $tR_n u$. Then either $s = t$ or s and t are at least n characters long and s and t begin with the same n characters, and either $t = u$ or t and u are at least n characters long and t and u begin with the same n characters. From this, we can deduce that either $s = u$ or both s and u are n characters long and s and u begin with the same n characters (because in this case we know that s , t , and u are all at least n characters long and both s and u begin with the same n characters as t does). Consequently, R_n is transitive. It follows that R_n is an equivalence relation. 

In Examples 6 and 7 we look at two relations that are not equivalence relations.

EXAMPLE 6 Show that the “divides” relation on the set of positive integers is not an equivalence relation.

Solution: By Examples 9 and 15 in Section 9.1, we know that the “divides” relation is reflexive and transitive. However, by Example 12 in Section 9.1, we know that this relation is not

symmetric (for instance, $2 \mid 4$ but $4 \nmid 2$). We conclude that the “divides” relation on the set of positive integers is not an equivalence relation. 

EXAMPLE 7 Let R be the relation on the set of real numbers such that xRy if and only if x and y are real numbers that differ by less than 1, that is, $|x - y| < 1$. Show that R is not an equivalence relation.

Solution: R is reflexive because $|x - x| = 0 < 1$ whenever $x \in \mathbf{R}$. R is symmetric, for if xRy , where x and y are real numbers, then $|x - y| < 1$, which tells us that $|y - x| = |x - y| < 1$, so that yRx . However, R is not an equivalence relation because it is not transitive. Take $x = 2.8$, $y = 1.9$, and $z = 1.1$, so that $|x - y| = |2.8 - 1.9| = 0.9 < 1$, $|y - z| = |1.9 - 1.1| = 0.8 < 1$, but $|x - z| = |2.8 - 1.1| = 1.7 > 1$. That is, $2.8R1.9$, $1.9R1.1$, but $2.8 \nR 1.1$. 

9.5.3 Equivalence Classes

Let A be the set of all students in your school who graduated from high school. Consider the relation R on A that consists of all pairs (x, y) , where x and y graduated from the same high school. Given a student x , we can form the set of all students equivalent to x with respect to R . This set consists of all students who graduated from the same high school as x did. This subset of A is called an equivalence class of the relation.

Definition 3

Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the *equivalence class* of a . The equivalence class of a with respect to R is denoted by $[a]_R$. When only one relation is under consideration, we can delete the subscript R and write $[a]$ for this equivalence class.

In other words, if R is an equivalence relation on a set A , the equivalence class of the element a is

$$[a]_R = \{s \mid (a, s) \in R\}.$$

If $b \in [a]_R$, then b is called a **representative** of this equivalence class. Any element of a class can be used as a representative of this class. That is, there is nothing special about the particular element chosen as the representative of the class.

EXAMPLE 8 What is the equivalence class of an integer for the equivalence relation of Example 1?

Solution: Because an integer is equivalent to itself and its negative in this equivalence relation, it follows that $[a] = \{-a, a\}$. This set contains two distinct integers unless $a = 0$. For instance, $[7] = \{-7, 7\}$, $[-5] = \{-5, 5\}$, and $[0] = \{0\}$. 

EXAMPLE 9 What are the equivalence classes of 0, 1, 2, and 3 for congruence modulo 4?

Extra Examples 

Solution: The equivalence class of 0 contains all integers a such that $a \equiv 0 \pmod{4}$. The integers in this class are those divisible by 4. Hence, the equivalence class of 0 for this relation is

$$[0] = \{\dots, -8, -4, 0, 4, 8, \dots\}.$$

The equivalence class of 1 contains all the integers a such that $a \equiv 1 \pmod{4}$. The integers in this class are those that have a remainder of 1 when divided by 4. Hence, the equivalence class of 1 for this relation is

$$[1] = \{\dots, -7, -3, 1, 5, 9, \dots\}.$$

The equivalence class of 2 contains all the integers a such that $a \equiv 2 \pmod{4}$. The integers in this class are those that have a remainder of 2 when divided by 4. Hence, the equivalence class of 2 for this relation is

$$[2] = \{\dots, -6, -2, 2, 6, 10, \dots\}.$$

The equivalence class of 3 contains all the integers a such that $a \equiv 3 \pmod{4}$. The integers in this class are those that have a remainder of 3 when divided by 4. Hence, the equivalence class of 3 for this relation is

$$[3] = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

Note that every integer is in exactly one of the four equivalence classes and that the integer n is in the class containing $n \bmod 4$. 

In Example 9 the equivalence classes of 0, 1, 2, and 3 with respect to congruence modulo 4 were found. Example 9 can easily be generalized, replacing 4 with any positive integer m . The equivalence classes of the relation congruence modulo m are called the **congruence classes modulo m** . The congruence class of an integer a modulo m is denoted by $[a]_m$, so $[a]_m = \{\dots, a - 2m, a - m, a, a + m, a + 2m, \dots\}$. For instance, from Example 9 we have $[0]_4 = \{\dots, -8, -4, 0, 4, 8, \dots\}$, $[1]_4 = \{\dots, -7, -3, 1, 5, 9, \dots\}$, $[2]_4 = \{\dots, -6, -2, 2, 6, 10, \dots\}$, and $[3]_4 = \{\dots, -5, -1, 3, 7, 11, \dots\}$.

EXAMPLE 10 What is the equivalence class of the string 0111 with respect to the equivalence relation R_3 from Example 5 on the set of all bit strings? (Recall that $sR_3 t$ if and only if s and t are bit strings with $s = t$ or s and t are strings of at least three bits that start with the same three bits.)

Solution: The bit strings equivalent to 0111 are the bit strings with at least three bits that begin with 011. These are the bit strings 011, 0110, 0111, 01100, 01101, 01110, 01111, and so on. Consequently,

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\}. \quad \text{◀}$$

EXAMPLE 11 Identifiers in the C Programming Language In the C programming language, an **identifier** is the name of a variable, a function, or another type of entity. Each identifier is a nonempty string of characters where each character is a lowercase or an uppercase English letter, a digit, or an underscore, and the first character is a lowercase or an uppercase English letter. Identifiers can be any length. This allows developers to use as many characters as they want to name an entity, such as a variable. However, for compilers for some versions of C, there is a limit on the number of characters checked when two names are compared to see whether they refer to the same thing. For example, Standard C compilers consider two identifiers the same when they agree in their first 31 characters. Consequently, developers must be careful not to use identifiers with the same initial 31 characters for different things. We see that two identifiers are considered the same when they are related by the relation R_{31} in Example 5. Using Example 5, we know that R_{31} , on the set of all identifiers in Standard C, is an equivalence relation.

What are the equivalence classes of each of the identifiers `Number_of_tropical_storms`, `Number_of_named_tropical_storms`, and `Number_of_named_tropical_storms_in_the_Atlantic_in_2017`?

Solution: Note that when an identifier is less than 31 characters long, by the definition of R_{31} , its equivalence class contains only itself. Because the identifier `Number_of_tropical_storms` is 25 characters long, its equivalence class contains exactly one element, namely, itself.

The identifier `Number_of_named_tropical_storms` is exactly 31 characters long. An identifier is equivalent to it when it starts with these same 31 characters. Consequently, every identifier at least 31 characters long that starts with `Number_of_named_tropical_storms` is equivalent to this identifier. It follows that the equivalence class of `Number_of_named_tropical_storms` is the set of all identifiers that begin with the 31 characters `Number_of_named_tropical_storms`.

An identifier is equivalent to the `Number_of_named_tropical_storms_in_the_Atlantic_in_2017` if and only if it begins with its first 31 characters. Because these characters are `Number_of_named_tropical_storms`, we see that an identifier is equivalent to `Number_of_named_tropical_storms_in_the_Atlantic_in_2017` if and only if it is equivalent to `Number_of_named_tropical_storms`. It follows that these last two identifiers have the same equivalence class. 

9.5.4 Equivalence Classes and Partitions

Let A be the set of students at your school who are majoring in exactly one subject, and let R be the relation on A consisting of pairs (x, y) , where x and y are students with the same major. Then R is an equivalence relation, as the reader should verify. We can see that R splits all students in A into a collection of disjoint subsets, where each subset contains students with a specified major. For instance, one subset contains all students majoring (just) in computer science, and a second subset contains all students majoring in history. Furthermore, these subsets are equivalence classes of R . This example illustrates how the equivalence classes of an equivalence relation partition a set into disjoint, nonempty subsets. We will make these notions more precise in the following discussion.

Let R be a relation on the set A . Theorem 1 shows that the equivalence classes of two elements of A are either identical or disjoint.

THEOREM 1

Let R be an equivalence relation on a set A . These statements for elements a and b of A are equivalent:

- (i) aRb (ii) $[a] = [b]$ (iii) $[a] \cap [b] \neq \emptyset$

Proof: We first show that (i) implies (ii). Assume that aRb . We will prove that $[a] = [b]$ by showing $[a] \subseteq [b]$ and $[b] \subseteq [a]$. Suppose $c \in [a]$. Then aRc . Because aRb and R is symmetric, we know that bRa . Furthermore, because R is transitive and bRa and aRc , it follows that bRc . Hence, $c \in [b]$. This shows that $[a] \subseteq [b]$. The proof that $[b] \subseteq [a]$ is similar; it is left as an exercise for the reader.

Second, we will show that (ii) implies (iii). Assume that $[a] = [b]$. It follows that $[a] \cap [b] \neq \emptyset$ because $[a]$ is nonempty (because $a \in [a]$ because R is reflexive).

Next, we will show that (iii) implies (i). Suppose that $[a] \cap [b] \neq \emptyset$. Then there is an element c with $c \in [a]$ and $c \in [b]$. In other words, aRc and bRc . By the symmetric property, cRb . Then by transitivity, because aRc and cRb , we have aRb .

Because (i) implies (ii), (ii) implies (iii), and (iii) implies (i), the three statements, (i), (ii), and (iii), are equivalent. 

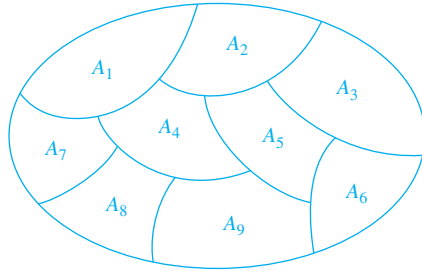


FIGURE 1 A partition of a set.

We are now in a position to show how an equivalence relation *partitions* a set. Let R be an equivalence relation on a set A . The union of the equivalence classes of R is all of A , because an element a of A is in its own equivalence class, namely, $[a]_R$. In other words,

$$\bigcup_{a \in A} [a]_R = A.$$

In addition, from Theorem 1, it follows that these equivalence classes are either equal or disjoint, so

$$[a]_R \cap [b]_R = \emptyset,$$

when $[a]_R \neq [b]_R$.

These two observations show that the equivalence classes form a partition of A , because they split A into disjoint subsets. More precisely, a **partition** of a set S is a collection of disjoint nonempty subsets of S that have S as their union. In other words, the collection of subsets A_i , $i \in I$ (where I is an index set) forms a partition of S if and only if

$$A_i \neq \emptyset \text{ for } i \in I,$$

$$A_i \cap A_j = \emptyset \text{ when } i \neq j,$$

and

$$\bigcup_{i \in I} A_i = S.$$

(Here the notation $\bigcup_{i \in I} A_i$ represents the union of the sets A_i for all $i \in I$.) Figure 1 illustrates the concept of a partition of a set.

EXAMPLE 12 Suppose that $S = \{1, 2, 3, 4, 5, 6\}$. The collection of sets $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ forms a partition of S , because these sets are disjoint and their union is S . ◀

We have seen that the equivalence classes of an equivalence relation on a set form a partition of the set. The subsets of S in this partition are the equivalence classes. Conversely, every partition of a set can be used to form an equivalence relation. Two elements are equivalent with respect to this relation if and only if they are in the same subset of S in the partition.

To see this, assume that $\{A_i \mid i \in I\}$ is a partition on S . Let R be the relation on S consisting of the pairs (x, y) , where x and y belong to the same subset A_i in the partition. To show that R is an equivalence relation we must show that R is reflexive, symmetric, and transitive.

We see that $(a, a) \in R$ for every $a \in S$, because a is in the same subset of S as itself. Hence, R is reflexive. If $(a, b) \in R$, then b and a are in the same subset of S in the partition, so that

Recall that an *index set* is a set whose members label, or index, the elements of a set.

$(b, a) \in R$ as well. Hence, R is symmetric. If $(a, b) \in R$ and $(b, c) \in R$, then a and b are in the same subset X of S in the partition, and b and c are in the same subset Y of S of the partition. Because the subsets of S in the partition are disjoint and b belongs to X and Y , it follows that $X = Y$. Consequently, a and c belong to the same subset of S in the partition, so $(a, c) \in R$. Thus, R is transitive.

It follows that R is an equivalence relation. The equivalence classes of R consist of subsets of S containing related elements, and by the definition of R , these are the subsets of S in the partition. Theorem 2 summarizes the connections we have established between equivalence relations and partitions.

THEOREM 2

Let R be an equivalence relation on a set S . Then the equivalence classes of R form a partition of S . Conversely, given a partition $\{A_i \mid i \in I\}$ of the set S , there is an equivalence relation R that has the sets A_i , $i \in I$, as its equivalence classes.

Example 13 shows how to construct an equivalence relation from a partition.

EXAMPLE 13 List the ordered pairs in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ of $S = \{1, 2, 3, 4, 5, 6\}$, given in Example 12.

Solution: The subsets of S in the partition are the equivalence classes of R . The pair $(a, b) \in R$ if and only if a and b are in the same subset of the S in the partition. The pairs $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 1)$, $(2, 2)$, $(2, 3)$, $(3, 1)$, $(3, 2)$, and $(3, 3)$ belong to R because $A_1 = \{1, 2, 3\}$ is an equivalence class; the pairs $(4, 4)$, $(4, 5)$, $(5, 4)$, and $(5, 5)$ belong to R because $A_2 = \{4, 5\}$ is an equivalence class; and finally the pair $(6, 6)$ belongs to R because $\{6\}$ is an equivalence class. No pair other than those listed belongs to R . 

The congruence classes modulo m provide a useful illustration of Theorem 2. There are m different congruence classes modulo m , corresponding to the m different remainders possible when an integer is divided by m . These m congruence classes are denoted by $[0]_m$, $[1]_m$, $\dots$, $[m-1]_m$. They form a partition of the set of integers.

EXAMPLE 14 What are the sets in the partition of the integers arising from congruence modulo 4?

Solution: In Example 9 we found the four congruence classes, $[0]_4$, $[1]_4$, $[2]_4$, and $[3]_4$. They are the sets

$$\begin{aligned} [0]_4 &= \{\dots, -8, -4, 0, 4, 8, \dots\}, \\ [1]_4 &= \{\dots, -7, -3, 1, 5, 9, \dots\}, \\ [2]_4 &= \{\dots, -6, -2, 2, 6, 10, \dots\}, \\ [3]_4 &= \{\dots, -5, -1, 3, 7, 11, \dots\}. \end{aligned}$$

These congruence classes are disjoint, and every integer is in exactly one of them. In other words, as Theorem 2 says, these congruence classes form a partition. 

We now provide an example of a partition of the set of all strings arising from an equivalence relation on this set.

EXAMPLE 15 Let R_3 be the relation from Example 5. What are the sets in the partition of the set of all bit strings arising from the relation R_3 on the set of all bit strings? (Recall that sR_3t , where s and t are bit strings, if $s = t$ or s and t are bit strings with at least three bits that agree in their first three bits.)

Solution: Note that every bit string of length less than three is equivalent only to itself. Hence $[\lambda]_{R_3} = \{\lambda\}$, $[0]_{R_3} = \{0\}$, $[1]_{R_3} = \{1\}$, $[00]_{R_3} = \{00\}$, $[01]_{R_3} = \{01\}$, $[10]_{R_3} = \{10\}$, and $[11]_{R_3} = \{11\}$. Note that every bit string of length three or more is equivalent to one of the eight bit strings 000, 001, 010, 011, 100, 101, 110, and 111. We have

$$[000]_{R_3} = \{000, 0000, 0001, 00000, 00001, 00010, 00011, \dots\},$$

$$[001]_{R_3} = \{001, 0010, 0011, 00100, 00101, 00110, 00111, \dots\},$$

$$[010]_{R_3} = \{010, 0100, 0101, 01000, 01001, 01010, 01011, \dots\},$$

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\},$$

$$[100]_{R_3} = \{100, 1000, 1001, 10000, 10001, 10010, 10011, \dots\},$$

$$[101]_{R_3} = \{101, 1010, 1011, 10100, 10101, 10110, 10111, \dots\},$$

$$[110]_{R_3} = \{110, 1100, 1101, 11000, 11001, 11010, 11011, \dots\},$$

$$[111]_{R_3} = \{111, 1110, 1111, 11100, 11101, 11110, 11111, \dots\}.$$

These 15 equivalence classes are disjoint and every bit string is in exactly one of them. As Theorem 2 tells us, these equivalence classes partition the set of all bit strings. 

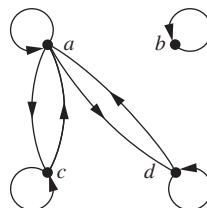
Exercises

- Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.
 - $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
 - $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
 - $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
 - $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
 - $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$
- Which of these relations on the set of all people are equivalence relations? Determine the properties of an equivalence relation that the others lack.
 - $\{(a, b) \mid a \text{ and } b \text{ are the same age}\}$
 - $\{(a, b) \mid a \text{ and } b \text{ have the same parents}\}$
 - $\{(a, b) \mid a \text{ and } b \text{ share a common parent}\}$
 - $\{(a, b) \mid a \text{ and } b \text{ have met}\}$
 - $\{(a, b) \mid a \text{ and } b \text{ speak a common language}\}$
- Which of these relations on the set of all functions from $\mathbf{Z}$ to $\mathbf{Z}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.
 - $\{(f, g) \mid f(1) = g(1)\}$
 - $\{(f, g) \mid f(0) = g(0) \text{ or } f(1) = g(1)\}$
 - $\{(f, g) \mid f(x) - g(x) = 1 \text{ for all } x \in \mathbf{Z}\}$
 - $\{(f, g) \mid \text{for some } C \in \mathbf{Z}, \text{ for all } x \in \mathbf{Z}, f(x) - g(x) = C\}$
 - $\{(f, g) \mid f(0) = g(1) \text{ and } f(1) = g(0)\}$
- Define three equivalence relations on the set of students in your discrete mathematics class different from the relations discussed in the text. Determine the equivalence classes for each of these equivalence relations.
- Define three equivalence relations on the set of buildings on a college campus. Determine the equivalence classes for each of these equivalence relations.
- Define three equivalence relations on the set of classes offered at your school. Determine the equivalence classes for each of these equivalence relations.
- Show that the relation of logical equivalence on the set of all compound propositions is an equivalence relation. What are the equivalence classes of $\mathbf{F}$ and of $\mathbf{T}$?
- Let R be the relation on the set of all sets of real numbers such that SRT if and only if S and T have the same cardinality. Show that R is an equivalence relation. What are the equivalence classes of the sets $\{0, 1, 2\}$ and $\mathbf{Z}$?
- Suppose that A is a nonempty set, and f is a function that has A as its domain. Let R be the relation on A consisting of all ordered pairs (x, y) such that $f(x) = f(y)$.
 - Show that R is an equivalence relation on A .
 - What are the equivalence classes of R ?
- Suppose that A is a nonempty set and R is an equivalence relation on A . Show that there is a function f with A as its domain such that $(x, y) \in R$ if and only if $f(x) = f(y)$.

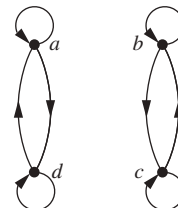
11. Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings of length three or more that agree in their first three bits is an equivalence relation on the set of all bit strings of length three or more.
12. Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings of length three or more that agree except perhaps in their first three bits is an equivalence relation on the set of all bit strings of length three or more.
13. Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings that agree in their first and third bits is an equivalence relation on the set of all bit strings of length three or more.
14. Let R be the relation consisting of all pairs (x, y) such that x and y are strings of uppercase and lowercase English letters with the property that for every positive integer n , the n th characters in x and y are the same letter, either uppercase or lowercase. Show that R is an equivalence relation.
15. Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation.
16. Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $ad = bc$. Show that R is an equivalence relation.
17. (*Requires calculus*)
 - a) Show that the relation R on the set of all differentiable functions from $\mathbf{R}$ to $\mathbf{R}$ consisting of all pairs (f, g) such that $f'(x) = g'(x)$ for all real numbers x is an equivalence relation.
 - b) Which functions are in the same equivalence class as the function $f(x) = x^2$?
18. (*Requires calculus*)
 - a) Let n be a positive integer. Show that the relation R on the set of all polynomials with real-valued coefficients consisting of all pairs (f, g) such that $f^{(n)}(x) = g^{(n)}(x)$ is an equivalence relation. [Here $f^{(n)}(x)$ is the n th derivative of $f(x)$.]
 - b) Which functions are in the same equivalence class as the function $f(x) = x^4$, where $n = 3$?
19. Let R be the relation on the set of all URLs (or Web addresses) such that xRy if and only if the Web page at x is the same as the Web page at y . Show that R is an equivalence relation.
20. Let R be the relation on the set of all people who have visited a particular Web page such that xRy if and only if person x and person y have followed the same set of links starting at this Web page (going from Web page to Web page until they stop using the Web). Show that R is an equivalence relation.

In Exercises 21–23 determine whether the relation with the directed graph shown is an equivalence relation.

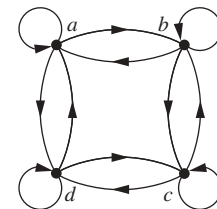
21.



22.



23.



24. Determine whether the relations represented by these zero-one matrices are equivalence relations.

a)
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

25. Show that the relation R on the set of all bit strings such that sRt if and only if s and t contain the same number of 1s is an equivalence relation.
26. What are the equivalence classes of the equivalence relations in Exercise 1?
27. What are the equivalence classes of the equivalence relations in Exercise 2?
28. What are the equivalence classes of the equivalence relations in Exercise 3?
29. What is the equivalence class of the bit string 011 for the equivalence relation in Exercise 25?
30. What are the equivalence classes of these bit strings for the equivalence relation in Exercise 11?
 - a) 010 b) 1011 c) 11111 d) 01010101
31. What are the equivalence classes of the bit strings in Exercise 30 for the equivalence relation from Exercise 12?
32. What are the equivalence classes of the bit strings in Exercise 30 for the equivalence relation from Exercise 13?
33. What are the equivalence classes of the bit strings in Exercise 30 for the equivalence relation R_4 from Example 5 on the set of all bit strings? (Recall that bit strings s and t are equivalent under R_4 if and only if they are equal or they are both at least four bits long and agree in their first four bits.)
34. What are the equivalence classes of the bit strings in Exercise 30 for the equivalence relation R_5 from Example 5 on the set of all bit strings? (Recall that bit strings s and t are equivalent under R_5 if and only if they are equal or they are both at least five bits long and agree in their first five bits.)

35. What is the congruence class $[n]_5$ (that is, the equivalence class of n with respect to congruence modulo 5) when n is
- a) 2? b) 3? c) 6? d) -3 ?
36. What is the congruence class $[4]_m$ when m is
- a) 2? b) 3? c) 6? d) 8?
37. Give a description of each of the congruence classes modulo 6.
38. What is the equivalence class of each of these strings with respect to the equivalence relation in Exercise 14?
- a) *No* b) *Yes* c) *Help*
39. a) What is the equivalence class of $(1, 2)$ with respect to the equivalence relation in Exercise 15?
- b) Give an interpretation of the equivalence classes for the equivalence relation R in Exercise 15. [*Hint*: Look at the difference $a - b$ corresponding to (a, b) .]
40. a) What is the equivalence class of $(1, 2)$ with respect to the equivalence relation in Exercise 16?
- b) Give an interpretation of the equivalence classes for the equivalence relation R in Exercise 16. [*Hint*: Look at the ratio a/b corresponding to (a, b) .]
41. Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?
- a) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$
- b) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$
- c) $\{2, 4, 6\}, \{1, 3, 5\}$ d) $\{1, 4, 5\}, \{2, 6\}$
42. Which of these collections of subsets are partitions of $\{-3, -2, -1, 0, 1, 2, 3\}$?
- a) $\{-3, -1, 1, 3\}, \{-2, 0, 2\}$
- b) $\{-3, -2, -1, 0\}, \{0, 1, 2, 3\}$
- c) $\{-3, 3\}, \{-2, 2\}, \{-1, 1\}, \{0\}$
- d) $\{-3, -2, 2, 3\}, \{-1, 1\}$
43. Which of these collections of subsets are partitions of the set of bit strings of length 8?
- a) the set of bit strings that begin with 1, the set of bit strings that begin with 00, and the set of bit strings that begin with 01
- b) the set of bit strings that contain the string 00, the set of bit strings that contain the string 01, the set of bit strings that contain the string 10, and the set of bit strings that contain the string 11
- c) the set of bit strings that end with 00, the set of bit strings that end with 01, the set of bit strings that end with 10, and the set of bit strings that end with 11
- d) the set of bit strings that end with 111, the set of bit strings that end with 011, and the set of bit strings that end with 00
- e) the set of bit strings that contain $3k$ ones for some nonnegative integer k , the set of bit strings that contain $3k + 1$ ones for some nonnegative integer k , and the set of bit strings that contain $3k + 2$ ones for some nonnegative integer k .
44. Which of these collections of subsets are partitions of the set of integers?
- a) the set of even integers and the set of odd integers
- b) the set of positive integers and the set of negative integers
- c) the set of integers divisible by 3, the set of integers leaving a remainder of 1 when divided by 3, and the set of integers leaving a remainder of 2 when divided by 3
- d) the set of integers less than -100 , the set of integers with absolute value not exceeding 100, and the set of integers greater than 100
- e) the set of integers not divisible by 3, the set of even integers, and the set of integers that leave a remainder of 3 when divided by 6
45. Which of these are partitions of the set $\mathbf{Z} \times \mathbf{Z}$ of ordered pairs of integers?
- a) the set of pairs (x, y) , where x or y is odd; the set of pairs (x, y) , where x is even; and the set of pairs (x, y) , where y is even
- b) the set of pairs (x, y) , where both x and y are odd; the set of pairs (x, y) , where exactly one of x and y is odd; and the set of pairs (x, y) , where both x and y are even
- c) the set of pairs (x, y) , where x is positive; the set of pairs (x, y) , where y is positive; and the set of pairs (x, y) , where both x and y are negative
- d) the set of pairs (x, y) , where $3 \mid x$ and $3 \mid y$; the set of pairs (x, y) , where $3 \mid x$ and $3 \nmid y$; the set of pairs (x, y) , where $3 \nmid x$ and $3 \mid y$; and the set of pairs (x, y) , where $3 \nmid x$ and $3 \nmid y$
- e) the set of pairs (x, y) , where $x > 0$ and $y > 0$; the set of pairs (x, y) , where $x > 0$ and $y \leq 0$; the set of pairs (x, y) , where $x \leq 0$ and $y > 0$; and the set of pairs (x, y) , where $x \leq 0$ and $y \leq 0$
- f) the set of pairs (x, y) , where $x \neq 0$ and $y \neq 0$; the set of pairs (x, y) , where $x = 0$ and $y \neq 0$; and the set of pairs (x, y) , where $x \neq 0$ and $y = 0$
46. Which of these are partitions of the set of real numbers?
- a) the negative real numbers, $\{0\}$, the positive real numbers
- b) the set of irrational numbers, the set of rational numbers
- c) the set of intervals $[k, k + 1]$, $k = \dots, -2, -1, 0, 1, 2, \dots$
- d) the set of intervals $(k, k + 1)$, $k = \dots, -2, -1, 0, 1, 2, \dots$
- e) the set of intervals $(k, k + 1]$, $k = \dots, -2, -1, 0, 1, 2, \dots$
- f) the sets $\{x + n \mid n \in \mathbf{Z}\}$ for all $x \in [0, 1)$
47. List the ordered pairs in the equivalence relations produced by these partitions of $\{0, 1, 2, 3, 4, 5\}$.
- a) $\{0\}, \{1, 2\}, \{3, 4, 5\}$
- b) $\{0, 1\}, \{2, 3\}, \{4, 5\}$
- c) $\{0, 1, 2\}, \{3, 4, 5\}$
- d) $\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}$